# ROYAL CANADIAN NAVY
# QUALIFICATION STANDARD AND PLAN (QSP)

## Above Water Warfare Director (AWWD)

Naval Warfare Officer (NWO) - 00207
Naval Electronic Sensor Operator (NES OP) - 00115

NQUAL - ALBU

MITE ID – 122143

# RECORD OF AMENDMENTS

The training authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC (A)) is the designated development centre responsible for the maintenance of training. Amendments to PO standard are to be requested through NTDC (A) and approved by CNPTG. Amendments to Lesson Guidance are to be approved by CO NTDC (A).

| Revision | Approval Date | Authority |
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| 2.00 | 23/09/2019 | NPTG HQ |
| 2.01 | 18/11/2020 | NTDC(A) |
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| | | |

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# FOREWORD

1. This training is based on the specification found at :

   a) A-PD-055-002/PP-001 Maritime Surface and Sub-surface (MARS) 00207 Job-Based Occupational Specification dated 01 Mar 2015;

   b) A-PD-055-002/PP-002 Naval Electronic Sensor Operator (NES Op) 00115 Job Based Occupational Specification dated 26 Aug 2014;

   c) A-PD-055-003/PQ-001 Above Water Warfare Director (AWWD) Occupational Specialty Specification dated 11 May 2016;

2. A trainee will achieve the qualification NQUAL **ALBU** when the standards listed in all Performance Objectives have been met.

3. A trainee will achieve MITE Course ID 122143 when standards listed have been met.

4. The designated TEs for this training is indicated below:

| TE | Phase / Module | MITE course ID | NQual |
|---|---|---|---|
| NFS(A) / NFS (P) | All | 122143 | ALBU |
| | | | |
| | | | |

# CHAPTER 1: TRAINING MANAGEMENT DETAILS

## AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Above Water Warfare Director (AWWD), NQUAL ALBU. This QSP is designed to assist NTDC (A) in developing, Naval Fleet Schools in conducting and evaluating, and NPTG in validating the training.

## COURSE DURATION

2. Course duration is calculated in accordance with (IAW) NPTORD 4500-22, Annex B. For this course, a period of instruction is 50 minutes in duration. The recommended schedule is based on **maximum load**. Scheduling cells are to increase/decrease course length based on number of students enrolled on the course.

| **Course Duration Calculation** | |
|---|---|
| **Allocation of ATT** | **Periods** |
| Sum of all EOs/POs | 456 |
| Enabling Checks (ECs) | 91 |
| Performance Checks (PCs) | 153 |
| **Actual Training Time (ATT) Sub-total** | 88 **trg days** |
| **Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22, Annex B Table 3.1 with the exception that all periods are 50 min duration (BAT, TAT & PT)** | |
| Base Administrative Time (BAT) | 23 |
| Training Administrative Time (TAT) | 23 |
| Training Travel Time (TTT) | 0 |
| Physical Fitness (PT) | 62 |
| **Sub-total** | 14 **trg days** |
| **Course Duration Calculation (ATT+BAT+TTT+TAT+PT)** **5400 minutes** **Partial days must be rounded up** | 102 **trg days** |



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## REQUIRED COURSE STAFFING

3. Identify key personnel required to conduct the training.

| Position | Rank | Qualifications | Qty | PO / EO |
|---|---|---|---|---|
| Course Director | Lt(N) or above | AWWD | 1 | All |
| AWWD Instructor | Lt(N)/PO1 | AWWD | 1 (**2 if 8+ students**) | All |
| EW Instructor | PO2 or above | RQ-PO2 NESOP | 1 | PO 001/004 |
| Radar Instructor | PO2 or Above | RQ-PO2 NCIOP | 1 | PO 001/004 |
| Ballistics SME | N/A | N/A | 1 | PO 001 |
| FC Instructor | MS or Above | RQ-MS NESOP | 1 | PO 001 |
| TDL 300 Instructor | Any | TDL 300 | 1 | PO 001 |
| MTP Instructor | Lt(N) or Above | Post-ORO | 1 | PO 002 |
| Target Cell Instructor | Any | Target Cell | 1 | PO 002 |
| Fleet Training Instructor | Any | CCFL/P, Sea Training or NFR as Avail | 1 | PO 002 |
| SETT Instructor | MS or Above | RQ-MS NCIOP | 1 | PO 002 |
| Trinity SME | Any | Trinity Staff | 1 | PO 003 |
| ROE Instructor | Any | JAG Staff | 1 | PO 003 |
| Mission Scenario Evaluator | Lt(N) or Above | Post-ORO | 1 | PC 003 |
| Air Asset SME | Capt. or Above | ACSO or MFC | 1 | PO 004 |
| LRPA SME | Capt. Or Above | LRPA ACSO | 1 | PO 004 |

## COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40.

a. Maximum – 8;

b. Minimum – 4; and

c. Optimum – 6.

## PREREQUISITES

5. Prerequisites. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

d. NESOP candidates for this training shall:

(1) Be a qualified Electronic Warfare Supervisor (ALBB); and

(2) Hold a security clearance of Level 2.

e. NWO candidates for this training shall:

(1) Be of the rank of Sub-Lieutenant or above;

(2) Be recommended by their Commanding Officer; and

(3) Hold a security clearance of Level 2.

## COURSE REPORTING

6. Upon completion of the course, students shall receive a paper record of completion by means of form CF 377 Course Report and Form CF 289 (or locally produced equivalent) Certificate of Military Achievement IAW DAOD 31-9 and NPTORD 4500-51.

## PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

7. PLARs shall be conducted IAW NAVORD 4500-2.

## SPECIAL INSTRUCTIONS

Nil.

# CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE



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## PO 001 – Operate AWW Weapons and Sensors (D0060)

1. **Performance**: The AWWD will Operate AWW Weapons and Sensors (D0060).

2. **Conditions**:

   a. Given:

      (1) Halifax-class operations room, including full sensor and weapons suite;
      (2) Drills and procedures manual; and
      (3) References.

   b. Denied: Nil

   c. Environment:

      (1) At sea or alongside (simulated); and
      (2) Day or night.

3. **Standard**: The AWWD shall operate AWW weapons and sensors (D00060) by:

   a. Directing the Use of AWW Weapons, Sensors and Associated Equipment (T0004) to include:

      (1) Operating C4I systems (F00210);
      (2) Operating Combat Systems Equipment (B0030);
      (3) Operating ASuW Weapons and Sensors (D00610);
      (4) Supervising Fire Control systems operations (D00130);
      (5) Operating AWW control systems (T0003);
      (6) Optimizing EW sensor employment (C00660);
      (7) Identifying system performance deficiencies, malfunctions or failures (BT0040);
      (8) Ensuring technical readiness of the AWW system (T0007); and
      (9) Assessing AWW operations and exercises (T006) by:
         (a) Conducting team training events (G0090);
         (b) Conducting individual training events (G00020);
         (c) Assisting in team training events (G00180);
         (d) Assisting in individual training events (G00010); and
         (e) Assessing team training performance (G00140).

   b. Conducting Naval Fire Support to include:
      (1) Naval Gun Fire Support; and
      (2) Strike Missions.

4. **Training Limitations**: CMS version in the synthetic environments ashore may not be the same version in use on ship.

EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.01– Direct the AWW Team (Theory)

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | |
| **Time** | **Classification** | **Instructor** |
| 150 mins | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Degrees of Readiness<br>2) States of Preparedness and Weapon States<br>3) Battle Orders/Mission Orders<br>4) Roles and responsibilities of the AWW Team<br>5) Closing-Up and Clearing-Away<br>6) Policy Orders<br>7) Weapons Control Orders | Individual Study<br>Collaborative Problem Solving<br>Tutorial |

| Activity Steps | |
|---|---|
| | 1) Students to conduct the following readings:<br>a. SSOs Chapter 6.10.1 and 6.10.2 (20 mins)<br>b. NCPM 1000 Chapters 1 and 2. (30 mins)<br>c. Sample battle orders. (20 mins)<br>2) Instructor will provide each student with a template of the Ops room and outstations with each station numbered, but with names of each position blank. Students will be asked to write the name all of the positions in the Ops Room next to each station number. (10 mins)<br>3) Instructor will then ask each student in turn to name one of the positions and describe their duties. Both instructor and other students will use this opportunity to clarify or amplify duties and responsibilities if not known by the student being asked. This process will continue until all positions in the Ops room have been discussed. (40 mins)<br>4) The instructor will then discuss LOs 5-7, using the whiteboard to demonstrate the timeline that start with closing up, and concludes with making weapons ready. They shall discuss each step of the way, what and who is involved with each step, and the importance of clear and consistence weapons control orders (including who can issue them). (30 mins) |

| Resources | References |
|---|---|
| 1) SSOs<br>2) NCPM 1000<br>3) Sample Battle Orders<br>4) Ops Room Layout Template | 1) SSOs<br>2) NCPM 1000<br>3) Battle Orders |

| Notes to Instructor |
|---|
| Instructor should begin discussion with members of the AWW team, skipping the UWW team if time does not permit. |



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EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.02– Employ Radars and IFF

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After 001.01.01 |
| **Time** | **Classification** | **Instructor** |
| 250 min | SECRET | RQ-PO2 NCIOP<br>Assistance from AWWD |

| Learning Objectives | Method |
|---|---|
| 1) Basic Radar and EW Theory (Signal generation, Wave propagation, pulse components, PW/PRI/PRF, bands/freqs/ranges/strengths environmental effects, , 3D vs 2D, PD vs Phased Array, etc.)<br>2) Range Prediction<br>3) What is Pulse Train Analysis (PTA)<br>4) RCN Sensor Capabilities and Limitations<br>a. SMART-S<br>b. SG-180<br>c. NAV Radars<br>5) Sensor Modes and Employment in Doctrine<br>6) TWS<br>7) Integration with CMS including Data Fusion<br>8) Doctrine and ASMD WRT EPM/HPDs<br>9) IFF Modes of Operations (including Mode 5) | Lecture<br>3-Star<br>Think-Pair-Share |

| Activity Steps | |
|---|---|
| | 1) Students will read the equipment handouts for the SG-180, SMART-S, and IFF systems. (100 min)<br>2) Using presentations already developed by the NTDC (A) MIW section, the instructor will conduct a lecture on basic radar theory, followed by a lecture on LOs 1-9. Critical concepts to cover are include in LO1 (100 min)<br>3) Students will develop with a list of the 3 radar or IFF capabilities that they felt will be the most critical to their duties as AWWD. (10 min)<br>4) Students will then pair-up and discuss their individual lists, and develop with a new set of agreed upon top 3 capabilities. (10 min)<br>5) In pairs, students will then present their lists to the class. The other students and the instructor will comment on and discuss their lists. (30 min) |

| Resources | References |
|---|---|
| 1) PP Presentation<br>2) Equipment Handouts | 1) Radar Manual(s) |

| Notes to Instructor |
|---|
| |



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EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.03 – Employ Tactical Data Links

| Conditions | | Sequence |
|---|---|---|
| Classroom | | After 001.01.02 |
| **Time** | **Classification** | **Instructor** |
| 200 min | UNCLASS | Guest Lecturer with TDL 300 Qualification |

| Learning Objectives | Method |
|---|---|
| 1) What is C4ISR;<br>2) What are COP/CTP/RMP/RAP;<br>3) What is TDL, what are the versions, and how do they work;<br>4) What are the differences link messages (11/16/22);<br>5) How are link networks controlled (PUs, JUs, NUs);<br>6) What is the DLPS and where is it;<br>7) What are the transmitting methods and caps/lims of each method (UHF/HF/JREAP);<br>8) How is Link integrated with CMS, and how is it affected by Doctrine; and<br>9) What are Command Messages and how do you utilize them. | Student Readings<br>Lecture<br>Essence |

| Activity Steps | |
|---|---|
| | 1) Students will read the equipment handout on Link. (20 min)<br>2) Using presentations already developed by the NTDC (A) MIW section, the instructor will conduct a lecture on LOs 1-9. (125 min)<br>3) In their own words, students will be asked to write a very brief (max 100 words) explanation of how Link contributes to the RMP and why it is so important to AWW. (10 min)<br>4) Students will then pass their explanation to another student in the class. Students will then be asked to reduce then length of their given explanation to 60 words. (15 min)<br>5) Students will then pass their reduced explanation to yet another student in the class. Students will then be asked to further reduce then length of their reduced explanation to 40 words. (15 min)<br>6) Students will then read their explanations to the class. (15 min) |

| Resources | References |
|---|---|
| 1) TDL Lecture<br>2) Equipment Handout Link | 1) ADatP-33 |

| Notes to Instructor |
|---|
| Remain in the lecture to provide insight into how this specifically applies to the AWWD |

EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.04 – Employ Electronic Support Systems

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After 001.01.03 |
| **Time** | **Classification** | **Instructor** |
| 150 min | SECRET | EW Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Electronic Signature Management Equipment<br>a. Basic Description, Caps and Lims (detectable ranges and frequencies, arcs and weaknesses, effect of environmental and detection)<br>b. Unique/Discrete ESM<br>c. Correlation to libraries (MEWD/EPL)<br>d. Integration with CMS (ID Fusion/MTDF)<br>e. Doctrine in ASMD<br>2) Communications Intercept Equipment<br>a. Basic Description, Caps and Lims (How it collects and direction finding)<br>b. Monitoring of Encrypted vs Uncovered Traffic<br>c. Integration with CMS<br>3) ELINT Collection Equipment<br>a. Basic Description, Caps and Lims<br>b. Unique identifies, direction finding, mutual interference and emitter conflicts<br>c. Purpose, process, and destination of collects<br>4) SigInt Equipment (Satellite) | Lecture<br>Card-Sort |

| Activity Steps | |
|---|---|
| | 1) Students will read the equipment handouts for ELISRA, Sea Search, and the SRD 504/505. (50 min)<br>2) With the aid of a PowerPoint presentation, the instructor will conduct a lecture covering LOs 1-4. (80 min)<br>3) The class will be broken up into two groups. Each group will be given a set of cards (or sticky notes), each labeled with one of the following: (5 min)<br>a. Electronic Signal Detected<br>b. Electronic Signal Analysed<br>c. Electronic Signal entered into MEWD<br>d. Electronic Signal Identified<br>e. Electronic Signal briefed to EWS<br>f. Electronic Signal transmitted to CMS<br>g. Electronic Signal briefed to AWWD<br>h. Electronic Signal associated<br>i. Zippo Called |



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| | |
|---|---|
| | 4) Each group will then organize their cards into chronological order. When they are finished, the instructor will review their order and debrief them as necessary. (15 min) |

| Resources | References |
|---|---|
| 1) Power-Point presentation<br>2) Cards x 2<br>3) Tape | 1) Elisra Operators Manual<br>2) SRD-504/505 Manual<br>3) Sea Search Manual<br>4) MARTI 5010/5090/5120 |

| Notes to Instructor |
|---|
| |

EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.05 – Employ Decoys

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After 001.01.04 With EO 001.01.06 |
| **Time** | **Classification** | **Instructor** |
| 150 min | SECRET | EW Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Chaff:<br>a. Basic Description, Caps and Lims<br>b. IR/CHAFF/Distraction/Seduction<br>c. DUERAS vs. OMNITRAP<br>d. Wind/bloom time/RCS<br>e. Integration with CMS<br>f. De-conflicting Softkill<br>g. MASS Threat Files vs CMS SKTTs/SKTDFs<br>2) DLF:<br>a. Basic Description, Caps and Lims<br>b. Seduction vs ASMs<br>c. Wind/sea state effect<br>d. Course/speed requirements<br>e. Mutual interference with MASS<br>f. Integration with CMS | Self-Study<br>Lecture<br>Table-Topics |

| Activity Steps | |
|---|---|
| | 1) Students will read the equipment handouts for MASS and DLF. (50 min)<br>2) With the aid of training videos and a power-point presentations, the instructor will conduct a lecture covering LOs 1 and 2. (70 min)<br>3) The students will then take part in table-topics activity.<br>4) The names of each LO will be written on a piece of paper and placed in a hat/bucket. One at a time, students will take turns drawing topics from the hat. As soon as they draw their topic from the hat, they will be required to describe their topic for a minimum of 30 seconds. Students will repeat this process until all topics have been drawn. (30 min) |

| Resources | References |
|---|---|
| 1) PowerPoint presentation<br>2) Training aid videos<br>3) Hat and sticky-notes/paper<br>4) Equipment Handouts | 1) MARTI 4785<br>2) MASS Handbook<br>3) NCPM 1000 Chapter 9 |

| Notes to Instructor |
|---|
| Instructors can add additional topics to the hat if desired. |

EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.06 – Employ Electronic Attack Systems

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | With EO 001.01.05 |
| **Time** | **Classification** | **Instructor** |
| 200 min | SECRET | EW Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Basics of Electronic Attack, including Jamming, Anti-Jamming, and interactions between EA, ES and EP.<br>2) Electronic Jammers:<br>a. Function of a jammer vs. ASMs, vs. A/C<br>b. How RAMSES jams a target<br>c. Jam immediate/Freq<br>d. Bands/Freq/coverage/Strength<br>e. Effect on EW spectrum/organic sensors<br>f. RAMSES TSDB<br>g. Integration with CMS/Doctrine for ASMD | Lecture<br>Case Study |

| Activity Steps | |
|---|---|
| | 1) Students will read the equipment handout for RAMSES. (20 min)<br>2) With the aid of training videos and a power-point presentations, the instructor will conduct a lecture covering LOs 1 and 2. (100 min)<br>3) Students will read the Wikipedia articles on the Battles of Latakia and Baltim during the Yom Kippur War, followed by the declassified document "Assessment of the Weapons and Tactics Used in the October 1973 Middle East War" pages 99-107. (50 min)<br>4) Considering all of the lesson learned in this lesson and lesson 001.01.05, the instructor will lead a discussion of the two naval battles. Questions to be asked and considered during the discussion include, but are not limited to: (30 min)<br>a. What tactics made the Israeli navy so effective?<br>b. What equipment made the Israeli navy so effective?<br>c. How might the Egyptians have acted differently to mitigate the advantages of the Israeli navy's EW systems?<br>d. If a similar engagement were to take place today, would the outcomes be the same?<br>e. Are modern EW systems as effective today at defending the ship? |

| Resources | References |
|---|---|
| 1) PowerPoint presentation<br>2) Training aid videos<br>3) Wikipedia article print-outs<br>4) Case Study Article (declassified report) | 1) MARTI 5010 |

| Notes to Instructor |
|---|
| |



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EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.07 – Employ Electro-Optic/Infra-Red Sensors

| Conditions | | Sequence |
|---|---|---|
| SEACOT (Primary)<br>Classroom (Secondary) | | After 001.01.06 |
| **Time** | **Classification** | **Instructor** |
| 50 min | SECRET | EW Instructor |

| Learning Objectives | Method |
|---|---|
| 1) EO/IR Camera (SIRIUS) Capabilities and Limitations<br>2) Setup, Calibration, and De-Cluttering<br>3) Modes of Operation (Air-Surveillance, Air Alert, Surrounding Observation, Stand-by, etc.)<br>4) Integration with CMS<br>5) Employment in Doctrine | Student reading<br>Lecture/Demo |

**Activity Steps**

1) Students will read equipment handout on SIRIUS (20 min).
2) With the instructor console connected the projector, the instructor will demonstrate the functioning of SIRIUS covering the content of LOs 1-5 in the process. (30 min)
3) Some areas for inclusion in the presentation/discussion include:
    a. What does SIRIUS look for that allows it to track targets?
    b. How does it maintain target “lock”?
    c. Why is SIRIUS optimal for SS missiles?
    d. What factors of a SS missile flight profile contribute to SIRIUS being effective?
    e. How is SIRIUS controlled by CMS?

| Resources | References |
|---|---|
| 1) Instructor Station with Projector<br>2) SIRIUS equipment handout | 1) C-58-420-000/MB-001 SIRIUS Manual<br>2) Combat System User Manual (CSUM) |

**Notes to Instructor**



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EO 001.01 – Direct the Use of AWW Weapons and Sensors

# 001.01.08 – Operate Guns (Theory)

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After 001.01.07 |
| **Time** | **Classification** | **Instructor** |
| 200 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Main Armament<br>a. Basic description, caps, and lims (Cutouts/Elevation and depression/ Firing rates, reload, capacity, ranges)<br>b. Gun Integration GCC;<br>c. Ammo types (3P/PFHE/NFVT/HCER/BLP/PRAP);<br>d. Magazines/Feed system/MV Radar;<br>2) CIWS<br>a. Basic description, caps, and lims<br>b. Search and Track Radars;<br>c. Range rates, Target CPA, First hit ranges, burst settings;<br>d. Modes (AAW Auto/Manuel/Surface-active-passive-Optronic-FC Bypass/Air Ready/Standby)<br>e. Control Local/Remote;<br>f. Pre-Action Calibration and settings required;<br>g. EMCOM and Air OPs/MUTINT; and<br>h. Engage/No engage Zones (AAW vs surface tracking)<br>i. CMS Integration | Individual Study<br>Research<br>Comparison presentation |

| Activity Steps | |
|---|---|
| | 1) Students will read the handouts on the Halifax-Class Main Armament and CIWS. (30 mins)<br>2) Instructor will discuss LOs 1-2 with students (20 mins)<br>3) Students are then assigned another nations gun and CIWS system to conduct basic research and draw comparisons between the foreign nations armament compared to Canadian main armament and CIWS systems. (100 mins)<br>4) Students to present their comparison charts to the class. (50 mins) |

| Resources | References |
|---|---|
| 1) Equipment Handouts<br>2) DWAN/CSNI computer for research<br>3) Flip chart paper and markers | 1) 57mm and CIWS Equipment Manuals<br>2) Open Source, Jane's fighting ships |

| Notes to Instructor |
|---|
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EO 001.01 – Direct the Use of AWW Weapons and Sensors

# 001.01.09 – Operate Surface to Surface Missiles (Theory)

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After 001.01.08 |
| **Time** | **Classification** | **Instructor** |
| 200 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Basic description, caps, and lims (active ASM/Fire and forget)<br>2) Operating Modes and Track Sources<br>3) Seeker head/Target discrimination<br>4) Re-attack and reacquisition<br>5) Chaff discrimination<br>6) Crypto, GPS and INS<br>7) HOJ<br>8) Flight profile (SS and CTS)<br>9) Weather and Sea state effect on seeker/flight<br>10) Boost/midcourse/terminal<br>11) HAFO, Pre-search Skim, Terminal Skim/Booster separation/standard/popup/Way Points (Normal and CTS)<br>12) CMS integration<br>13) Duds | Individual Study<br>Lecture |

| Activity Steps | |
|---|---|
| | 1) Students are to read MARTI 3012 – Tactical Employment of Harpoon Anti-Ship Missile. While reading this MARTI, students are each develop a list of 20 questions and answer that they believe represent the most critical functions and capabilities of the HARPOON. Students may also make use of the HARPOON handout. Once students are done their readings and making their quiz questions, they will submit them to the instructor. (100 min)<br>2) Students will then be divided into two equal groups, and will take part in a quiz competition. The instructor will select one question from amongst all of the submitted questions, and ask it aloud to the class. The first team to answer correctly gets a point. First team to 21 points wins. (50 min)<br>3) The instructor will then spend the remainder of this lesson elaborating on any learning objectives that were not sufficiently captured. (50 min) |

| Resources | References |
|---|---|
| 1) MARTI 3012<br>2) HARPOON Equipment Handout | 1) CFCD 106 – MARTI 3012 |

| Notes to Instructor |
|---|
| Lesson must be geared toward basic understanding and system manipulation. Tactics to be covered in PO 004. Specific focus on LO 11, as this is a noted deficiency in the fleet. |



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EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.10 – Ballistic Theory

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After 001.01.09 |
| **Time** | **Classification** | **Instructor** |
| 150 min | SECRET | Guest Lecturer/AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Basic Ballistics Theory<br>a) Effect of Wind on Rounds over range<br>b) Effect of Weather (Rain/Snow etc.)<br>c) Effect of Temperature of the round<br>d) Effect of Air Temperature | Lecture |

| Activity Steps | |
|---|---|
| | 1) A SME guest lecture will conduct a lecture covering ballistic theory. (150 mins)<br>2) Units with SMEs include: CFMWC, CAF Artillery Units |

| Resources | References |
|---|---|
| See instructor note | Nil |

| Notes to Instructor |
|---|
| If SME unavailable instructor can use open source material to cover concepts. LSC will produce a recorded brief which will be added to courseware once available |



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EO 001.01 – Direct the Use of AWW Weapons and Sensors

# 001.01.11 – Operate Surface to Air Missiles (Theory)

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After 001.01.10 |
| **Time** | **Classification** | **Instructor** |
| 150 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Basic description, caps, and lims (nature of Semi Active Command line of sight)<br>2) Miscellaneous settings (salvo size and spacing, TGT, HOJ, self-destruct, etc.)<br>3) Phases of Flight/Boost/Midcourse/terminal;<br>4) ESSM modes (Mid-course guidance uplink /HAW (Home All the Way)/ Delayed Illumination/ THAW (True Home All the Way)<br>5) Limitations on flight due to speed/seeker frequency;<br>6) Telemetric settings<br>7) CWI and Rear Ref<br>8) DUD types<br>9) EIC and ICP<br>10) Command destruct/Hold fire/Cease fire<br>11) CMS integration and assignment order<br>12) Doctrine in ASMD<br>13) Test/train (when and how to use it) | Individual Study<br>Treasure Hunt |

| Activity Steps | |
|---|---|
| | 1) Instructor will provide to students the treasure hunt handout and explain the treasure hunt. (5 min).<br>2) Students will read the hand-out on the Evolved Sea Sparrow Missile and CWI. While reading, they will be required to complete (fill-in) their treasure hunt handouts. (95 min)<br>3) With the aid of training videos, the instructor will demonstrate various ESSM modes and employment methods and elaborate on any of the key learning objectives. (50 mins) |

| Resources | References |
|---|---|
| 1) Treasure Hunt Handout<br>2) Training Aid Videos | 1) MARTI 4781 |

| Notes to Instructor |
|---|
| LSC will produce a recorded brief which will be added to courseware that will provide guidance on both ESSM V1 andV2 once available |



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EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.12 – Equipment Familiarization and Tour

| Conditions | | Sequence |
|---|---|---|
| Dartmouth Dockyard Annex/Naval Armament Depot or West Coast Gun Bay<br>Halifax-Class (secondary) | | After 001.01.11 |
| **Time** | **Classification** | **Instructor** |
| 200 min (includes travel time) | Unclassified | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) RCN Equipment Familiarization<br>2) RCN Training Asset Familiarization | Tour |

| Activity Steps | |
|---|---|
| | 1) Students will conduct a tour of DDA/Gun Bay. Students will have the opportunity to examine and in some cases operate actual RCN weapon systems. Students will also have the opportunity to discuss the equipment theory they have learned thus far with CSE personnel. |

| Resources | References |
|---|---|
| 1) Nil | 1) Nil |

| Notes to Instructor |
|---|
| If for whatever reason DDA/NAD-Gun Bay is unable to support the tour, or if certain pieces of equipment normally functioning at these locations are not functioning, all or portions of the tour can be moved to an alongside CPF, if practical. |

EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.13 – Direct the AWW Team (Practical)

| Conditions | | Sequence |
|---|---|---|
| SEACOT (Primary)<br>SEAWOLF (Secondary) | | After 001.01.12 |
| **Time** | **Classification** | **Instructor** |
| 50 mins | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Closing-Up and Clearing Away the Watch<br>2) Assuming Policy | EDI |

| Activity Steps | |
|---|---|
| | 1) The instructor will demonstrate the process of closing-up, clearing away, and assuming policy. (10 mins)<br>2) Students will then pair up and take turns closing-up, clearing-away, and assuming policy, with one student acting as the SWC, and the other students acting as other members of the AWW team, including ORO. (40 mins) |

| Resources | References |
|---|---|
| 1) NCPM 1000 | 1) SSOs<br>2) NCPM 1000 |

| Notes to Instructor |
|---|
| Students should be closing up in the 2nd Degree of AWW readiness. Instructor and students are free to assume whichever policy they chose; however, in order for students to receive the greatest exposure to the full policy report, the chosen policy should incorporate all weapon systems. |



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EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.14 – Operate CMS (Practical)

| Conditions | | Sequence |
|---|---|---|
| SEACOT (Primary)<br>SEAWOLF (Secondary) | | After 001.01.13 |
| **Time** | **Classification** | **Instructor** |
| 200 Min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) How to perform basic system-setup, how to manipulate the TACPLOT, Hook contacts, navigate TACPLOT mini-panels, navigate Aux Display Panels, and manage Weapon/Sensor Adapters<br>2) How to use, modify, and enable/disable SID, TDAs, Zones, and Threat Ref Pts.<br>3) What are and how to view track data and assess/modify ID Fusions and Multi-Track Data Fusion (MTDF) in both the TACPLOT and Aux display<br>4) What are TEWA and the planners, and how to manipulate the Swim Lanes<br>5) How to modify the planners and other TEWA settings<br>6) How to change CMS Modes (SA, AA, AE)<br>7) How to CMS Manual Engagement Methods (No-Support, Manual, Urgent)<br>8) How to select/approve EPMs<br>9) How to enable/change Doctrine (MSD/HPD)<br>10) SKTT/HKTT, SKTDF/HKTDF and PK Tables<br>11) Red and Green System Status Indicators (Meatballs) | Student Readings<br>EDI<br>Treasure Hunt |

| Activity Steps | |
|---|---|
| | 1. Students are provided a copy of MARTI 4790 to read and complete the MARTI related treasure hunt questions. (30 min)<br>2. Each student will close-up at a CMS console. The instructor, using their own console connected to the projector, will then guide the students through all of the areas of CMS, focusing on (but not limited to) the areas outlined in the LOs. (135 min)<br>3. Students will then pair-up. The instructor will provide each student with a "treasure hunt" list of settings, features, and inputs that they must navigate to and activate/deactivate/fill-out. (35 min) |

| Resources | References |
|---|---|
| 1) Basic CMS system setup<br>2) Treasure Hunt List<br>3) MARTI 4790 | 1) CMS User's Manual<br>2) CMS Equipment Handout<br>3) MARTI 4790 |

## Notes to Instructor



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EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.15 – Operate Fire Control Radars (Practical)

| Conditions | | Sequence |
|---|---|---|
| SEACOT | | After 001.01.14 |
| **Time** | **Classification** | **Instructor** |
| 100 min | SECRET | AWWD Instructor<br>RQMS 5B NESOP |

| Learning Objectives | Method |
|---|---|
| 1) Basic description, caps, and lims (bands and freqs)<br>2) Support for Main Armament and SAMs<br>3) EP Measures (PD/MTI/FA)<br>4) Tracking modes (OPTRO/LASER/IR/ ACTIVE)<br>5) Integration with the Gun Ctrl Computer (GCC)<br>6) Integration with LCP for ESSM<br>7) Environmental vulnerabilities (Nulls)<br>8) Integration with CMS/Doctrine;<br>9) Powering-On (Start Methods, Radar Allow X-Mit)<br>10) Designate FCR (Radar, Optro, Laser)<br>11) SABOD<br>12) Drills and Voice Procedures for Primary and Secondary, including tracking runs | EDI |

| Activity Steps | |
|---|---|
| | 1. Students will read the equipment handout on the CEROS 200 (functional description and modes of operation only). (10 min)<br>2. Using both CMS and the simulated FC console, the instructor will demonstrate and explain the basic capabilities and functions of the Fire Control Radar system. (LOs 1-8; 40 min)<br>3. With the instructor console connected to the projector, the instructor will demonstrate how to operate the Fire Control Radar (FCR) using the TACPLOT mini-panels, SIDs, and Fire Control adapter panel in the Aux display. The instructor should emphasize the functionality of the SABOD function, and explain how it interacts with TEWA (15 min).<br>4. Each student will then close-up at a CMS console. Beginning with a shut-down FC system, each student will power-on their FC radars, and enable all necessary features until they obtain green “meatballs”. (5 min)<br>5. The instructor will then place various air and surface contacts in the vicinity of the student’s ships, giving them the opportunity to lock-up the contacts using the various manual methods of assigning the FC radars (FCR designate from the adapter panel, SABOD from the SID, etc.). (30 min) |

| Resources | References |
| --- | --- |
| 1) CEROS Equipment Handout<br>2) NCPM 1000 | 1) NCPM 1000<br>2) CEROS Equipment Manual |

| Notes to Instructor |
| --- |
| Instructor to emphasize the importance of the FCR in height finding, while also explaining that ROE can preclude the use of FCRs in such a manner. |

EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.16 – Operate Guns (Practical)

| Conditions | | Sequence |
|---|---|---|
| SEACOT (Primary)<br>SEAWOLF (Secondary) | | After 001.01.15 |
| **Time** | **Classification** | **Instructor** |
| 450 min | SECRET | AWWD Instructor and MS NESOP |

| Learning Objectives | Method |
|---|---|
| 1) Pre-action Calibration<br>2) Gun Patterns<br>3) Probability of Kill ($P_K$)<br>4) CMS Integration/Doctrine in ASMD<br>5) Main Armament drills and voice procedures in Primary<br>a. Surface (No-Support and Urgent)<br>b. Warning Shots<br>c. AA (No-Support and Urgent)<br>d. AA Semi-Auto (with and without Auto-Fire)<br>6) Main Armament drills and voice procedures in Secondary<br>a. Pre-Action Calibration<br>b. Surface<br>7) CIWS drills and voice Procedures<br>a. Pre-Action Calibration<br>b. AAW Auto<br>c. AAW Manual<br>d. Surface<br>8) Layered Main Armament and CIWS engagement | EDI<br>Fishbowl |

| Activity Steps | |
|---|---|
| | LO 1 -4<br>1) Using the instructor console connected to the projector, the instructor will discuss the concepts of LOs 1-4<br>2) The instructor will then demonstrate the conduct of primary drills and voice procedures, covering the engagements listed in LO 5. (30 min)<br>3) Students will then each close-up at a CMS display. The instructor will then walk-them through each of the same engagements, with the students following along at their own consoles. (30 min)<br>4) The instructor will then insert various air and surface contacts into the system, and students will proceed at their own pace, practicing each of the engagements types. (90 min)<br>LO 5/6/7<br>5) Using support manning for the FC and CIWS consoles, the instructor will demonstrate secondary drill and voice procedures for Main Armament PACs and Surface gunnery. (20 min) |



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| | |
|---|---|
| | 6) Two students at a time will then take turns conducting these same engagements with their FC teams (instructor should have sufficient support staff to enable 2 students to conduct engagements at a one time). Students not conducting the engagements shall observe the practicing students while they await their own turns. (20 min per student – total 80 min)<br>7) This process will be repeated for the engagements listed in LOs 7 and 8. (200 min total) |

| Resources | References |
|---|---|
| 1) Support Manning FC Sup/CIWS<br>2) NCPM 1000 | 1) NCPM 1000<br>2) MARTI 4370 |

| Notes to Instructor |
|---|
| |

EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.17 – Operate Surface to Air Missiles (Practical)

| Conditions | | Sequence |
|---|---|---|
| SEACOT (Primary)<br>SEAWOLF (Secondary) | | After 001.01.16 |
| **Time** | **Classification** | **Instructor** |
| 200 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Surface to Air Missile drills and voice procedures:<br>a. Urgent<br>b. No-Support<br>c. Semi-Auto (With and without auto-fire)<br><br>2) Layered SAM and Main Armament engagements<br>a. No-Support<br>b. Manual<br>c. Semi-Auto (with and without auto-fire) | EDI |

| Activity Steps | |
|---|---|
| | LO 1/2<br>1) Using the instructor console connected to the projector, the instructor will demonstrate the conduct of primary drills and voice procedures, covering the engagements listed in LO 1. (20 min)<br>2) Students will then each close-up at a CMS display. The instructor will then walk-them through each of the same engagements, with the students following along at their own consoles. (20 min)<br>3) The instructor will then insert various air and surface contacts into the system, and students will proceed at their own pace, practicing each of the engagements types. (60 min)<br>4) Steps 1-3 will then be repeated for LO 2. (100 min) |

| Resources | References |
|---|---|
| 1) NCPM 1000 | 1) NCPM 1000 |

| Notes to Instructor |
|---|
| Instructor to ensure students have a good grasp on manual plan settings and not rely solely on auto alone. |

EO 001.01 – Direct the Use of AWW Weapons and Sensors

## 001.01.18 – Operate Surface to Surface Missiles (Practical)

| Conditions | | Sequence |
|---|---|---|
| SEACOT (primary)<br>SEAWOLF (secondary) | | After 001.01.17 |
| **Time** | **Classification** | **Instructor** |
| 150 min | SECRET | AWWD Instructor |

| Leaning Objectives | Method |
|---|---|
| 1) Select Change Operating Modes and Track Source (Remote, Mixed, Local)<br>2) Verify Missile Inventory<br>3) Designate Target (CMS and AHWCS)<br>4) Create Plan, including:<br>• DTOT/STOT/Ripple, GPS vs INS, re-attack, HAFO, flight altitude, terminal skim vs. pop-up, seeker-head search bias, way-points<br>5) Assign/De-assign Missiles<br>6) Conducting a BF report<br>7) Manipulating the HPLEP<br>8) HARPOON misfire procedures<br>9) Firing from secondary firing position (FCER3) | Individual Study<br>EDI |

| Activity Steps | |
|---|---|
| | 1) Students will read NCPM 1000 Chapter 6 – Parts 1 and 3 (20 mins)<br>2) With the instructor console connected to the projector, the instructor will demonstrate simple HARPOON engagement planning with AHWCS. The instructor should have a simple surface picture with contacts surrounding the TG at ranges from 10-70nm. While planning their shots, the instructor shall ensure that they cover LOs 1 through 7. *see notes to instructor (25 min)<br>3) The instructor will then explain and demonstrate the appropriate response to a HARPOON misfire. (15 min)<br>4) Each student will close-up at a CMS console. The instructor will then direct each student through setting-up and executing a HARPOON engagement against a given target. (15 min)<br>5) At their own pace, students will then practice setting-up and executing HARPOON engagements against any one of the surface contact. BF reports will be given in the blind. While this is happening, the instructor will take turns with each student, one at a time, mentoring them through a misfire procedure. (70 min)<br>6) The instructor will finish the lesson by explaining that there exists an ability to operate AHWCS from FCER3, but that it does not operate exactly the same as the AHWCS in the Operations Room. As there is no simulation of *this* system in the training establishments, *students* should be encouraged to investigate this system and learn how to use it on their own time once they are posted to a ship. (5 min) |

| Resources | References |
| --- | --- |
| 1) NCPM 1000 | 1) NCPM 1000<br>2) AHWCS User Guide |

## Notes to Instructor

The aim of this lesson is not to impart upon the students the tactical employment of the various HARPOON settings in AHWCS (i.e. HAFO, terminal manoeuvre, etc.). Students need only be shown *how* to implement these functions, but not necessarily *why*. The "*why"* (i.e. the tactical decision making behind these features) will be covered in EO 004.03.04.

Explain the difference between DD:MM:SS and DD: MM.mmmm

# EC 001.01 – Direct the Use of AWW Weapons, Sensors and Associated Equipment

| Scope | Conditions | Sequence |
|---|---|---|
| EO 001.01 | SEACOT (primary)<br>SEAWOLF (secondary) | After 001.01.18 |
| **Time** | **Classification** | **Instructor** |
| 800 min | SECRET | AWWD Instructor |

| Assessment Steps | |
|---|---|
| | Input: Student is closed up with support manning detailed in NWTP 2, where they must execute the duties of the AWWD during peacetime firing.<br>Performance: students will conduct a 90 min run where they must perform the drills for:<br>• CUCA & Assume Policy<br>• PAC (main armament)<br>• PAC (CIWS)<br>• Basic main armament surface firing (no support)<br>• Warning Shots<br>• CIWS surface firing<br>• Basic main armament AA Firing (no support)<br>• Basic CIWS AA Firing<br>• ESSM Urgent (Air or SASS)<br>• ADEP<br>• Surface to Surface Missile Engagement (must include manipulation of HAFO, waypoint manipulation)<br>The scenario <u>must include</u> one error of each of the following categories:<br>1. A system setup error (ex. Radar power, gun alert states, weapon assignment, force recoil, radar allow xmit, etc.)<br>2. An incorrect drill/voice procedure conducted by team member<br>3. Equipment fault (ex. RADAR fail, veto panel unresponsive, gun hydraulic failure, gun hoist issue)<br>4. Checkfire, range foul or dangerous proximity situation<br>5. Misfire/checkload (weapons stoppage)<br>6. An a/c attack profile that will not allow for a successful engagement on the aircraft (CPA too large, altitude too high)<br>Output: Trainee must achieve 32/39 SATs on non-critical factors **AND** all critical factors to pass this EC.<br><br>Feedback: Immediate verbal and written using PC 001 checklist. 10 min allocated to reset the scenario and conduct debrief. |

| Resources | References |
|---|---|
| 1) NCPM 1000<br>2) Scenarios<br>3) TEM | 1) NCPM 1000<br>2) CFCD 106<br>3) CSUMS<br>4) NAVORD 1000-1<br>5) NAVORD 4995-0 |

| Notes to Instructor |
|---|
| Support manning IAW NWTP 2<br>No assistance will be provided to students during the assessments.<br>If student conducts a critical failure during an assessment run, the run will continue as a developmental opportunity (significant feedback notes to be taken and provided using PC checklist) |

| | |
|---|---|
| **Limitations** | NIL |
| **Impact of Failure** | Further instruction and re-assess |

| Item | EC 001.01 – DIRECT THE USE OF AWW WEAPONS, SENSORS AND ASSOCIATED EQUIPMENT | SAT | UNSAT | Remarks |
|---|---|---|---|---|
| **1. Conduct Close Up procedures** | | | | |
| Conducts CUCA IAW NCPM 1000 | | ☐ | ☐ | |
| Assumes Policy IAW NCPM 1000 | | ☐ | ☐ | |
| Address system setup error injected by Instructor | | ☐ | ☐ | |
| **2. Conduct Pre Action Calibrations** | | | | |
| Conducts main Armament PAC IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts CIWS PAC IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | | ☐ | ☐ | |
| **3. Conduct basic main Armament firing against surface target (no support)** | | | | |
| Conducts deflection checks IAW NCPM 1000 | | ☐ | ☐ | |
| Assess tracking IAW NCPM 1000 | | ☐ | ☐ | |
| Apply spotting corrections into CMS as given | | ☐ | ☐ | |
| Inputs Salvo size into CMS IAW Salvo Policy | | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | | ☐ | ☐ | |
| **4. Conduct Warning Shots** | | | | |
| Conducts Warning Shot System manipulation IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts Warning Shot voice procedures IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | | ☐ | ☐ | |
| **5. Conduct CIWS surface firing against surface target** | | | | |
| Assess tracking IAW NCPM 1000 | | ☐ | ☐ | |
| Conduct CIWS surface firing IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | | ☐ | ☐ | |
| **6. Conduct basic main Armament AA Firing (ADEP)** | | | | |
| Assess tracking IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts system manipulation IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts voice procedures IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | | ☐ | ☐ | |
| **7. Conduct basic CIWS AA Firing** | | | | |

| | | | |
|---|---|---|---|
| Assess tracking IAW NCPM 1000 | ☐ | ☐ | |
| Conduct CIWS AA firing IAW NCPM 1000 | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | ☐ | ☐ | |
| **8. Conduct ESSM (ADEP/TEWA)** | | | |
| Conducts system manipulation IAW NCPM 1000 | ☐ | ☐ | |
| Conducts voice procedures IAW NCPM 1000 | ☐ | ☐ | |
| Conducted ESSM misfire/DUD procedures IAW NCPM 1000 | | | |
| Conducts SITREPS on appropriate nets | ☐ | ☐ | |
| **9. Conduct ADEP (advanced AA layered firings)** | | | |
| Conducts system manipulation IAW NCPM 1000 | ☐ | ☐ | |
| Conducts voice procedures IAW NCPM 1000 | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | ☐ | ☐ | |
| **10. Conducts SS missile engagement** | | | |
| Conducts system manipulation IAW NCPM 1000 | ☐ | ☐ | |
| Conducts BF voice procedures IAW NCPM 1000 | ☐ | ☐ | |
| Conducts BULLDOGS AWAY voice procedures IAW NCPM 1000 | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | ☐ | ☐ | |
| **12. Respond to Faults** | | | |
| Corrected drill/voice procedure of team | ☐ | ☐ | |
| Identified system fault | ☐ | ☐ | |
| Briefed fault to ORO using SAR format | ☐ | ☐ | |
| **13. Safety **** CRITICAL****** | | | |
| ***Adhered to CO/ORO direction for weapons employment** | ☐ | ☐ | |
| ***Adhered to positive weapon control orders** | ☐ | ☐ | |
| ***Conducted checkfire, range foul or dangerous proximity situation action IAW NAVORD 4995-0** | ☐ | ☐ | |
| ***Conducted main armament Cold Gun Misfire IAW NCPM 1000** | ☐ | ☐ | |
| ***Conducted main armament Hot Gun Misfire with failed refire IAW NCPM 1000** | ☐ | ☐ | |
| ***Maintains Safe CIWS ARS throughout serial** | ☐ | ☐ | |

| | | | |
|---|---|---|---|
| ***Conducted CIWS cold or hot gun bore loaded** | ☐ | ☐ | |
| ***Conducted harpoon misfire/DUD procedures IAW NCPM 1000** | ☐ | ☐ | |

**Score: /39 + /8 CRITICAL**

Trainee must achieve 32/39 SATs on non-critical factors **AND** all critical factors to pass this PC.

| Positive Feedback | Improvement Feedback |
|---|---|
| | |

| Overall Assessment – Circle One | | | |
|---|---|---|---|
| **Met Standard** | | **Did Not Meet Standard** | |
| **Assessor**<br>I have discussed the results of this assessment with the candidate. | | **Candidate**<br>I have been debriefed on the result of this assessment | |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

PROTECTED B (when complete)

EO 001.02 Conducting Naval Fire Support

## 001.02.01 – Naval Gun Fire Support (NFS-G)

| Conditions | | Sequence |
|---|---|---|
| SEACOT (primary)<br>SEAWOLF (secondary) | | After EC 001.01 |
| **Time** | **Classification** | **Instructor** |
| 300 min | UNCLAS | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Types of Naval Gun Fire Support (Indirect/direct /suppression)<br>2) GURF Report<br>3) FOOs, Calls for Fire, and Airspace Clearance<br>4) Purpose of X-Battery<br>5) AWW team roles in NFGS<br>6) How to employ CMS in NGFS, including how to input NGFS targets in CMS | Individual Study<br>EDI<br>Fishbowl |

| Scenario | |
|---|---|
| **Activity Steps** | 1) Students will read the following: (100 min)<br>a. NCPM 1000: Ch. 7 p4<br>b. ATP 04: 0101-0215/0222/0301-0302/Annex A, D/Glossary<br>2) If available, the instructor will introduce the students to the X-Battery software and explain how X-battery is employed in NGFS. (15 min)<br>3) With the instructor console connected to the projector, the instructor will demonstrate an NGFS engagement. The demonstration should cover the topics outlined in LOs 1-6, and shall include all of the following: (35 min)<br>a. Providing a Guns Up Ready to Fire (GURF) report;<br>b. Receiving a call for fire and briefing the ORO<br>c. Inputting target in CMS<br>d. Engaging target with appropriate engagement mode<br>e. Receiving corrections from the FOO (Grid or Lat/Long)<br>f. Applying corrections from X-battery<br>4) The instructor will then have several students (minimum 3) run through several different NGFS engagements (one at a time) at the instructor console connected to the projector, pointing out how the student is proceeding to the other trainees on the projection and focusing the students on the correct procedures. (15 mins per student – 45 min total)<br>5) Each student will then close up at a CMS console, and each will provide a GURF report to the instructor. The instructor will then run the class through several NGFS engagements that involve multiple students firing at the same time. Due to manning restrictions, each student will have to use the same set of simulated X-Battery corrections. (100 mins) |

| Resources | References |
|---|---|
| 1) X-Battery Software<br>2) NGFS Scenario | 1) ATP 04<br>2) NCPM 1000<br>3) CFCD 106 |

## Notes to Instructor

Since this is a simulated engagement, the corrections from the “FOO” will be equally fictitious. Therefore, the training scenario can be executed without an actual X-Battery laptop. The instructor can simply “simulate” that the corrections that would normally be received from the EWS.

EO 001.02 Conducting Naval Fire Support

## 001.02.02 – Targeting

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom / Fleet Target Services | | After 001.02.01 |
| **Time** | **Classification** | **Instructor** |
| 400 min | SECRET | Guest Lecturer |

| Learning Objectives | Method |
|---|---|
| 1) Target life cycle<br>2) Elements of target org<br>3) JAG, INT, Image Analysts<br>4) Role of collateral Damage Estimates<br>5) Allied Integration<br>6) Allied weapons/TLAM/ SLAM/ Drones | Self-Study<br>Guided Discussion (Guest Lecturer)<br>Table Top |

**Activity Steps**

1) Students are to read MARTI 3012 Annex K, and CFJP 3-9 chapter 1, 3, 4 and 6. (50 min)
2) <u>Self-Study:</u>
   - Preface, page V, para 03;
   - What is Targeting, page 1-1, paras 0101 and 0102;
   - What is a Target, page 1-2, paras 0105, 0106, 0108, 0109, and 0110;
   - Targeting Methodologies, page 1-6, paras 0118, 0119, 0120;
   - Targeting Criteria, page 1-9 and 1-10, para 0127;
   - Self Defence and Targeting, page 2-10, para 0240;
   - Targeting Duties and Responsibilities, page 3-1, para 0301; and
   - Targeting Authority Framework, page 3-1, para 0303.

<u>Guided Discussion:</u>
Instructor is to conduct a guided discussion with students to ensure they have an understanding of the basic knowledge contained in the above readings. These readings will give the students some basic knowledge prior to the targeting presentation given the next day during 005.03.02. Ensure the students are informed that the NFS (H)/CTS process will be taught later in the course.

<u>Table-Top</u>
3) Students are to run through a table top exercise led by the guest lecturer. Scenario should be drawn from the class mission (ARBITER, VERANGIAN, etc.). They will explore how the info is presented and applicable when advising the CO. The instructor is encouraged to draw the chosen scenario on the whiteboard to enhance student understanding.

| Resources | References |
| --- | --- |
| 1) Example TEMs<br>2) MCC Targeting Cell presentation on targeting<br>3) MARTI 3012 Annex K<br>4) CFJP 3-9 | 1) CFCD 106<br>2) CFJP 3-9 |

| Notes to Instructor |
| --- |
| |

EO 001.02 Conducting Naval Fire Support

## 001.02.03 – Strike Missions (NFS-H)

| Conditions | | Sequence |
|---|---|---|
| SEACOT (primary)<br>SEAWOLF (secondary) | | After 001.02.02 |
| **Time** | **Classification** | **Instructor** |
| 150 | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Designate Target (CMS and AHWCS)<br>2) How to read a Target Engagement Message (TEM)<br>3) Create CTS plan, including LAT/LONG, TAWP, PLWP, TAA, and Altitude<br>4) Create Harpoon Plan, including:<br>a. DTOT/STOT/Ripple, GPS, HAFO, flight altitude, dive angles, way-points<br>5) Verifying Crypto and Wind<br>6) Conducting 9-line verification with the CO | Individual Study<br>EDI |

| Scenario | |
|---|---|
| **Activity Steps** | 1) Students are to read: (50 mins)<br>a. MARTI 3012 Annex K<br>b. NCPM 1000 NFS-H Procedures<br>2) Each student will be given a copy of an exercise TEM. With the instructor console connected to the projector, the instructor will demonstrate planning for a strike mission in AHWCS with the HARPOON in CTS mode. Each student will be closed-up at their own consoles and will follow along with the instructor. While planning their shots, the instructor shall ensure that they cover LOs 2 through 5. The instructor will complete the tutorial by demonstrating the 9-line verification. (20 min)<br>3) Students will pair-up, and a new target engagement message will be given to each student (each pair should receive 2 different TEMs). One at a time, each student will plan their CTS engagement using the provided TEM. Once complete, the student will conduct the 9-line verification, with the other student acting as CO. The students will then switch roles and repeat. A sufficient number of exercise TEMs must be available to enable each student to perform a minimum of at least 2 unique engagement plans. (80 mins) |

| Resources | References |
| --- | --- |
| 1) Target Engagement Messages (5 different versions minimum) | 1) CFCD 106<br>2) MARTI 3012 K |

| Notes to Instructor |
| --- |
| |

# EC 001.02 – Conduct Naval Fire Support

| Scope | Conditions | Sequence |
|---|---|---|
| EO 001.03 | SEACOT (primary)<br>SEAWOLF (secondary) | After 001.02.03 |
| **Time** | **Classification** | **Instructor** |
| 350 min | SECRET | AWWD Instructor |

| | |
|---|---|
| **Assessment Steps** | Input: Student is closed-up at policy, with support manning detailed in NWTP 2, where they must execute the duties of the AWWD during a strike mission.<br><br>Performance: students will conduct a 30 min run where they must perform the drills for Naval Fire Support – Harpoon. Students will be given a Target Engagement Message, and must create an engagement. Once complete, the student will perform a 9-line verification with the instructor acting as CO.<br><br>Output: Trainee must achieve 4/5 SATs on non-critical factors **AND** all critical factors to pass this EC.<br><br>Feedback: Immediate verbal and written using PC 001 checklist. 10 min allocated to reset the scenario and conduct debrief. |

| Resources | References |
|---|---|
| 1) TEMs<br>2) Warning Orders | 1) MARTI 3012 Annex K<br>2) NCPM 1000 |

| Notes to Instructor |
|---|
| |

| | |
|---|---|
| **Limitations** | |
| **Impact of Failure** | Further instruction and re-assess |

| Item | EC 001.02 – NAVAL FIRE SUPPORT | SAT | UNSAT | Remarks |
|---|---|---|---|---|
| **1. Conducts Naval Fire Support – Harpoon** | | | | |
| Conducts system manipulation IAW NCPM 1000 | | ☐ | ☐ | |
| Conduct 9 line verification with CO | | ☐ | ☐ | |
| Ensure flight corridor/background shipping are correct | | ☐ | ☐ | |
| Verify crypto and environmentals | | ☐ | ☐ | |
| Conducts BUZZARDS AWAY voice procedures IAW NCPM 1000 | | ☐ | ☐ | |
| **2. Safety **** CRITICAL****** | | | | |
| ***Adhered to CO/ORO direction for weapons employment** | | ☐ | ☐ | |
| ***Adhered to positive weapon control orders** | | ☐ | ☐ | |
| ***Missile geometry displayed in CMS** | | ☐ | ☐ | |

**Score: / 5** **CRITICALS: / 3**

Trainee must achieve 4/5 SATs on non-critical factors **AND** all critical factors to pass this EC.

| Positive Feedback | Improvement Feedback |
|---|---|
| | |

| Overall Assessment – Circle One | | | |
|---|---|---|---|
| **Met Standard** | | **Did Not Meet Standard** | |
| **Assessor**<br>I have discussed the results of this assessment with the candidate. | | **Candidate**<br>I have been debriefed on the result of this assessment | |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

## PC 001.01 – Operate AWW Weapons and Sensors (Written)

| Scope | Conditions | Sequence |
|---|---|---|
| PO 001 | Classroom | After EC 001.02 |
| **Time** | **Classification** | **Instructor** |
| 50 min | SECRET | AWWD Instructor |

| | |
|---|---|
| **Scenario** | Ship is executing the Tiered readiness plan and your ship will conduct weapons system certification. |
| **Assessment Steps** | Input: Students will be provided with a hardcopy of the AWWD Individual Weapon System Certification written exam.<br><br>Performance: Students must complete the exam within the allotted time.<br><br>Output: Trainee must achieve an 80% on the exam to pass this PC.<br><br>Feedback: Immediate verbal and written using PC 001 checklist. 10 min allocated to reset the scenario and conduct debrief. |

| Resources | References |
|---|---|
| 1) Nil | 1) CFCD 102<br>2) NAVORD 4995-0<br>3) NCPM 1000<br>4) CMS Manuals |

| Notes to Instructor |
|---|
| Students must not have access to references during this exam.<br>Instructors shall be in the class during the exam.<br>Students will not be permitted to leave the classroom until they have completed their exams. |

| | |
|---|---|
| **Limitations** | NIL |
| **Impact of Failure** | IAW NPTORD 4500-54 |

# PO 001 – Consolidation

| Conditions | | Sequence |
|---|---|---|
| SEACOT (Primary)<br>SEAWOLF (Secondary) | | After PC 002 |
| **Time** | **Classification** | **Instructor** |
| 3200 min | Up to SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Build and Maintain the RMP<br>2) Manage the AWW Team<br>3) Defend Against Missile Threats<br>4) Defend Against Surface Threats<br>5) Defend Against Air Threats | EDI |

| Activity Steps | |
|---|---|
| | Input: The students, as a class, will produce and deliver a pre-firing brief prior to commencing PC 001B. Student is closed up with support manning detailed in NWTP 2, where they must execute the duties of the AWWD during peacetime firing.<br><br>Performance: IAW PC 001B<br><br>Students should have the opportunity to complete a minimum of 2 developmental runs (with instructor assessing the student with the PC checklist). Any additional time can be used for either group mentorship or additional developmental runs. **Note**: PC 001B has the Main Armament firing with the Planner (ADEP/TEWA) vice no support for EC 001.01<br><br>Output: IAW PC 001B<br><br>Feedback: Immediate verbal and written using PC 001 checklist. |

| Resources | References |
|---|---|
| 1) IAW PC 001 | 1) NCPM 1000 |

# PC 001.02– Operate AWW Weapons and Sensors (Practical)

| Scope | Conditions | Sequence |
|---|---|---|
| PO 001 | SEACOT (primary)<br>SEAWOLF (secondary) | After PO 001<br>Consolidation |
| **Time** | **Classification** | **Instructor** |
| 1600 min | SECRET | AWWD Instructor |

| | |
|---|---|
| **Scenario** | Ship is executing the Tiered readiness plan and your ship will conduct weapons system certification. |
| **Assessment Steps** | Input: Student is closed up with support manning detailed in NWTP 2, where they must execute the duties of the AWWD during peacetime firing.<br><br>Performance: students will conduct a 120 min run where they must perform the drills for:<br>• CUCA & Assume Policy<br>• PAC (main armament)<br>• PAC (CIWS)<br>• Basic main armament surface firing (no support)<br>• Warning Shots<br>• CIWS surface firing<br>• Basic main armament AA Firing (no support)<br>• Basic CIWS AA Firing<br>• ESSM Urgent (Air or SASS)<br>• ADEP<br>• Surface to Surface Missile Engagement (must include manipulation of HAFO, waypoint manipulation)<br>• Naval Fire Support – Harpoon<br>• Naval Fire Support – Main armament<br>The scenario must include one error of each of the following categories:<br>• A system setup error (ex. Radar power, gun alert states, weapon assignment, force recoil, radar allow xmit, etc.)<br>• An incorrect drill/voice procedure conducted by team member<br>• Equipment fault (ex. RADAR fail, veto panel unresponsive, gun hydraulic failure, gun hoist issue)<br>• checkfire, range foul or dangerous proximity situation<br>• Misfire/checkload (weapons stoppage)<br>• An a/c attack profile that will not allow for a successful engagement (CPA too large, altitude too high)<br>Output: Trainee must achieve 36/44 SATs on non-critical factors **AND** all critical factors to pass this PC. |

| | |
|---|---|
| | Feedback: Immediate verbal and written using PC 001 checklist. 10 min allocated to reset the scenario and conduct debrief. Debriefs are to be conducted with the assistance of the ORO to ensure thoroughness and impartiality. |

| Resources | References |
|---|---|
| 2) Pre-Firing Brief<br>3) NCPM 1000<br>4) Scenarios<br>5) TEM | 1) NCPM 1000<br>2) CFCD 106<br>3) CSUMS<br>4) NAVORD 11000-1<br>5) NAVORD 4995-0 |

| Notes to Instructor |
|---|
| Support manning IAW NWTP 2<br>No assistance will be provided to students during the assessments.<br>If student conducts a critical failure during an assessment run, the run will continue as a developmental opportunity (significant feedback notes to be taken and provided using PC checklist)<br>A second assessor must be available for subsequent PC attempts. |

| | |
|---|---|
| **Limitations** | No Assistance will be provided to students during assessments. |
| **Impact of Failure** | IAW NPTORD 4500-54 |

| Item | PC 001 – EXECUTE AWW OPERATIONS AND EXERCISES | SAT | UNSAT | Remarks |
|---|---|---|---|---|
| **1) Conduct Close Up procedures** | | | | |
| Conducts CUCA IAW NCPM 1000 | | ☐ | ☐ | |
| Assumes Policy IAW NCPM 1000 | | ☐ | ☐ | |
| Address system setup error | | ☐ | ☐ | |
| **2) Conduct Pre Action Calibrations** | | | | |
| Conducts main Armament PAC IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts CIWS PAC IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | | ☐ | ☐ | |
| **3) Conduct basic main Armament firing against surface target** | | | | |
| Conducts deflection checks IAW NCPM 1000 | | ☐ | ☐ | |
| Assess tracking IAW NCPM 1000 | | ☐ | ☐ | |
| Apply spotting corrections into CMS as given | | ☐ | ☐ | |
| Inputs Salvo size into CMS IAW Salvo Policy | | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | | ☐ | ☐ | |
| **4) Conduct Warning Shots** | | | | |
| Conducts Warning Shot System manipulation IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts Warning Shot voice procedures IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | | ☐ | ☐ | |
| **5) Conduct CIWS surface firing against surface target** | | | | |
| Assess tracking IAW NCPM 1000 | | ☐ | ☐ | |
| Conduct CIWS surface firing IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | | ☐ | ☐ | |
| **6) Conduct basic main Armament AA Firing (no support)** | | | | |
| Assess tracking IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts system manipulation IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts voice procedures IAW NCPM 1000 | | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | | ☐ | ☒ | |

| | | | |
|---|---|---|---|
| **7) Conduct basic CIWS AA Firing** | | | |
| Assess tracking IAW NCPM 1000 | ☐ | ☐ | |
| Conduct CIWS AA firing IAW NCPM 1000 | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | ☐ | ☐ | |
| **8) Conduct ESSM (ADEP/TEWA)** | | | |
| Conducts system manipulation IAW NCPM 1000 | ☐ | ☐ | |
| Conducts voice procedures IAW NCPM 1000 | ☐ | ☐ | |
| Conducted ESSM misfire/DUD procedures IAW | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | ☐ | ☐ | |
| **9) Conduct ADEP (advanced AA layered firings)** | | | |
| Conducts system manipulation IAW NCPM 1000 | ☐ | ☐ | |
| Conducts voice procedures IAW NCPM 1000 | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | ☐ | ☐ | |
| **10) Conducts SS missile engagement** | | | |
| Conducts system manipulation IAW NCPM 1000 | ☐ | ☐ | |
| Conducts BF voice procedures IAW NCPM 1000 | ☐ | ☐ | |
| Conducts BULLDOGS AWAY voice procedures IAW NCPM 1000 | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | ☐ | ☐ | |
| **11) Conducts Naval Fire Support – Harpoon** | | | |
| Conducts system manipulation IAW NCPM 1000 | ☐ | ☐ | |
| Conduct 9 line verification with CO | ☐ | ☐ | |
| Conducts BUZZARDS AWAY voice procedures IAW NCPM 1000 | ☐ | ☐ | |
| Conducts SITREPS on appropriate nets | ☐ | ☐ | |
| **12) Respond to Faults** | | | |
| Corrected drill/voice procedure of team | ☐ | ☐ | |
| Identified system fault | ☐ | ☐ | |
| Briefed fault to ORO using SAR format | ☐ | ☐ | |
| **13) Safety **** CRITICAL****** | | | |
| ***Adhered to CO/ORO direction for weapons employment** | ☐ | ☐ | |
| ***Adhered to positive weapon control orders** | ☐ | ☐ | |

| | | | |
|---|---|---|---|
| ***Conducted checkfire, range foul or dangerous proximity situation IAW NAVORD 4995-0** | ☐ | ☐ | |
| ***Conducted main armament Cold Gun Misfire IAW NCPM 1000** | ☐ | ☐ | |
| ***Conducted main armament Hot Gun Misfire with failed refire IAW NCPM 1000** | ☐ | ☐ | |
| ***Conducted CIWS cold or hot gun bore loaded** | ☐ | ☐ | |
| ***Conducted harpoon misfire/DUD procedures IAW NCPM 1000** | ☐ | ☐ | |
| ***Maintains safe CIWS ARS throughout serial** | ☐ | ☐ | |

**Score: /44 + /8 CRITICAL**

Trainee must achieve 36/44 SATs on non-critical factors **AND** all critical factors to pass this PC.

| Positive Feedback | Improvement Feedback |
|---|---|
| | |

| Overall Assessment – Circle One | | | |
|---|---|---|---|
| **Met Standard** | | **Did Not Meet Standard** | |
| **Assessor**<br>I have discussed the results of this assessment with the candidate. | | **Candidate**<br>I have been debriefed on the result of this assessment | |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

PROTECTED B (when complete)

## PO 002 – Ensure the Operational Readiness of AAW Teams (T0008)

5. **Performance**: The AWWD will ensure the operational readiness of AAW teams (T0008).

6. **Conditions**:

   a. Given:

      (1) Access to DWAN and CSNI;
      (2) References and templates.

   b. Denied:

      (1) Nil.

7. **Environment**:

      (1) At sea or alongside; and
      (2) Day or night.

8. **Standard**: The AWWD shall ensure the operational readiness of AAW teams by:

   a. Planning/Designing training exercises (G0050) by:

      (1) Selecting training scenarios (G00110); and
      (2) Coordinating the preparation of surface targets for gunnery exercises (D00020);

   b. Preparing operational training exercises (B00200) by:

      (1) Executing AWW related admin duties (T0009); and
      (2) Drafting firing orders (BT0120);

   c. Advising command on anti-air/anti-surface warfare safety (BT0320/BT0330).

9. **Training Limitations**: Nil.

EO 002.01 - Plan/Design training exercises

## 002.01.01 – Introduction to the Maritime Tactical Planner (MTP)

| Conditions | | Sequence |
|---|---|---|
| Unclass Classroom | | After PC 001.01 |
| **Time** | **Classification** | **Instructor** |
| 150 min | Unclass | Guest Lecturer (OROCO/ORO CCD Manager) |

| Learning Objectives | Method |
|---|---|
| 1) What is the Maritime Technical Planner (MTP)<br>2) How the MTP improves planning<br>3) How to use the MTP<br>4) How Risk Factors into CAF Operations | Lecture |

| Activity Steps | |
|---|---|
| | 1) Using PowerPoint, the instructor will conduct a lecture on the MTP, and demonstrate how it can be used to enable effective planning of AWW training and exercises.<br>2) Using the Risk Management Document discuss with the students the Risk Management Process and how the inputs at the Director level can affect operation |

| Resources | References |
|---|---|
| 1) PowerPoint Presentation<br>2) MTP Template<br>3) Risk Management for CF Ops Joint Doctrine Publication | 1) Joint Doctrine Manual, Risk Management for CF Operations |

| Notes to Instructor |
|---|
| As Maritime Technical Planner is in development, LO 4 can be utilized for the entirety of the lesson if MTP not available. |

EO 002.01 - Plan/Design training exercises

## 002.01.02 – Combat Readiness Requirements (CRRs)

| Conditions | | Sequence |
|---|---|---|
| Unclass Classroom | | After 002.01.01 |
| **Time** | **Classification** | **Instructor** |
| 50 min | Unclass | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1. Using CFCD 102/129<br>2. AWW Applicable CRRs<br>3. Training requirements (personnel, equipment, facilities, time, resources)<br>4. Validity period/maintenance<br>5. Identifying authority for CRR completion<br>6. Utilizing ORM | lecture<br>Student presentations |

**Activity Steps**

1. Instructor to introduce the publication CFCD 102 and ORM. (10 min)
    a. Explain use of publication (tracking of unit readiness)
    b. Identify that Sea Training Group is the authority/maintainer of CFCD 102
    c. Demonstrate how to determine the applicable chapters
    d. Indicate how to disseminate CRR status (inform IMD)
    e. Indicate where to access ORM and information contained within
2. Students are then paired up and assigned (minimum) 2 AWW CRRs per group and asked to identify findings to include: (20 min)
    a. CRR title
    b. Purpose
    c. Readiness level
    d. Validity period
    e. Training requirements (resources, personnel, etc.)
    f. Standard (assessment criteria)
    g. Assessment Authority
3. Each group to present their findings to the class (15 min)
4. Instructor to reaffirm the purpose of CRRs and how this will be applicable in their role as the Training PO. (5 min)

| Resources | References |
|---|---|
| 1) DWAN computer/student<br>2) Projector<br>3) CFCD 102 (web based) | 1) CFCD 102/129<br>2) Readiness Training Guide<br>3) ORM |

**Notes to Instructor**

EO 002.01 - Plan/Design training exercises

## 002.01.03 – References for Planning Training Exercises

| Conditions | | Sequence |
|---|---|---|
| Unclass Classroom | | After 002.01.02 |
| **Time** | **Classification** | **Instructor** |
| 50 min | Unclass | AWWD Instructor<br>(Guest Instructor Target Cell) |

| Learning Objectives | Method |
|---|---|
| 1) Utilize AXP 2 in serial planning<br>2) Determine AWW training requirements<br>3) Determining target requirements for serials<br>4) Roll of Target Cell<br>5) Determining OPAREAs required for serial<br>6) Where to find target information for a Fleet planned exercise | Paired research/presentation lecture |

| Scenario | |
|---|---|
| **Activity Steps** | 1) Students are paired up and are assigned a training scenario. They must read AXP 2 section relevant to their specific training scenario. Students will prepare a brief presentation on the details contained in AXP 2 they would be required to know/use when planning a training exercise. (20 min)<br>2) Student pairs are to present their training exercise requirements to the class, 2 min per group. (10 min)<br>3) Guest Instructor to provide lecture on how to determine targets that can be used in various OPAREAS. They will demonstrate where to find the information (CFCD 102 and Target Cell) and how to select the applicable target. This lecture will indicate OPAREA requirements for range selection. (20 min) |

| Resources | References |
|---|---|
| 1) Pictures of all Avail Targets<br>2) Target Cell Lecture/PPT to include caps/lims | 1) AXP 2 (Unclass)<br>2) Commercial references for targets as available |

| Notes to Instructor |
|---|
| |

EO 002.01 - Plan/Design training exercises

## 002.01.04 – Ship/Fleet Training

| Conditions | | Sequence |
|---|---|---|
| Unclass Classroom | | After 002.01.03 |
| **Time** | **Classification** | **Instructor** |
| 50 min | Unclass | Guest lecture from CCFL/P or Sea Training/NFR |

| Learning Objectives | Method |
|---|---|
| 1) Readiness level training requirements<br>2) Ships Tiered Readiness Program<br>a. MRI 1 & 2<br>b. OTT 1 & 2<br>c. SRIs<br>3) Fleet Training requirements (CREST) | lecture |

| Activity Steps | |
|---|---|
| | 1) Guest lecturer to present on the LOs identified (40 min)<br>a. Ships Tiered Readiness Program<br>b. Fleet Training Requirements<br>2) Show and explain example for Fleet Training messages as well as a ships training calendar (10 min) |

| Resources | References |
|---|---|
| 1) DWAN computer<br>2) Projector<br>3) example messages for fleet training<br>4) example ships training calendar | 1) NAVORD 1000-1<br>2) NAVORD 3137-2<br>3) SSOs<br>4) CFCD 102<br>5) CFCD 129<br>6) Readiness Training Guide |

| Notes to Instructor |
|---|
| Liaise in advance with guest lecture to ensure their presentation meets LO requirements. |

EO 002.01 - Ensure the operational readiness of AAW teams

# 002.01.05 – SETT/SORTI/SAF SIM

| Conditions | | Sequence |
|---|---|---|
| SEACOT (primary)<br>SEAWOLF (secondary) | | After 002.01.04 |
| **Time** | **Classification** | **Instructor** |
| 100 min | SECRET | AWWD Instructor<br>RQ-PO1 NCI OP |

| Learning Objectives | Method |
|---|---|
| 1) SETT/SORTI/SAF SIM capabilities<br>2) Training objectives that can be met using SETT/SORTI/SAF SIM<br>3) Team responsibilities in utilizing synthetic training<br>4) Weapon status requirements during synthetic training | Lecture<br>T-Chart (List)<br>Demonstration/lecture |

**Activity Steps**

1) Instructor to describe SETT/SORTI/SAF SIM. They will explain what the system can do, and what the limitations of the system include. (10 min)
2) Students are then put into 2 groups where they must create a list of the training objectives that can be met using the synthetic training systems. (5 min)
3) Students are to then present their list of training objectives, comparing the comprehensive lists. (5 min)
4) Instructor will then indicate what the various roles are when putting the system in SETT/SORTI (tech, operators, etc.). They will demonstrate how to create a contact and manipulate that contact to show how complex the process is. (15 mins)
5) The instructor will then demonstrate how to manoeuvre contacts, and fire weapons. Throughout, the instructor will then field any questions/discussion points the students may have about synthetic training. (15 min)
6) Students are to then practice manipulating the system in SAF SIM, creating and maneuvering contacts, and fire weapons. (50 min)

| Resources | References |
|---|---|
| 1) flip chart and markers<br>2) Projector | 1) SAFSIM User Guide |

**Notes to Instructor**



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EO 002.01 - Ensure the operational readiness of AAW teams

## 002.01.06 – Serial CONOPS Brief

| Conditions | | Sequence |
|---|---|---|
| Unclass Classroom | | After 002.01.05 |
| **Time** | **Classification** | **Instructor** |
| 100 min | Unclass | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) CAF 7-Slide Presentations | Lecture<br>Student practical exercice |

| Activity Steps | |
|---|---|
| | 1) Instructor will demonstrate the CAF 7-slide brief template, explaining its purpose and sections (15 min)<br>2) Instructor will conduct a CONOPs brief using the CAF 7-slide template and an exercise serial from AXP-02 (10 min)<br>3) Students will select an AWW CRR from CFCD 102. Students must then use AXP-02 to select an exercise serial that will satisfy the requirements of the CRR (10)<br>4) Student will prepare a CONOPs brief using the CAF 7-slide template and the exercise serial they selected from AXP-02 (50 min)<br>5) Student will submit their presentations to their instructor. Instructor to provide feedback to each student (hard copy or electronic based on classroom assets). |

| Resources | References |
|---|---|
| 1) CAF 7-slide brief template<br>2) AXP-02<br>3) CFCD 102 | 1) CFCD 102<br>2) AXP-02 |

| Notes to Instructor |
|---|
| |



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## EC 002.01 – Plan/Design Training Exercises

| Scope | Conditions | Sequence |
|---|---|---|
| | Classroom with Projector | After 002.01.06 |
| **Time** | **Classification** | **Instructor** |
| 400 mins | SECRET | AWWD Instructor |

| Assessment Steps | |
|---|---|
| | Input: Students will select an AWW CRR from CFCD 102 and an exercise from AXP-02 that will satisfy their chosen CRR.<br><br>Performance: The student will have 200 mins to prepare a 7-slide briefing slide deck IAW CAF format and orally brief the conduct of their serial. Students will have 10 mins to brief their chosen serial.<br><br>Output: Trainee must achieve 10/13 SATs on all factors to pass this EC.<br><br>Feedback: Immediate verbal and written using EC 002.01 checklist. 15 min are allocated for questions, debrief, and for the next student to prepare their brief. |

| Resources | References |
|---|---|
| 1) CAF 7-slide brief template<br>2) CFCD 102<br>3) RTG<br>4) AXP-02<br>5) NAVORD 4995-0 | 1) CFCD 102<br>2) RTG<br>3) AXP-02<br>4) NAVORD 4995-0 |

| Notes to Instructor |
|---|
| |

| Limitations | |
|---|---|
| **Impact of Failure** | Further instruction and re-assess |

| Item | EC 002.01 PLAN/DESIGN TRAINING EXERCISES | SAT | UNSAT | Remarks |
|---|---|---|---|---|
| Develop a Concept of Operation for a Serial | | | | |
| 1. | Plan meets the requirements of the CRR presented IAW CFCD 102 | ☐ | ☐ | |
| 2. | Plan is achievable and safe IAW NAVORD 4995-0 | ☐ | ☐ | |
| 3. | Plan is presented in 10 minutes or less | ☐ | ☐ | |
| 4. | Briefing is clear and comprehensible | ☐ | ☐ | |
| 5. | Time required | ☐ | ☐ | |
| 6. | Op Area and requirements | ☐ | ☐ | |
| 7. | Restrictions | ☐ | ☐ | |
| 8. | Assets required | ☐ | ☐ | |
| 9. | Purpose of CRR | ☐ | ☐ | |
| 10. | End state | ☐ | ☐ | |
| 11. | Ship manoeuvring | ☐ | ☐ | |
| 12. | Ammunition required and | ☐ | ☐ | |
| 13. | CRR requirements | ☐ | ☐ | |
| • | | | | |

**Score: /13** Trainee must achieve 10/13 SATs to pass this EC.

| Positive Feedback | Improvement Feedback |
|---|---|
| | |

| EC 001.01 – Overall Assessment – Circle One | | | |
|---|---|---|---|
| **Met Standard** | | **Did Not Meet Standard** | |
| **Assessor**<br><br>I have discussed the results of this assessment with the candidate. | | **Candidate**<br><br>I have been debriefed on the result of this assessment | |
| **EC Administrator** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

EO 002.02 - Prepare operational training exercises

## 002.02.01 – Drafting Request for Service Support Messages

| Conditions | | Sequence |
|---|---|---|
| SECRET Classroom | | After EC 002.01 |
| **Time** | **Classification** | **Instructor** |
| 300 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Drafting RSS messages<br>2) Interpreting OPAREA Assignment message<br>3) Interpreting NOTAMs | Lecture<br>practical |

| Activity Steps | |
|---|---|
| | 1) Students are to read MARLANTORD 3255-1/MARPACORD3350-1 (50 min)<br>2) Instructor to lead students through understanding and drafting an RSS message (50 min)<br>3) Instructor to lead students through reading an OPAREA assignment message (20 min)<br>4) Students are to then use message drafter to develop a template that can be used for follow on activities (30 min)<br>5) Students are provided a scenario (firing serial) where they would be required to draft an RSS message. Instructor to provide feedback to each student (hard copy or electronic based on classroom assets).(150 min) |

| Resources | References |
|---|---|
| 1) CSNI/SECRET workstations with message drafter<br>2) Example Messages (EAST/WEST) RSS/OPAREA/NOTAM<br>3) MARLANTORD 44-4 | 1) AXP-02<br>2) MARPACORD 3350 -1<br>3) MARLANTORD 3255-1<br>4) CCFL Range Procedures<br>5)MARLANTORD 3255-1<br>6) AHP 6 |

| Notes to Instructor |
|---|
| Scenarios should be such that they provide a variety of participating units, firing type (air/surface), Op Areas and targets involved.<br>Should discuss how areas are requested in other parts of the world (VACAPES, PMRF, AHP 6 through NATO) |

EO 002.02 - Prepare operational training exercises

## 002.02.02 – Drafting TABORDs

| Conditions | | Sequence |
|---|---|---|
| SECRET Classroom | | After 002.02.01 |
| **Time** | **Classification** | **Instructor** |
| 250 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Using the Standard Order Table<br>2) Incorporating the CONOPs into the TABORDs (i.e. formation, screens, and target placement)<br>3) Drafting TABORDs | Experiential learning (self-guided) |

| Activity Steps | |
|---|---|
| | 1) Instructor to conduct a lecture explaining the purpose of TABORDs, demonstrating a completed TABORD, and describing how the AXP 02/ATP-01 Standing Order Tables are used in the creation of a TABORD (30)<br>2) Instructor to break class into pairs, and provide a scenario where they must draft an AWW TABORD message. Students are to use the references and knowledge gained from previous lessons to draft 1 X AWW TABORD message per team. (150 min).<br>3) Instructor to provide feedback on each teams message (hard copy or electronic based on classroom assets).<br>4) Students provided a new scenario and must draft a second AWW TABORD, this time individually. Instructor to provide feedback to each student (hard copy or electronic based on classroom assets) (100 min) |

| Resources | References |
|---|---|
| 1) CSNI/SECRET workstation with message drafter<br>2) Scenarios<br>3) Sample TABORD<br>4) ATP-01<br>AXP-02 | 1) ATP-01<br>2) AXP-02 (Unclass) |

| Notes to Instructor |
|---|
| Scenarios should be such that they provide a variety of participating units, firing type (air/surface), Op Areas and targets involved. |

## 002.02.03 – Contribute to FLEX development

| Conditions | | Sequence |
|---|---|---|
| Unclass classroom | | After 002.02.02 |
| **Time** | **Classification** | **Instructor** |
| 100 mins | Unclass | ORO/AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Time requirements for pre and post firing admin and maintenance requirements (i.e. setting up and launching targets)<br>2) De-conflicting with other serials (i.e. flight-ops)<br>3) Developing a flex | Lecture<br>Pair-Think-Share<br>Demo<br>Practical |

| Activity Steps | |
|---|---|
| | 1) Instructors will explain to the students how important it is to consider pre and post firing admin and maintenance, as well as conflicting ship activities, when developing a training serial and developing the flex (5 min)<br>2) Students will divide into pairs and prepare a list of items they feel will require time during pre and post firing periods (10 min)<br>3) Student pairs will present their findings orally to the rest of the class (10 min)<br>4) Student pairs will then prepare a list of at-sea activities that may interfere with a firing exercise (10 min)<br>5) Student pairs will present their findings orally to the rest of the class (10 min)<br>6) Instructor will then provide a sample flex that illustrates the various items that must be considered and included when planning a firing serial (5 min)<br>7) Using the scenario provided to them for their individual TABORD assignment in lesson 002.02.02, students will develop a flex for their assigned serial incorporating pre and post firing admin and maintenance relevant to their scenarios (30 min)<br>8) Instructor to provide feedback on each FLEX (hard copy or electronic based on classroom assets). |

| Resources | References |
|---|---|
| 1) FLEX template<br>2) Unclass Workstation<br>3) example Flex | 1) NAVORD 3000-2<br>2) NAVORD 4995-0 |

| Notes to Instructor |
|---|
| |



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EO 002.02 - Prepare operational training exercises

## 002.02.04 – Draft Firing Orders

| Conditions | | Sequence |
|---|---|---|
| Unclass classroom | | After 002.02.03 |
| **Time** | **Classification** | **Instructor** |
| 50 min | Unclass | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Drafting Firing Orders<br>2) Breakdown OPAREA assignment messages | Lecture<br>Think-pair-share<br>Lecture<br>Demo<br>practical |

| Activity Steps | |
|---|---|
| | 1) Instructor will discuss the importance of Firing Orders (5 min)<br>2) Students are then asked to individually consider what information needs to be included when drafting Firing Orders. (2 min)<br>3) Students are to pair with a peer, and discuss their list. (3 min)<br>4) Pairs are to then share their lists with the class, creating an overall list for the class. (5 min)<br>5) Instructor will confirm the information that must be contained in Firing Orders and where to find the various pieces of information (ammunition type to include amount, OPAREA assignment, timings, target types). They will utilize a sample Firing Order throughout to explain. (10 min)<br>6) Students are broken into pairs, where they will be provided a training package to include a flex and OPAREA assignment message from previous lesson. Students must create Firing Orders and determine with substantiation the type and amount of rounds and target type to be used. Instructor to walk throughout and provide feedback to students while they work. (25 min) |

| Resources | References |
|---|---|
| 1) Sample Firing Orders Template<br>2) Sample Firing Orders<br>3) OPAREA assignment message & Flex | 1) AXP-02 (Unclass) |

| Notes to Instructor |
|---|
| |



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EO 002.02 - Prepare operational training exercises

## EC 002.02 – Prepare Operational Training Exercises

| Scope | Conditions | Sequence |
|---|---|---|
| | Classroom | After 002.02.04 |
| **Time** | **Classification** | **Instructor** |
| 200 mins | SECRET | AWWD Instructor |

| | |
|---|---|
| **Assessment Steps** | Input: Students will continue to use the serial selected for EC 002.01 and must prepare the following pre-serial administration for their selected serial:<br>• RSS Message<br>• TABORDs/OCS Intentions<br>• Flex<br>• Firing Orders<br>Performance: Each student will have 200 min to prepare their products.<br><br>Output: Trainee must achieve 8/10SATs on all factors to pass this EC.<br><br>Feedback: Students will receive feedback within 48 hrs. of completing the EC deliverables IAW EC 002.02 checklist. |

| Resources | References |
|---|---|
| 1) Templates of all deliverables<br>2) Sample Serial/Scenario | 1. AXP 2<br>2. NAVORD 3000-2<br>3. NAVORD 4995-0<br>4. ATP-01<br>5. MARPACORD 3350-1<br>6. MARLANTORD 3255-1<br>7. CCFL Range Procedures<br>8. RTG |

| Notes to Instructor |
|---|
| |

| | |
|---|---|
| **Limitations** | |
| **Impact of Failure** | Further instruction and re-assess |

| Item | EC 002.02 - PREPARE OPERATIONAL TRAINING EXERCISES | SAT | UNSAT | Remarks |
|---|---|---|---|---|
| **1. Conduct Pre-Firing Admin** | | | | |
| RSS format IAW MARLANT/MARPAC ORDs | | ☐ | ☐ | |
| RSS included required assets (targets and/or aircraft) | | ☐ | ☐ | |
| RSS requested correct water/air space | | ☐ | ☐ | |
| RSS describes execution IAW AXP-2 and assigned serial | | ☐ | ☐ | |
| TABORD formatted IAW AXP-2 | | ☐ | ☐ | |
| TABORD described execution IAW AXP-2 and assigned serial | | ☐ | ☐ | |
| Firing Orders formatted IAW RTG | | ☐ | ☐ | |
| Firing Orders contain appropriate ammunition allocation | | ☐ | ☐ | |
| FLEX included all close up times and organizations, commencement and completion of all activities, and details as required for the OOW to prepare for the serial | | ☐ | ☐ | |
| Timings are consistent across all documents | | ☐ | ☐ | |

**Score:** **/ 10** Trainee must achieve 8/10 SATs on all factors to pass this EC.

| Positive Feedback | Improvement Feedback |
|---|---|
| | |

**Overall Assessment – Circle One**

| Met Standard | Did Not Meet Standard |
|---|---|
| **Assessor**<br>I have discussed the results of this assessment with the candidate. | **Candidate**<br>I have been debriefed on the result of this assessment |

| | | | |
|---|---|---|---|
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

EO 002.03 - Advise command on anti-air/anti-surface warfare safety

## 002.03.01 – Weapons Firing Safety

| Conditions | | Sequence |
|---|---|---|
| Unclass classroom | | After EC 002.02 |
| **Time** | **Classification** | **Instructor** |
| 100 min | Unclass | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Safety Firing Arcs<br>2) Out of Bounds Areas<br>3) Occasions to call Check Fire<br>a. Marine Mammal Mitigation<br>b. Range foul<br>c. Safety<br>4) Reactions to a Check Fire | Student Reading<br>Team Quiz |

| Activity Steps | |
|---|---|
| | 1) Instructor to conduct brief introduction on the importance of weapons firing safety. (2 min)<br>2) Students are to read NAVORD 4995-0 and NCPM 1000 Chapter 1 (instructions on Check Fire, Hold Fire, Cease Fire reactions). While reading they are to create 10 quiz questions on the content of the readings, including answers. (70 min)<br>3) Students are to submit their questions to the instructor.<br>4) Instructor will split the class in ½ and use the student created questions to conduct an oral quiz on the readings. Questions are to be asked to the entire class; first group to provide a correct response gains a point. Instructors are to use quiz responses (or lack thereof) as developmental opportunities to expand upon and clarify concepts. (28 min) |

| Resources | References |
|---|---|
| 1) DWAN workstations or SECRET Stand-alone computers<br>2) NAVORD 4995-0 (web based)<br>3) NCPM 1000 (web based) | 1) NAVORD 4995-0<br>2) NCPM 1000 |

| Notes to Instructor |
|---|
| |

EO 002.03 - Advise command on anti-air/anti-surface warfare safety

## 002.03.02 – Creating a Pre-Firing Brief

| Conditions | | Sequence |
|---|---|---|
| Unclass Classroom | | After 002.03.01 |
| **Time** | **Classification** | **Instructor** |
| 150 | Unclass | |

| Learning Objectives | Method |
|---|---|
| 1) RTG Pre-Firing Brief Template<br>2) Who is responsible for the content of each section<br>3) Required execution content IAW RTG template<br>4) Required safety content IAW RTG template | EDI |

| Activity Steps | |
|---|---|
| | 1) Instructor to go-through a sample pre-firing brief for a given firing serial, highlighting and explaining all of the different areas and who is responsible for each (50)<br>2) Instructor is to conduct a pre-firing brief presentation (25)<br>3) Students will then pair-up and create a pre-firing brief using the RTG template and one of the scenarios provided to them in previous lessons. (50)<br>4) Students will take turns presenting their briefs to each other, providing feedback to each other. (25)<br>5) Instructor to walk throughout and provide feedback to students while they work and present their presentations. |

| Resources | References |
|---|---|
| 1) Projector<br>2) Pre-made pre-firing brief<br>3) Pre-firing brief template | 1) NAVORD 4995-0<br>2) RTG Pre-Firing Brief Template<br>3) CFCD 102<br>4) AXP-02<br>5) MARLANTORD 3255-1<br>6) MARPACORD 3350-1<br>7) NCPM 1000<br>8) SSOs |

| Notes to Instructor |
|---|
| |

# EC 002.03 - Advise Command on Anti-Air/Anti-Surface Warfare Safety

| Scope | Conditions | Sequence |
|---|---|---|
| | classified classroom | After 002.03.02 |
| **Time** | **Classification** | **Instructor** |
| 550 mins | SECRET | AWWD Instructor |

| | |
|---|---|
| **Assessment Steps** | Input: Students will continue to use the serial selected for EC 002.01. Student will produce a pre-firing brief IAW the RTG format.<br><br>Performance: The student will have 200 mins to prepare their presentations, then 30 mins to deliver their pre-firing brief.<br><br>Output: Trainee must achieve 8/10 SATs on all factors to pass this EC.<br><br>Feedback: Immediate verbal and written using EC 002.01 checklist. 15 min are allocated for questions, debrief, and for the next student to prepare their brief. |

| Resources | References |
|---|---|
| 14. TABORDS etc. Ressources from EC 002.02<br>15. 2) projector<br>16. computer | 1) RTG<br>2) NAVORD 4995-0<br>3) CFCD 102<br>4) AXP-02<br>5) NCPM 1000 |

| Notes to Instructor |
|---|
| |

| Limitations | |
|---|---|
| **Impact of Failure** | Further instruction and re-assess. Should students fail this EC based solely on a lack of confident briefing style, students will be given the opportunity to re-present their brief, but shall not be required to create a new brief. |

| Item | EC 002.03 ADVISE COMMAND ON ANTI-AIR/ANTI-SURFACE WARFARE SAFETY | SAT | UNSAT | Remarks |
|---|---|---|---|---|
| **1. Deliver Pre-Firing Brief** | | | | |
| Execution presented is IAW AXP-02 and assigned serial, and included all required areas IAW the RTG template | | ☐ | ☐ | |
| | | ☐ | ☐ | |
| Briefed safety considerations IAW 4995-0 and assigned serial to include: Op area, weather, mission, execution and surface spotting | | ☐ | ☐ | |
| | | ☐ | ☐ | |
| | | ☐ | ☐ | |
| | | ☐ | ☐ | |
| | | ☐ | ☐ | |
| | | ☐ | ☐ | |
| Demonstrate confident briefing style<br>• Eye contact<br>• Articulates content clearly<br>• Moderate tone/speed<br>• Succinctly maintains focus on content<br>• Logical transitions in content | | ☐ | ☐ | |
| Accurately responds to follow on questions related to assigned serial | | ☐ | ☐ | |

**Score: / 10** Trainee must achieve 8/10 SATs to pass this EC.

| Positive Feedback | Improvement Feedback |
|---|---|
| | |

**Overall Assessment – Circle One**

| Met Standard | Did Not Meet Standard |
|---|---|
| **Assessor**<br>I have discussed the results of this assessment with the candidate. | **Candidate**<br>I have been debriefed on the result of this assessment |

| | | | |
|---|---|---|---|
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

# PC 002 - Ensure the Operational Readiness of AAW Teams

| Scope | Conditions | Sequence |
|---|---|---|
| PO 002 | Classroom<br>Briefing space with projector | After RC 002.03 |
| **Time** | **Classification** | **Instructor** |
| 600 mins total<br>400 preparation<br>200 briefings | NATO CONFIDENTIAL | AWWD Instructor |

| | |
|---|---|
| **Assessment Steps** | Input: Students will be assigned a CRR from CFCD 102 and firing exercise from AXP-02. Students must then produce:<br>• RSS Message<br>• TABORD<br>• Flex<br>• Firing Orders<br>• Pre-firing brief (including conduct & safety)<br>Performance: Each student will have 400 min to prepare the products assigned then 25 min to deliver the pre-firing brief.<br><br>Output: Trainee must achieve 9/12 SATs to pass this PC.<br><br>Feedback: Immediate verbal and written using PC 002 checklist. |

| Resources | References |
|---|---|
| 1) Templates (RSS, Flex, Firing Orders, TABORD, Pre-Firing Brief) | 1) NAVORD 4995-0<br>2) NAVORD 1000-1<br>3) CFCD 102<br>4) AXP-02<br>5) MARLANTORD 3255-1<br>6) MARPACORD 3350-1<br>7) RTG |

| Notes to Instructor |
|---|
| |

| | |
|---|---|
| **Limitations** | No support in product preparation, students are provided focused time allocated to exercise planning (where onboard ship would have competing priorities). |
| **Impact of Failure** | IAW NPTORD 4500-54 |

| Item | PC 002 - ENSURE THE OPERATIONAL READINESS OF AAW TEAMS | SAT | UNSAT | Remarks |
|---|---|---|---|---|
| **2. Conduct Pre-Firing Admin** | | | | |
| RSS format IAW MARLANT/MARPAC ORDs | | ☐ | ☐ | |
| RSS included required assets (targets and/or aircraft) | | ☐ | ☐ | |
| RSS requested correct water/air space IAW serial msg (MARLOAS/ WCFA assignment areas) | | ☐ | ☐ | |
| RSS describes execution IAW AXP-02 and assigned serial | | ☐ | ☐ | |
| TABORD formatted IAW AXP-02 | | ☐ | ☐ | |
| TABORD described execution IAW AXP-02 and assigned serial | | ☐ | ☐ | |
| Firing Orders formatted IAW RTG | | ☐ | ☐ | |
| Firing Orders contain ammunition allocation for serial | | ☐ | ☐ | |
| FLEX included all close up times and organizations, commencement and completion of all activities, and details as required for the OOW to prepare for the serial | | ☐ | ☐ | |
| Timings are consistent across all documents | | ☐ | ☐ | |
| **3. Deliver Pre-Firing Brief** | | | | |
| Execution IAW AXP-02 and assigned serial | | ☐ | ☐ | |
| Briefed safety considerations IAW 4995-0 and assigned serial | | ☐ | ☐ | |
| Demonstrate confident briefing style<br>• Eye contact<br>• Articulates content clearly<br>• Moderate tone/speed<br>• Succinctly maintains focus on content<br>• Logical transitions in content | | ☐ | ☐ | |
| Accurately responds to follow on questions related to mission | | ☐ | ☐ | |

**Score: / 12** The trainee must achieve 9/12 SATs to pass this PC.

| Positive Feedback | Improvement Feedback |
|---|---|
| | |

| Overall Assessment – Circle One | | | |
|---|---|---|---|
| **Met Standard** | | **Did Not Meet Standard** | |
| **Assessor**<br>I have discussed the results of this assessment with the candidate. | | **Candidate**<br>I have been debriefed on the result of this assessment | |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

PROTECTED B (when complete)

## PO 003 – Plan AWW Operations (BT0040/BT0030)

1. **Performance**: The AWWD will plan AWW operations (BT0040/BT0030)

2. **Conditions**:

   a. Given:

      (1) Access to DWAN and CSNI;
      (2) References and templates.

   b. Denied:

      (1) Nil.

3. **Environment**:

      (1) At sea or alongside; and
      (2) Day or night.

4. **Standard**: The AWWD shall plan AWW operations (BT0040/BT0030) by:

   a. Performing Threat Analysis (C00540) by:

      (1) Organizing intelligence (BT0490);
      (2) Evaluating intelligence information (B0090); and
      (3) Conducting threat analysis to include:
         (a) Collection;
         (b) Analysis; and
         (c) Dissemination of electronic data.
      (4) Compiling operational order of battle (OOB) (C00190);
      (5) Assessing AWW threat information (E0030); and
      (6) Determining the effectiveness of anti-air/anti-surface weapons and sensors (BT0390/BT0400).

   b. Preparing operational messages (BT0130) by:
      (1) Applying promulgated tactical info as it applies to internal and external organizations (C00040);
      (2) Interpreting EMCON polices (C00650);
      (3) Interpreting EW orders (C00530);and
      (4) Communicating operational information.

5. **Training Limitations:** Nil.

## PO 003 – Introduction to the Mission

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After PC 001 |
| **Time** | **Classification** | **Instructor** |
| 150 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Understanding the Geo-Political Situation<br>2) Understanding the Assigned Mission | Individual Study |

| Activity Steps | |
|---|---|
| | 1) Students will read the OPGEN and OPORD for the chosen mission scenario. (100 min)<br>2) The instructor will then discuss their mission and how the remainder of the course will be organized to prepare for and execute that mission. The instructor will attempt to answer any questions regarding the mission without going into too much detail of topics that will be covered throughout the remainder of this PO. (40 min)<br>3) The instructor will then explain that the students will use the primary threat nation (i.e. China for OP Arbiter or Russia/Syria for Op Varangian) for the remainder of the EOs and ECs, but that for their final PC that they will all be assigned a unique nation from within the AOO of the mission scenario in use. (5 min)<br>4) The instructor will then assign each student a nation from within the AOO for the mission scenario that they will use in their final PC. Students are free to conduct research and readings into that nation and its forces throughout the remainder of this PO. (5 min) |

| Resources | References |
|---|---|
| 1) OPGEN<br>2) OPORD | 1) Nil |

| Notes to Instructor |
|---|
| More than one student may be assigned the same nation, if necessary, based on the AOO. Instructors shall NOT assign to students the primary threat nation from the selected mission scenario (i.e. they shall not be assigned Russia or Syria if running Op VARANGIAN, nor will they be assigned China if running Op Arbiter, etc.).<br>Selected countries should provide as wide a variety as possible of air and surface platforms. |

EO 003.01 - Prepare Intelligence information

## 003.01.01 – Sources of Intelligence Data

| Conditions | | Sequence |
|---|---|---|
| DMOC (primary)<br>CSNI Lab (secondary) | | After PO 003 introduction |
| **Time** | **Classification** | **Instructor** |
| 200 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Identify the organization that provide fleet support to include:<br>a) TRINITY INT<br>b) METOC<br>c) Jane's Publications/CIA world fact book/DIS Threat Web<br>d) SAM handbook (1 CAD)<br>e) CFMWC<br>f) CFEWC/NEWC<br>g) Military Integrated Database<br>2) Interpret RFI/RFSs | Think, Pair, Share<br>Instructor led lecture<br>Student research<br>Flip Chart Presentation<br>Instructor led lecture<br>RFI Confirmation Activity<br>Instructor led Consolidation<br>Example |

| | |
|---|---|
| **Scenario** | Mission Scenario |
| **Activity Steps** | 1) Students to refer to the OPGEN and OPORD for the selected mission scenario (game). (50 min)<br>2) Instructor to pose to class "What organizations do we draw on for Intelligence data?" Make a list individually (2 min).<br>Students are then asked to pair up with a partner and share their lists. (3 min)<br>Students are then to present their "paired" lists and create one class list. (5 min)<br><br>3) Instructor to discuss the various units that provide intelligence data (LO 1) then lead students through navigating the various websites to include (15 min):<br>a. TRINITY INT<br>b. METOC<br>c. Jane's Publications/CIA world fact book/DIS Threat Web<br>d. SAM handbook (1 CAD)<br>e. CFMWC<br>f. CFEWC/NEWC<br>4) Students are then grouped (2-3 students per group) and assigned one of the websites to research and create a flip chart presentation on the information they would collect/gather using the assigned website. "What information can be found on this website, and what is the significance?" (30 min) |

5) Students to then present their assigned topics to the class and Instructor to provide any information that was not covered by student presentation. Instructor to bring up the website after each group presents, and demonstrate where to find info and its significance being specific to any information that was not presented by students. (up to 5 min presentation, 5 min instructor feedback and content delivery per group) 50 min total

| Anticipated Responses | |
|---|---|
| TRINITY INT | ORBAT<br>How to determine base locations and fleet<br>Foreign nation tactics |
| METOC | OFA<br>Potential Ranges<br>Historical Wx data |
| SAM handbook | Stand-off distances<br>SAM caps/lims |
| CFMWC | Canadian Doctrine<br>Allied Doctrine<br>References (ATP 28) |
| Jane's/DIAS Threat Web/CIA world fact book | Platform Data<br>Weapons Data<br>Country Data |
| CFEWC/NEWC | EW data<br>Platform fits<br>Weapons and sensors<br>Jamming information |

6) Instructor to discuss what an RFI is, what information is contained in RFIs and how to determine the relevant information. (10 min)
7) Each student provided an RFI and briefly write down/highlight what information they feel is relevant within that RFI. (3 key points). (10 min)
8) Students to then share their response and build a class list of the key information from the RFI provided. (20 min)

## Resources

1) CSNI access
2) Flip Charts & Markers
3) RFI example
4) RFI example worksheet
5) Instructor scenario

## References

1) NAVORD 2000-1
2) NAVORD 2000-2
3) Various Websites

## Notes to Instructor

Instructor to introduce various supporting units by locating them on DWAN/CSNI.
The instructor will with the class interpret the provided data by identifying relevant information

EO 003.01 - Prepare Intelligence information

## 003.01.02 – TRINITY support

| Conditions | | Sequence |
|---|---|---|
| TRINITY visit or Classified VTC room (west coast) | | After 003.01.01 |
| **Time** | **Classification** | **Instructor** |
| 200 min | Up to SECRET | TRINITY Guest Lecturer |

| Learning Objectives | Method |
|---|---|
| 1) TRINITY support provided<br>2) Current Events<br>3) How to access TRINITY support | Site Visit/lecture |

| Activity Steps | 1) Trinity brief on site or via classified VTC (west coast option). Topic to include: What they provide to deployed units, turnaround time for requests, pre sailing support etc. (200 min). |
|---|---|

| Resources | References |
|---|---|
| 1) N/A | 2) N/A |

| Notes to Instructor |
|---|
| For West Coast if time is an issue have ROJOC provide the briefing to students, as this is where you will go for support in your local Operating Areas, however they must be briefed on how TRINITY supports works. |

EO 003.01 - Prepare Intelligence information

## 003.01.03 – Drafting RFIs

| Conditions | | Sequence |
|---|---|---|
| Classified classroom CSNI Lab | | After 003.01.02 |
| **Time** | **Classification** | **Instructor** |
| 100 min | Up to SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Utilizing the RFI website/tool<br>2) Determining RFI requirements<br>3) Drafting RFIs<br>4) Submitting RFIs | Guided web based lesson practical |

| Activity Steps | |
|---|---|
| | 1) Instructor to lead students through the Trinity website to determine if the information is already in existence. They will also discuss how to look through the RFI log to determine pending Intelligence products before submitting their own. (25 min)<br>2) Instructor to then discuss what occasions they would need an RFI (what info they may be looking for) and how to use the RFI drafting/submission tool and what information is required to be completed. Of note: RFIs are not limited to Intelligence products, but may also apply to mission support products and geo specific threats. Instructor to touch on RSS submissions (service support during a mission). (25 min)<br>3) Students are then paired up and asked to use the mission specific message traffic (OPGEN/OPORD) and determine their RFI requirements. Teams are to use the Trinity website to determine what products are already in existence that apply, and which information they will be required to submit a RFI for. Students are will then submit training RFIs to TRINITY. (30 min)<br>4) Instructor walk amongst the class and check for accuracy and provide feedback on their submissions. (20 min) |

| Resources | References |
|---|---|
| 1) CSNI workstations<br>2) Projector<br>3) Mission scenario OPGEN and OPORD | 1) Trinity webpage<br>2) NAVORD 2000-2 |

| Notes to Instructor |
|---|
| Use the Mission OPORD for the course mission.<br>Advise TRINITY that training RFIs will be submitted. |

EO 003.01 – Perform Threat Analysis

## 003.01.04 – Determine Effectiveness of Weapons & Sensors

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom CSNI Lab | | After 003.01.03 |
| **Time** | **Classification** | **Instructor** |
| 300 | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Determining SALVO Size<br>2) Determining Stand-Off Distances<br>3) Determining Enemy Emitters and Associated Weapons<br>4) Determining ESM Capability, Detection Ranges and Counter Detection Ranges<br>5) Evaluating Advantages and Disadvantages vs Threat | Student readings<br>Collaborative problem solving |

**Activity Steps**

1) Students are to read MARTI 3550/3555/3556/3560 and 3012 (H) HARPOON Engagement Planning and Salvo Sizes (50min)
2) Students will pair-up. Each pair will be assigned a surface platform that is not specifically listed in the MARTI. Using the MARTI as guidance, students will chose a salvo-size that they believe is appropriate for a mission kill on the assigned platform in both high-EA and low-EA environments. (15min)
3) Using the same surface platform, student pairs will use the SAM Handbook to choose what they believe is an appropriate Stand-Off Distance (STOF) based on the platform's AD capabilities. (15min)
4) Each pair will present and explain their salvo size and STOF choices to the class. (5 min per pair – 20 min total)
5) Students pairs will then compile a list of the of the radars, weapons, and ESM equipment for their assigned platform. Using all available resources, student will determine:
   a) An estimate of their radar detection ranges
   b) An estimate of whether or not Halifax class sensors are detectable by their ESM equipment, and if so, at what range
   c) What weapons systems of the assigned platform fall within these detection ranges (50 min)
6) Each pair will conduct an analysis, comparing Halifax class weapon and sensor caps and lims against the assigned platform's weapon and sensor caps and lims. Student will make an assessment of the advantages and disadvantages of the Halifax class against the assigned platform, addressing detection ranges, counter-detection ranges and weapon engagement ranges. (100 min)
7) Each pair will present their analysis to the classroom. (5 min per pair. – total 20 min)

| Resources | References |
| --- | --- |
| 1) CSNI Lab<br>2) Flip chart paper and markers | 3) SAM Handbook 1 CAD<br>4) CFCD 106 MARTIs 3012(H)/ 3550/3555/ 3556/3560<br>5) DIS Threat Web<br>6) PRO USER (NEDB)<br>7) MEWD |

## Notes to Instructor

Instructors should assign surface platforms with advanced AA, ASuW and Radar/EW capabilities to ensure that students have to conduct a thorough and critical analysis of their capabilities.

EO 003.01 - Perform Threat Analysis

## 003.01.05 – Compile OOB/ORBAT

| Conditions | | Sequence |
|---|---|---|
| DMOC or CSNI Lab | | After 003.01.04 |
| **Time** | **Classification** | **Instructor** |
| 100 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Breakdown EOB<br>2) Compiling an OOB/ORBAT | Lecture<br>Practical |

| Activity Steps | |
|---|---|
| | 1) Instructor is to provide a sample EOB, explain the components of the document and who is responsible for the creation of the document. (15 min)<br>2) Instructor will then explain the ORBAT template students will use, the components of the document and how to complete it. They must provide guidance on formatting, unit organization within the document and determining how to de-conflict information from differing sources. (15 min)<br>3) Students will then individually complete an ORBAT with a minimum or 1 surface unit and 1 A/C assigned by the instructor. (70 min)<br>4) Students to submit their ORBAT and instructor will review and provide feedback within 24hrs. |

| Resources | References |
|---|---|
| 1) CSNI workstation<br>2) Sample EOB<br>3) ORBAT template<br>4) CFCD 106<br>5) SAM handbook | 1) CFCD 106<br>2) SAM handbook |

| Notes to Instructor |
|---|
| |



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EO 003.01 - Perform Threat analysis

## 003.01.06 – Briefing Threat

| Conditions | | Sequence |
|---|---|---|
| Classified classroom | | After 003.01.05 |
| **Time** | **Classification** | **Instructor** |
| 50 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Build Threat Analysis Portion of Mission Brief | EDI Students will Imitate during EC 003.01 |

| Activity Steps | |
|---|---|
| | 1) The instructor will show the students an example of brief/presentation that contains an analysis of the threat. The instructor will use the primary threat ORBAT from the mission scenario in use, and show how much detail needs to go into the brief.<br>2) The instructor will demonstrate how much analysis needs to go into the presentation, specifically outline the TG advantages and disadvantages against the enemy forces. (50 min) |

| Resources | References |
|---|---|
| 1) Brief template with intel complete<br>2) ORBAT | 1) Nil |

| Notes to Instructor |
|---|
| |



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## EC 003.01 – Perform Threat Analysis

| Scope | Conditions | Sequence |
|---|---|---|
| | Classroom | After 003.01.06 |
| **Time** | **Classification** | **Instructor** |
| 400 min (240 min for prep; 160 min for briefing) | SECRET | AWWD Instructor |

| | |
|---|---|
| **Scenario** | Mission Scenario |
| **Assessment Steps** | Input: Using one of the primary threat nations of the mission scenario in use, students will create an ORBAT for their assigned nation to include 3 surface units and 2 A/C, then draft the briefing slides for the threat. Students will be provided a template for the ORBAT and the slides. Brief to include TG Advantages/Disadvantages vs assigned nation's platforms.<br><br>Performance: Student will have 240 mins to complete their ORBATs and threat slides. The students will present their brief to the AWWD instructor. The brief must be 15 min with an additional 5 minutes allocated for "CO's questions". (360 min total)<br><br>Output: The trainee must achieve 8/10 SATs to pass this EC.<br><br>Feedback: Students will receive feedback within 48 hrs of completing the EC deliverables. |

| Resources | References |
|---|---|
| 1) ORBAT template<br>2) Mission brief template | 1) CFCD 106<br>2) Jane's Publications |

| Notes to Instructor |
|---|
| Instructor should attempt, if at all possible, to avoid having students brief the same platforms. However, depending on the size of the naval and air forces of the threat nation (for example, Iran), there may not be a sufficient variety to make this happen. |

| | |
|---|---|
| **Limitations** | |
| **Impact of Failure** | Further instruction and re-assess |

| Item | EC 003.01 - PERFORM THREAT ANALYSIS | SAT | UNSAT | Remarks |
|---|---|---|---|---|
| **1. Create ORBAT** | | | | |
| ORBAT includes minimum number of surface units and aircraft | | ☐ | ☐ | |
| ORBAT is consistent with provided template | | ☐ | ☐ | |
| ORBAT includes all required platform data (length, speed, weapons/sensors, etc.) | | ☐ | ☐ | |
| ORBAT includes required salvo size to be used against assigned nation's surface units | | ☐ | ☐ | |
| ORBAT includes STOF for friendly a/c against assigned nation's surface units | | ☐ | ☐ | |
| **2. AWW Mission Brief** | | | | |
| Identified Surface Threat capabilities/limitation | | ☐ | ☐ | |
| Identified Air Threat capabilities/limitations | | ☐ | ☐ | |
| Identified TG Advantages/Disadvantages vs assigned nation's platforms | | ☐ | ☐ | |
| **3. Conduct Mission Brief** | | | | |
| Demonstrate confident briefing style<br>• Eye contact<br>• Articulates content clearly<br>• Moderate tone/speed<br>• Succinctly maintains focus on content<br>• Logical transitions in content | | ☐ | ☐ | |
| Accurately responds to follow on questions related to mission | | ☐ | ☐ | |
| **Score: / 10** | Trainee must achieve 8/10 SATs on all factors. | | | |

| Positive Feedback | | Improvement Feedback | |
|---|---|---|---|
| | | | |
| **Overall Assessment – Circle One** | | | |
| **Met Standard** | | **Did Not Meet Standard** | |
| **Assessor**<br>I have discussed the results of this assessment with the candidate. | | **Candidate**<br>I have been debriefed on the result of this assessment | |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

PROTECTED B (when complete)

EO 003.02 - Prepare Operational Messages

## 003.02.01 – Interpret Command Structure & Strategic Guidance

| Conditions | | Sequence |
|---|---|---|
| Classified classroom | | After EC 003.01 |
| **Time** | **Classification** | **Instructor** |
| 150 min | Up to SECRET | ORO CO |

| Learning Objectives | Method |
|---|---|
| 1) Interpret command structure (Task Force, Task Group, Task Unit, Task Element design)<br>2) Hierarchy of Commands/Orders/References<br>3) TACON/TACOM/OPCON/OPCOM<br>4) Transfer of Command Authority (TOCA) | Student readings<br>Student presentations<br>Lecture<br>Essence |

| Activity Steps | |
|---|---|
| | 1) Students will conduct readings : ATP-01 Vol 1, CFCD 129, Command Structure (110 min)<br>2) Instructor will split the class into 3 groups. Groups are assigned one of the following topics: command structure, hierarchy of commands/orders, and TACON/TACOM/OPCON/OPCOM Using Essence method each group will reduce content to no more than 10 salient words to describe assigned topic. (20 min)<br>3) One member of from each group, will present the content from the group's assigned topic. (10 min)<br>4) Instructor to discuss Transfer of Command Authority, CHOP procedures and the significance of TOCA. (10 min) |

| Resources | References |
|---|---|
| 1) SECRET workstation<br>2) References<br>3) Flip chart & Markers | 1) ATP-01 Vol 1<br>2) CFCD 129 |

| Notes to Instructor |
|---|
| |



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EO 003.02 - Prepare operational messages

## 003.02.02 – Rules of Engagement

| Conditions | | Sequence |
|---|---|---|
| Classified classroom | | After 003.02.01 |
| **Time** | **Classification** | **Instructor** |
| 100 min | Up to SECRET | MARLANT/MARPAC JAG officer |

| Learning Objectives | Method |
|---|---|
| 1) Purpose of ROE<br>2) Legal basis for ROE<br>3) Control of force (Self-defence, use of force, escalation of force)<br>4) Political sensitivity of ROE authorization<br>5) Importance of Warnings on ROE | Student reading<br>Guest lecture |

| Activity Steps | |
|---|---|
| | 1) Students are to conduct readings from CAF Use of Force Manual (50 min)<br>2) Guest lecture to present a lesson on LO (40 min)<br>3) Instructor or guest lecturer to explain the importance of conducting warnings in relation to ROE (10 min) |

| Resources | References |
|---|---|
| 1) 1 Workstation per student<br>2) DWAN workstation<br>3) projector | 1) CAF Use of Force Manual |

| Notes to Instructor |
|---|
| Ensure you contact guest lecturer well in advance providing the LOs to be covered. |



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EO 003.02 - Prepare Operational Messages

## 003.02.03 – Interpret Operational Messages

| Conditions | | Sequence |
|---|---|---|
| Classified classroom | | After 003.02.02 |
| **Time** | **Classification** | **Instructor** |
| 150 min | Up to SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) OPORD/Warning Orders/OPGEN/ OPTASKs/OPTASK SCC<br>2) Interpret EMCON policies effect on weapons & sensors<br>3) Interpret EW orders<br>4) How to use APP 11 | Jigsaw<br>lecture |

| Activity Steps | |
|---|---|
| | 1) Instructor to divide class into 4 groups. Each group will be assigned one of the following topics:<br>• OPORD/Warning Order<br>• OPGEN<br>• OPTASKs<br>• OPTASKEW and EMCON<br>2) Students are to work within their groups to become "experts" on the assigned topic(s). The goal is not for the students to become experts on the content of an operational message, but experts on their *purpose* and design (importance and hierarchy in relation to other messages). They must conduct readings as assigned using standing messages from mission phase scenarios. (80 min).<br>3) Student will then re-organize into new groups where each group has at least one "expert" from each topic.<br>4) Within their new groups, students will teach each other the topic that they have become experts in. (50 min)<br>5) The instructor will then bring-up APP-11 message drafting tool and demonstrate how to navigate the program, highlighting all of the relevant fields for an OTPASK AAW and OPTASK ASuW. (20 min) |

| Resources | References |
|---|---|
| 1) APP-11 Message Drafting Tool<br>2) Mission Scenario Messages | 1) APP-11 |

| Notes to Instructor |
|---|
| For details on how to conduct a Jigsaw activity, reference AIT manual. |

EO 003.02 - Prepare operational messages

## 003.02.04 – Interpreting OPTASK AAW/AD

| Conditions | | Sequence |
|---|---|---|
| Classified classroom | | After 003.02.03 |
| **Time** | **Classification** | **Instructor** |
| 100 min | Up to SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Interpret data from OPTASK AAW/AD | Student readings<br>Modified 3 STAR<br>Think-Pair-Share |

| Activity Steps | |
|---|---|
| | 1) Students are to conduct readings from Allied Publications, to include mission scenario OPTASK AAW (65 min)<br>2) Upon completion of readings students are to create a list of their TOP 5 most important/significant pieces they believe are contained in an OPTASK AAW. (10 min)<br>a. Duties (to include general understanding and AAW specific duties)<br>b. CAP + AAW Stns<br>c. Ctrl types (AEW/EW)<br>d. AREA/Circles/Sectors<br>e. Surveillance Areas<br>f. Weapon Coordination<br>g. Safe Sectors<br>h. MEZ/FEZ/COZ<br>i. Alerts (Weapons/AC)<br>j. Zippo Areas<br>k. STOF<br>l. AAW Ref Pts.<br>m. ID GATES and ID Safety Zones<br>n. Force Marshalling<br>o. Warnings<br>3) Students are to then pair up, share their top 5 list and collaborate to create a TOP 3 (10 min).<br>4) Students are to then present their list to the class, providing substantiation/reasoning for their selections. Instructor to be engaged in the presentations, providing their opinion on the substantiations (15 min) |

| Resources | References |
| --- | --- |
| 1) CSNI computer/SECRET standalone (13)<br>2) Projector<br>3) OPTASK AAW message<br>4) OPTASK AAW Supp message<br>5) OPGEN/OPORD | 1) APP 11<br>2) ATP-1 Vol 1<br>3) AJP-3 Series<br>4) CFCD 106<br>5) APP-1 |

| Notes to Instructor |
| --- |
| |

EO 003.02 - Prepare Operational Messages

# 003.02.05 – Interpreting OPTASK ASuW/SCC

| Conditions | | Sequence |
|---|---|---|
| Classified classroom | | After 003.02.04 |
| **Time** | **Classification** | **Instructor** |
| 100 min | Up to SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Interpret data from OPTASK ASuW/ ASuW (Supp) | Student readings<br>Modified 3 STAR<br>Think-pair-share |

**Activity Steps**

1) Students are to conduct readings from Allied Publications to include mission scenario OPTASK ASuW (65 min)
2) Upon completion of readings students are to create a list of their TOP 5 most important/significant pieces they believe are contained in an OPTASK ASuW. (10 min)
    i) Duties (to include general understanding and ASuW specific duties)
    ii) Screens
    iii) Sags/Surf Plans
    iv) TRU Priorities
    v) Unit ctrl (Area/Circ/Zones)
    vi) TRU
    vii) FPB
    viii) Surveillance Zones
    ix) Alert States
    x) Engagement instructions/Salvos
    xi) Warnings
3) Students are to then pair up, share their top 5 list and collaborate to create a TOP 3 (10 min).
4) Students are to then present their list to the class, providing substantiation/reasoning for their selections. Instructor to be engaged in the presentations, providing their opinion on the substantiations (15 min)

| Resources | References |
|---|---|
| 1) CSNI computer/SECRET stand alone<br>2) Projector<br>3) OPTASK ASuW message<br>4) OPTASK ASuW Supp message<br>5) OPORD/OPGEN | 1) APP 11<br>2) ATP-1 Vol 1<br>3) CFCD 106<br>4) APP-1 |

**Notes to Instructor**



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EO 003.02 - Prepare operational messages

## 003.02.06 – Identify and Classify Contacts

| Conditions | | Sequence |
|---|---|---|
| Classified classroom | | After 003.02.05 |
| **Time** | **Classification** | **Instructor** |
| 150 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) What is Recognition Confidence Levels (RCL)<br>2) What is Identification Criteria (IDCRIT)<br>3) Making IDCRIT mission specific | Student readings<br>Lecture<br>Study period<br>quiz |

| | |
|---|---|
| **Scenario** | Mission Scenario |
| **Activity Steps** | 1) Students are to read References as required (30 mins).<br>2) The instructor will then present the IDCRITs and RCLs that were developed for the selected mission scenario IAW the mission OPTASK messages. Instructor to elaborate on the specific information of the messages that pertain to the lesson and explain the relevance of the details. (20 min)<br>3) Students are then provided a study period to review IDCRITs and RCLs in preparation for a quiz. (50 min)<br>4) Student will then complete a 10 question quiz where they are required to provide the baseline definitions for IDCRITs and an explanation of the RCLs IAW the references. The instructor will correct and provide feedback on the quiz within 24hrs. Student must achieve 100% on the quiz. If they do not achieve 100%, they will repeat the quiz until they achieve 100%. (50 min) |

| Resources | References |
|---|---|
| 1) RCL and IDCRIT Quiz<br>2) OPTASK AAW & ASuW<br>3) DS solution for RCL & IDCRIT quiz | 1) ATP 31 Pg. 8-7 8207<br>2) STANAG 1241<br>3) APP-01 Annex A<br>4) FOTI 0611 |

| Notes to Instructor |
|---|
| Definition is not required to be “word for word” from the reference but must clearly align with the definition content. |



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EO 003.02 - Prepare Operational Messages

## 003.02.07 – Drafting OPTASK Supp Messages

| Conditions | | Sequence |
|---|---|---|
| Classified classroom | | After 003.02.06 |
| **Time** | **Classification** | **Instructor** |
| 100 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) OPTASK Supp messages drafting<br>2) Utilizing message drafter | Experiential Learning/Practical |

| Scenario | |
|---|---|
| **Activity Steps** | 1) Students are provided the standing Scenario OPTASK messages and the OPTASK Supp template (from EC 003.02). They are to draw the information from the standing messages to populate the Supp. Using List of Potential Changes/Amplification for Standing Messages Instructor to provide any assistance the students may require in using message drafter software as they may not have a lot of experience with this tool. Walk throughout and provide feedback to students on format. (100 min) |

| Resources | References |
|---|---|
| 1) SECRET stand-alone/CSNI workstation<br>2) Message drafter software<br>3) OPTASK AAW and ASuW<br>4) List of Potential Changes/Amplification for Standing Messages | 1) APP 11<br>2) Standing Messages |

| Notes to Instructor |
|---|
| |



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EO 003.02 - Prepare Operational Messages

## EC 003.02 – Prepare Operational Messages

| Scope | Conditions | Sequence |
|---|---|---|
| | Classroom | After 003.02.07 |
| **Time** | **Classification** | **Instructor** |
| 200 min | SECRET | AWWD Instructor |

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|---|---|
| **Scenario** | Mission Scenario |
| **Assessment Steps** | Input: Student will continue to use the threat nation and platforms assigned during EC 003.01. Students will envision a hypothetical scenario where their TG may be required to engage those specific naval and air forces. Students will produce a reduced OPTASK AWW Supp or OPTASK ASuW Supp (instructor to assign) based on this hypothetical engagement and the reduced OTPASK Supp template.<br><br>Performance: Student will have 200 mins to complete their OPTASK Supp.<br><br>Output: The trainee must achieve 3/3 SATs to pass this EC.<br><br>Feedback: Students will receive feedback within 48 hrs of completing the EC deliverables. |

| Resources | References |
|---|---|
| 1) Standing OPTASK AAW/ASuW | 1) APP-11<br>2) Standing Messages |

| Notes to Instructor |
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| Instructors should aim to evenly assign Supps (AAW or ASuW) amongst the students. |

| | |
|---|---|
| **Limitations** | |
| **Impact of Failure** | Further instruction and re-assess |

| Item | EC 003.02 – PREPARE OPERATIONAL MESSAGES | SAT | UNSAT | Remarks |
|---|---|---|---|---|
| **1. Draft an OPTASK Supp message** | | | | |
| OPTASK completed IAW provided template | | ☐ | ☐ | |
| Amendments entered | | ☐ | ☐ | |
| OPTASK absent spelling errors and all fields are IAW APP-11 formatting | | ☐ | ☐ | |

**Score: /3** The trainee must achieve 3/3 SATs to pass this EC.

| Positive Feedback | Improvement Feedback |
|---|---|
| | |

**Overall Assessment – Circle One**

| Met Standard | Did Not Meet Standard |
|---|---|
| **Assessor**<br>I have discussed the results of this assessment with the candidate. | **Candidate**<br>I have been debriefed on the result of this assessment |

| | | | |
|---|---|---|---|
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

# PC 003 – Plan AWW Operations

| Scope | Conditions | Sequence |
|---|---|---|
| PO 003 | Classified classroom | O/C of PO 003 |
| **Time** | **Classification** | **Instructor** |
| 600 min | Up to SECRET | AWWD Instructor<br>Post ORO NWO to receive brief |

| | |
|---|---|
| **Scenario** | Mission Scenario |
| **Assessment Steps** | Input: Student will be assigned a nation from within the AOO of the mission scenario in use. This nation must be different from the one the students were assigned for EC 003.01 and EC 003.02. Using all research tools available, students will then create either an OPTASK AWW Supp or OPTASK ASuW Supp based on a hypothetical conflict between the student's TG and the assigned nation. The student will then create and deliver an AWW portion of a mission planning brief. Students will be provided the brief template.<br><br>Performance: Student will have 400 mins to complete their OPTASK Supp and their mission brief. The brief must include all information required as part of the EC 003.01 brief, as well as:<br>• Assigned nation's ORBAT (minimum of 3 surface ships and 2 A/C)<br>• Advantages/Disadvantages vs assigned nation's platforms<br>• Tactical decisions outlined in the OPTASK Supp based on the assigned nation's capabilities<br><br>The students will present their brief to a Post ORO NWO, with the AWWD Instructor as assessor. The brief must be 15 min with an additional 5 minutes allocated for "CO's questions". (200 min total)<br><br>Output: The trainee must achieve 7/8 SATs to pass this PC.<br><br>Feedback: Immediate verbal and written feedback for the briefing using PC 003 checklist. Students will receive feedback for their written assignment (OPTASK Supp) within 24 hrs of submission. |

| Resources | References |
|---|---|
| 1) Brief Template<br>2) DWAN/SECRET Computer<br>3) Projector | 1) Trinity Webpage |



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| Notes to Instructor |
| --- |
| An example would be a student being assigned Egypt or Israel if the mission scenario is Op Arbiter. More than one student may be assigned the same nation if necessary based on the AOO. Selected countries should provide a wide breadth of air and surface platforms. |

| Limitations | |
| --- | --- |
| **Impact of Failure** | IAW NPTORD 4500-54 |

| Item | PC 003 - PLAN AWW OPERATIONS | SAT | UNSAT | Remarks |
|---|---|---|---|---|
| **1. Brief Intelligence Information** | | | | |
| Provided basic overview of nation data (population, location, geo-political situation) | | ☐ | ☐ | |
| Identified impact of anticipated Pattern of Life | | ☐ | ☐ | |
| Identified assigned nation role in conflict | | ☐ | ☐ | |
| **2. Brief Threat Information** | | | | |
| Identified Surface Threat capabilities/limitation | | ☐ | ☐ | |
| Identified Air Threat capabilities/limitations | | ☐ | ☐ | |
| Identified own ship Advantages/Disadvantages vs threat platforms (so what of caps/lims) | | ☐ | ☐ | |
| **3. Brief Operational Messages information to include:** | | | | |
| Demonstrate confident briefing style<br>• Eye contact<br>• Articulates content clearly<br>• Moderate tone/speed<br>• Succinctly maintains focus on content<br>• Logical transitions in content | | ☐ | ☐ | |
| Accurately responds to follow on questions related to mission | | ☐ | ☐ | |

**Score: / 8**

The trainee must achieve 7/8 SATs to pass this PC.

| Positive Feedback | Improvement Feedback |
|---|---|
| | |

| Overall Assessment – Circle One | | | |
|---|---|---|---|
| **Met Standard** | | **Did Not Meet Standard** | |
| **Assessor**<br>I have discussed the results of this assessment with the candidate. | | **Candidate**<br>I have been debriefed on the result of this assessment | |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

PROTECTED B (when complete)



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## PO 004 – Execute Duties of an AWW Director (T0001)

1. **Performance**: The AWWD will execute the duties of an AWW Director (T0001)

2. **Conditions**:

   a. Given:

      (1) Halifax-class operations room;
      (2) Halifax-class sensor and weapons suite;
      (3) Drills and procedures manual; and
      (4) References.

   b. Denied:

      (1) Nil

   c. Environment:

      (1) At sea or alongside (simulated); and
      (2) Day or night.

3. **Standard**: The AWWD shall execute the duties of an AWW Director (T0001) by:

   a. Maintaining Recognized Maritime Picture (RMP) (F00190) by:

      (1) Coordinating the employment of aircraft (BT0230) by:
         (a) Coordinating the control of aircraft (BT0220); and
         (b) Tactically employing aircraft assigned to the force (E00070)
      (2) Coordinating the compilation of RMP (E00050);
      (3) Supervising the compilation of the anti-surface warfare picture (BT0270);
      (4) Identifying maritime, air, and land based platforms;
      (5) Contributing to the RMP by:
         (a) Communicating using tactical voice procedures (A00060);
         (b) Authenticating communication transmissions (A00010);
         (c) Performing communications checks (A00020) and
         (d) Configuring communications terminal for operations (A00030).
      (6) Amplifying track information (F0010);
      (7) Tracking high interest vessels (C00670);
      (8) Adhering to EMCON restrictions (B00130);
      (9) Reacting to changes in the EW environment (C00580);
      (10) Coordinating ships ES, EA and EP operations (C00110);
      (11) Classifying RADAR contacts (B0010);
      (12) Performing area of interest (AOI0 monitoring C00450); and
      (13) Advising command on AWW tactics (T0002) by:
         (a) Coordinating the control of sensors (BT0020);

(b) Coordinating the employment of sensors (BT0210);
(c) Coordinate the control of weapons BT0180);
(d) Coordinate the employment of weapons (BT0190); and
(e) Recommending tactical employment of sensor (C00460).

b. Defending against missile threats by:
(1) Performing detect to engage sequence;
(2) Directing Electronic Warfare (EW) (BT0520); and
(3) Directing ships ES, EA and EP operations (C00520);

c. Defending against surface threats by:

(1) Implementing anti-surface warfare tactics (BT0300);
(2) Coordinating anti-surface warfare ops (BT0070); and
(3) Coordinating ASuW (E00040).

d. Defending against air threats by:

(1) Implementing anti-air warfare tactics (BT0290)
(2) Supervising the compilation of the anti-air warfare pictures (BT0260);
(3) Assessing jamming (C00050);
(4) Coordinating radar anti-jamming procedures (D0012);
(5) Coordinating anti-air warfare ops (BT0060); and
(6) Coordinate the local AAW (E00060).

**Training Limitations**:

a. CMS version in the synthetic environments ashore may not be the same version in use on ship;
b. Equipment is SEACOT are often simulated and do not always operate in the same manner as equipment aboard ship; and
c. SEAWOLF does not have either an ELISRA or SRD-505 simulation in any form, thereby reducing the realism of EW operations.

EO 004.01 - Maintain Recognized Maritime Picture

## 004.01.01 – Employ Air Assets and CAP

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After PC 003 |
| **Time** | **Classification** | **Instructor** |
| 200 min | SECRET | Guest Lecturer (LRPA ACSO or RCAF MFC) |

| Learning Objectives | Method |
|---|---|
| 1) Understanding the capabilities of various air assets, including weapons and sensors, mission endurance, and concept of employment<br>2) A/C mission types (i.e. SSSC, ASW, etc.)<br>3) How a/c manages stand-off distances and risk levels<br>4) How a/c joins are conducted the different types of tasking orders (and how they are conveyed – voice/link)<br>5) Understand Air Coordination (ACC, LAAWC/SAAWC, Airspace Control, CROWN, FLYPRO, ACO/ATO)<br>6) CROWN, FLYPROs, ATOs and ACOs<br>7) Understanding CAP:<br>a. CAP Mission Types (DCA vs SUCAP)<br>b. Common CAP a/c<br>c. Who employs/directs CAP (ATC vs MFC)<br>d. CAP Flight Patterns<br>e. MEZ/FEZ/COZ and OILFIED/KOBOLD<br>f. Key code words when CAP is being employed | Guest Lecture<br>Tutorial |

| Scenario | |
|---|---|
| **Activity Steps** | 1) Guest Lecturer to provide a brief on the description, general capabilities, and importance to the SWC of the following air assets:<br>a) LRPA/MPA<br>b) CP-140<br>c) P3C<br>d) Aircraft Carrier<br>e) AWACS<br>f) E2 Hawkeye<br>g) Cyclone<br>h) F/A-18<br>2) Guest lecturer to continue his brief by covering the subjects of LOs 2-6. (150 min)<br>3) Using the whiteboard with model ships and a/c, the instructor will design various scenarios with complex air and surface pictures, and explain to the students how various air assets may be employed to achieve the goal of building the RMP (i.e. what type of a/c could be found in a TG and what missions they can be directed to perform). |

| | |
|---|---|
| | 4) The instructor will then discuss LO 7, again using the whiteboard to demonstrate where various things that a SWC can expect to encounter when CAP is being employed in their AO. |

| Resources | References |
|---|---|
| 1) Secret Workstation<br>2) Projector<br>3) Whiteboard and Markers | 1) |

| Notes to Instructor |
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| |

EO 004.02 – Maintain the Recognized Maritime Picture

## 004.01.02 – Over The Horizon Targeting (OTHT)

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After 004.01.01 |
| **Time** | **Classification** | **Instructor** |
| 50 min | SECRET | Guest Lecturer (LRPA ACSO)<br>AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) What is OTHT and what are TRUs<br>2) Types of TRUs (A/C, surface units, shore based, etc.)<br>3) Types of TRU sensors (radar, visual, acoustic)<br>4) Reporting Methods (Link vs Voice) and Reference Points<br>5) How to assess damage<br>6) Effect of Reporting Method on PAcq | Lecture<br>Three Star |

| | |
|---|---|
| **Scenario** | Mission Scenario |
| **Activity Steps** | 1) Continuing form lesson 004.01.01, the instructor will discuss LOs 1-5. They should use whatever resources are available, including the whiteboard and model ships, to demonstrate how a TRU vastly increasing the range at which the TG can maintain situational awareness of the RMP. (25 min)<br>2) The instructor will then discuss the concept of DAMAS, including presenting the students with DAMAS categories, explaining who can conduct DAMAS, and how they do it. (10 min)<br>3) When discussing Probability of Acquisition (PAcq), the AWWD instructor shall explain how PAcq is affected by the accuracy and currency (frequency) of target data updates (e.g. in HARPOON planning), and emphasize the negative impact that reporting over voice has on PAcq. (10 min)<br>4) Students are asked to write down the 3 most critical items they learned in the lesson. Then once everyone has done so, share it with a peer and discuss why they feel their points are the most critical. (5 min) |

| Resources | References |
|---|---|
| 1) Fleet-in-a-Box | 1) CFCD 106 3012 Annex K<br>2) CFJP 3-9 Chapters 1, 3, 4, 6 |

| Notes to Instructor |
|---|
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EO 004.01 - Maintain Recognized Maritime Picture

## 004.01.03 – Utilize ESM Information

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After 004.01.02 |
| **Time** | **Classification** | **Instructor** |
| 150 min | Secret | EW Instructor (RQ-PO2 NESOP) |

| Learning Objectives | Method |
|---|---|
| 1) EMCON<br>2) Trouts, Rents, and Criss-crosses<br>3) Associated vs Unassociated ESM<br>4) AOPs<br>a) Types (Rectangular, Circular)<br>b) Expansion Rates<br>c) Sanitizing | T-Chart<br>Interactive Lecture |

| Activity Steps | |
|---|---|
| | 1) Students are broken into 2 teams, using a flip chart they are to each create a table (t-chart) indicating what they believe are the positives and negatives of a restrictive EMCON state. (10 min)<br>2) Students are to then present their lists to the class (5 min)<br>3) Using the whiteboard to provide realistic tactical scenarios, the instructor will explain and demonstrate to the students the concepts and importance of: (50 min)<br>a) The function of EMCON<br>b) How EMCON is utilised in maritime operations<br>c) How does a restrictive EMCON state help you tactically<br>d) What ways are there to mitigate the negatives of a restrictive EMCON state<br><br>4) Using the whiteboard, instructor is to explain and demonstrate Trout, Rent, and Criss-Crosses reports. Instructor is to explain the purpose of each and who initiates them, and will demonstrate the voice procedures associated with each.<br>5) Instructor to explain the difference between an associated and unassociated ESM<br>6) Instructor to demonstrate how unassociated ESM results in an AOP<br>7) Instructor to explain how AOPs are created, the rationale behind the type and expansion rate<br>8) Instructor to explain how AOPs are sanitized (with radar, visual clearing, and acoustics). (100 min) |



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| Resources | References |
| --- | --- |
| 1) Whiteboard and markers<br>2) Fleet-in-a-box<br>3) CMS 3D Printed Magnetic Game Pieces<br>4) flip chart with markers | 1) |

| Notes to Instructor |
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EO 004.01 - Maintain Recognized Maritime Picture

## 004.01.04 – Resolve Procedures and Warnings

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After 004.01.03 |
| **Time** | **Classification** | **Instructor** |
| 150 min | Secret | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) How to conduct a resolve procedure<br>2) How to make a contact assessment<br>3) How to use check-print lines<br>4) How to report internally and externally after an assessment<br>5) How to issue warnings<br>6) How to interpret responses to warnings | Individual Study<br>Whiteboard<br>EDI<br>Team quiz |

| Scenario | |
|---|---|
| **Activity Steps** | 1) Students are to read MARTIs 4410 and 4450. Students are to create 2 questions each, with the answers while doing their readings and hand in to instructor. (60 min)<br>2) Using the whiteboard, Instructor to explain the resolve procedure, including when it is initiated, who is involved, and anticipated responses from those involved. (15 min)<br>3) Instructor will then discuss the various possible assessments (IDCRIT) that the SWC can make based on what information they obtained through the resolve procedure. (5 min)<br>4) Instructor will create different scenarios where air contact and associated information (bearing, range, speeds, etc.) are written on the white board. One at a time, students will conduct resolve procedures using the provided information, including providing a final assessment of the IDCRIT for the contact. The instructor will provide the responses for the EWS, ARRO and ES OP. Each student should have at least 1 opportunity, but the instructor may conduct additional training (time permitting). (5 min per student – total 40 min)<br>5) Instructor will then discuss what follow-on actions that are required as a result of a completed resolve and assessment (report external, issue warnings, brief ORO). They will explain the use of check-print lines, warning organization (i.e. who issues warnings), the issuing of warning IAW ROE (i.e. air vs surface warnings), and the process of upgrading from one warning to the next. (15 min)<br>6) Instructor must then explain the importance of responses to warning, including prompting the ORS for responses, briefing those responses to the ORO, and how warning responses may affect the IDCRIT and plan of action regarding the contact. (5 min)<br>7) Instructor to then use the student created questions to do a final confirmation activity, asking the questions of the class and getting them to answer based off their learning. (10 min) |

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|---|---|
| | NOTE: resolve procedures will be evaluated in EC004.01, however, employing the warning organization will be evaluated in EC004.04. |

| Resources | References |
|---|---|
| 1) White Board, Markers<br>2) CMS 3D Printed Magnetic Game Pieces | 1) MARTI 4410<br>2) MARTI 4450 |

## Notes to Instructor

When discussing warnings, instructor to ensure they make it clear the differences and similarities between Surface-to-Air, Surface-to-Surface, and Air-to-Surface warnings. MARTI brevity words have the caveat that only the amplification is classified. Remind students of security diligence concerning this item.

EO 004.01 - Maintain Recognized Maritime Picture

## 004.01.05 – Communications

| Conditions | | Sequence |
|---|---|---|
| SEACOT (primary)<br>SEAWOLF (secondary) | | After 004.01.04 |
| **Time** | **Classification** | **Instructor** |
| 200 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Configure communications terminals for ops<br>2) Perform communications checks<br>3) Net management, difference between C&R and Coord nets, open and directed nets, red/green nets, UHF, HF, SATCOM nets<br>4) Deliver picture compilation reports and returns to include:<br>a) Voice reports/contact classification, "I make" reports, contact upgrades, faded, regain;<br>b) Chat/contact reports, templates, OPTASK formats;<br>5) Communicate using tactical voice procedures<br>6) Authenticate communications transmissions | Student readings<br>EDI |

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|---|---|
| **Scenario** | Mission Scenario |
| **Activity Steps** | 1) Students will read the following: (50 min)<br>a. CFCD 114: s509<br>b. APP-01: 0801-0830<br>c. CFCD 106: MARTI 4230<br>2) The instructor will discuss and demonstrate the aspects of LO 1-3 using the SHINCOM terminals in SEACOT. (15 min)<br>3) With the instructor console connect to the projector and utilizing a simple surface and air picture, the instructor will demonstrate various voice reports between the SWC, ORO and PWC that would occur has part of resolving the air picture and building the RMP (LO 4a). The focus is not on building the RMP, but demonstrating to the students proper net management, and ensuring that reports are delivered both internally and externally. (15 min)<br>**Instructor must discuss the importance of keeping the ORS informed, especially when the ORO is unavailable to receive a report<br>4) The instructor will then demonstrate the same process, but demonstrate the use of Chat to achieve the same effect (LO 4b). (15 min)<br>5) Students will then pair-up at CMS consoles. The instructor will ask one student at a time to close-up at the instructor's console, where they will be required to deliver internal and external reports to the instructor. The instructor will select which contacts the student must make reports for (i.e. resolve an air contact, report it to the ORO, and conduct an assessment and/or a change/amplification of IDCRIT/RCL report to the PWC). During this activity, the instructor will act as both ORO and PWC. While this is taking place, the remaining students (still in |

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|---|---|
| | pairs) will make reports to each other without the use of SHINCOM (as the nets will be in use by the instructor and the student being mentored). (70 min)<br>6) Once each student has had an opportunity for one-on-one mentorship with the instructor, the instructor will demonstrate to the students the process of authenticating communications. (15 min)<br>7) Once again in pairs, student will close-up at CMS consoles. Each console will be assigned a 3-letter call-sign and be provided with an exercise authentication table. Using the call-signs on the external net, the instructor will practice performing authentication procedures with each student. While waiting their turns, students will practice authenticating with one another. (20 min) |

| Resources | References |
|---|---|
| 1) Exercise authentication tables | 1) CFCD 114<br>2) ATP-01<br>3) CFCD 106 |

| Notes to Instructor |
|---|
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EO 004.01 - Maintain Recognized Maritime Picture

## 004.01.06 – ID and Data Fusion

| Conditions | | Sequence |
|---|---|---|
| SEACOT (primary)<br>SEAWOLF (secondary) | | After 004.01.05 |
| **Time** | **Classification** | **Instructor** |
| 200 min | SECRET | RQ-PO2 NCIOP |

| Learning Objectives | Method |
|---|---|
| 1) ID sources: IFF/Kinematics/GCCS/Stone ghost/Grey Goose/SATCOM<br>2) IFF interrogation, lethal interrogation, Modes 1/2/3/C/S/4/5;<br>3) Understand MTDF and CMS inputs<br>4) What does Auto-IDENT do how does it affect CMS<br>5) How link contributes to the RMP.<br>6) Dealing with Dual Tracking, Unfused Contacts, and ID Conflicts | Individual Study<br>EDI |

| | |
|---|---|
| **Scenario** | Mission Scenario |
| **Activity Steps** | 1) Students will read: (100 min)<br>a. ADatP-33: 4.7.4/5.5.6.9/5.7.2.11/5.8-5.8.7.3.12/5.10.2-5.10.6<br>b. CFCD 106: MARTI 1400 p35/3600 p18, 22/4230 p48/4265 p9/4600 p25/4731 p17/4740 p7/4741 p5/<br>c. ATP-31: 3206/6219/7305-7306/8925-8929/9204<br>d. CSUMs: 5.3.2.1.7.1-5.3.2.1.7.1.6<br>2) Each student will close-up at a CMS console. Their ships will all be in close proximity (NMT 20nm between ships) on LINK 16 so they can share picture data and so that organic sensor data match. The instructor will walk the students through the various panels and display in CMS that deal with contact ID and IFF, focusing on LOs 1-3. (25 min)<br>3) The instructor will introduce contacts with varying IFF modes, altitudes, speeds and courses. The instructor will then instruct one ship to assign one ID set and another ship to assign a different ID set. The first ship will link the contact, the second will not. This will be repeated with different ID characteristics to demonstrate the presence or lack of ID conflicts. The instructor shall explain to the students how ID conflicts negatively affect picture compilation, and how we deal with ID conflicts. (15 min)<br>4) The instructor will show how different IFF modes are displayed and what happens when one radar is interrogating all modes and the other radar is off. They will then look at what happens when both radars interrogate all modes. (15 min)<br>5) The instructor will then fly two contacts in close proximity with IFF on one and not on the other, then the trainees will change one to hostile and one to friend. Then place the system in Auto-Ident. The instructor will then provide ID details to the trainees to update and link out to observe system change. The instructor shall |

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| | explain the consequences for the SWC and picture compilation when CMS has Auto-IDENT active. (15 min)<br>6) Once complete, the instructor will let the trainees work on their own to resolve the picture while they inject varying contacts and criteria. (30 min) |

| Resources | References |
|---|---|
| 1) | 1) ADatP-33<br>2) CFCD 106<br>3) ATP-31<br>4) CSUMs |

| Notes to Instructor |
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EO 004.01 – Maintain Recognized Maritime Picture

## 004.01.07 – RMP Management

| Conditions | | Sequence |
|---|---|---|
| SEACOT (primary)<br>SEAWOLF (secondary) | | After 004.01.06 |
| **Time** | **Classification** | **Instructor** |
| 1000 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Applying RCL and IDCRIT<br>2) Managing the AWW team<br>3) Communicating with the ORO<br>4) Tasking the SAC<br>5) Employing air assets<br>6) Building the RMP | Practical<br>Fishbowl |

| | |
|---|---|
| **Scenario** | Mission Scenario |
| **Activity Steps** | 1) With the instructor console connected to the projector, and with support manning IAW NWTP vol 2, the instructor will conduct a full demonstration of RMP building. The instructor will include several threat platforms (air and surface) from the chosen mission scenario. The scenario will require instructor to localize the enemy forces while maintaining the overall RMP. During this process, the instructor must ensure that they demonstrate: (50 min)<br>a) Surface Contacts of varying RCL and IDCRIT<br>b) Air Contacts of varying IDCRIT<br>c) Commercial Air<br>d) Resolve procedures<br>e) Issue of Warnings<br>f) Employment of the TRU (and tasking the SAC)<br>g) ESM Rackets<br>h) AOPs<br>i) All associated voice reports<br>j) Threat reduction<br>2) Students will then take turns practicing RMP building with support manning detailed in NWTP 2. If sufficient manning is available, more than 1 student can be practicing at a time; however, instructors should endeavour to focus their attention and mentorship on one student at a time.<br>3) All LOs from EO 004.01 should be exercised during these runs. However, these runs will be the first real opportunity for students to hone their skills at understanding and managing the AWW team. Of particular note, the students must be explained the relationship between the SWC and the SAC, explained how the SWC directs and tasks the SAC, and what kind of control an RCN SAC has when tasking a/c for a given mission. |

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| | 4) Students will conduct a minimum of 2 x 50 min runs where they must build a surface and air picture with all contacts updated to include RCLs and ID criteria in CMS. (950 min)<br>5) Feedback will be provided using EC 004.01 checklist. |

| Resources | References |
|---|---|
| 1) RMP Building Scenarios | 1) APP-01 Annex A<br>2) ATP-31 8207<br>3) FOTI 0611<br>4) ATP-1D |

| Notes to Instructor |
|---|
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# EC 004.01 - Maintain Recognized Maritime Picture (RMP)

| Scope | Conditions | Sequence |
|---|---|---|
| EO 004.01 | SEACOT (primary)<br>SEAWOLF (secondary) | After 004.01.07 |
| **Time** | **Classification** | **Instructor** |
| 400 min | SECRET | AWWD Instructor |

| | |
|---|---|
| **Assessment Steps** | Input: Student is closed-up with policy assumed and support manning detailed in NWTP 2, where they must execute the duties of the AWWD in maintaining RMP during mission runs.<br><br>Performance: Students will conduct a 50 min run where they must build a surface picture with all contacts updated to include RCLs and ID criteria in CMS. Scenario must include:<br>a) 1 x TRU<br>b) 5 x Surface Contacts (Requiring ID Crit and RCLs to be applied)<br>c) 5 x Air Contacts (ESM or IFF provided)<br>d) Should be a combination of Military and Civilian Contacts<br><br>Output: Trainee must achieve 18/23 SATs to pass this EC.<br><br>Feedback: Immediate verbal and written using EC 004.01 checklist. |

| Resources | References |
|---|---|
| 1) Scenarios for RMP | 1) CFCD 106 |

| Notes to Instructor |
|---|
| Students may create a tote or use memory, as long as they are able to update CMS. |

| | |
|---|---|
| **Limitations** | |
| **Impact of Failure** | Further instruction and re-assess |



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| Item | EC 004.01 - MAINTAIN RECOGNIZED MARITIME PICTURE (RMP) | SAT | UNSAT | Remarks |
|---|---|---|---|---|
| **1. Pre-Run Set-Up** | | | | |
| CMS configured IAW locally produced set-up sheet or battle orders | | ☐ | ☐ | |
| Correct MSD/HPD selected | | ☐ | ☐ | |
| Communication nets selected and set-up | | ☐ | ☐ | |
| Briefed AWW team on Alpha threat, mission priority, primary threat, and team expectations | | ☐ | ☐ | |
| Briefed all delegations, ensuring duties and responsibilities were understood | | ☐ | ☐ | |
| **2. Build and Maintain the Recognized Maritime Picture** | | | | |
| Ensured all contacts are updated with all ID and multi-link fields | | ☐ | ☐ | |
| Conducted assessment of all contacts using RCL/IDCRIT IAW OPTASK messages | | ☐ | ☐ | |
| Directed the AWW Team in evaluating all contacts | | ☐ | ☐ | |
| Resolved and assigned RCL/IDCRIT for all air contacts IAW OPTASK messages | | ☐ | ☐ | |
| All contacts reported IAW OPTASK | | ☐ | ☐ | |
| Ensured all ESM rackets are associated IAW Doctrine | | ☐ | ☐ | |
| Sanitized all AOPs | | ☐ | ☐ | |
| All air contacts with IFF in air lanes are assessed and monitored | | ☐ | ☐ | |
| **3. Tactically Employ Assigned Air Assets** | | | | |
| Gave direction to SAC regarding air tasking | | ☐ | ☐ | |
| Employment of Air Asset supports the tactical situation and achievement of the mission | | ☐ | ☐ | |

| | | | |
|---|---|---|---|
| Briefed CO on status of air asset at join, assumption of mission, alteration of mission | ☐ | ☐ | |
| Gave direction to SAC on SAM Standoff (STOF) IAW doctrine | ☐ | ☐ | |
| **4. Direct and Lead the AWW Team** | | | |
| Provided direction to the team and set priorities | ☐ | ☐ | |
| Followed Warfare Commander direction | ☐ | ☐ | |
| Used voice procedures IAW ACP-125 | ☐ | ☐ | |
| Used SAR format when briefing ORO | ☐ | ☐ | |
| Briefed ORO on potential threats before they become imminent threats | ☐ | ☐ | |

**Score: / 22**

The trainee must achieve 18/22 SATs to pass this EC.

| Positive Feedback | Improvement Feedback |
|---|---|
| | |

| Overall Assessment – Circle One | | | |
|---|---|---|---|
| **Met Standard** | | **Did Not Meet Standard** | |
| **Assessor**<br>I have discussed the results of this assessment with the candidate. | | **Candidate**<br>I have been debriefed on the result of this assessment | |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

PROTECTED B (when complete)

EO 004.02 – Defend Against Missile Threats

## 004.02.01 – Anti-Ship Missiles

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom<br>CSNI Lab | | After EC 004.01 |
| **Time** | **Classification** | **Instructor** |
| 200 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) ASM Seeker Types<br>2) ASM Features and Characteristics<br>3) ASM Support Infrastructure and Hardware<br>4) ASM Targeting Methods | Jigsaw<br>Student research/presentations |

**Activity Steps**

1) Instructor will break the class into 4 groups. Each group will be assigned a topic from the following lists. They are to research their topic with the provided course material, and references IOT present their content to the class. They must include the details below the title in their presentations. Groups will have 50 min to conduct their research. (50 min)

**A. Seeker Types**

i) ARM Seekers;
ii) Active Radar Seekers;
iii) Infrared Seekers;
iv) Dual mode Seekers;
v) Command Line of Sight guidance;
vi) Electro-Optical/TV guidance; and
vii) Laser guidance.

**B. Features and Characteristics**

i) HOJ;
ii) Chaff discrimination;
iii) Seeker search pattern;
iv) Manoeuverability/Flight profiles
v) Re-attack
vi) SOJ
vii) Reacquire

**C. Infrastructure & Hardware**

i) Coastal defense Cruise Missile launchers;
ii) Transporter Erecter Launches + Radar TELAR;
iii) Canisters, canisters integrated to missiles, quad-octuple cell design;
iv) Fire Control System and combat system integration
v) Submarine launched

**D. Targeting Methods**

i) Target Reporting Units
ii) Satellite
iii) Link
iv) UAV;
v) Asymmetric;
vi) GPS;
vii) Forward observers; and
viii) RADAR installations.

2) Groups will then be rearranged (IAW AIT manual for Jigsaw activity) and instruct their content to their groups. (50 min)

3) Each student will then be individually assigned an anti-ship missile. They will research their given missile, determining the characteristics above for that missile. (50 min)

4) Students will then present their information to the rest of the class. They must include in their presentation a visual description (using the whiteboard) of how they expect that the missile would be employed against their TG. (50 min total)

## Resources

1. SECRET workstation (CSNI Lab)
2. Projector
3. Courseware info pack/research

## References

1) DI Threat Web
2) Jane's Publications
3) CFCD 106
4) open source information

## Notes to Instructor

Missiles assigned to students should come from the mission scenario in use, and should represent a variety of seeker types (if possible).
For instruction on how to lead a Jigsaw, see AIT manual.

EO 004.02 – Defend Against Missile Threats

## 004.02.02 – ZIPPOs

| Conditions | | Sequence |
|---|---|---|
| Classified classroom | | After 004.02.01 |
| **Time** | **Classification** | **Instructor** |
| 100 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Missile Groups<br>2) ZIPPO Criteria<br>3) ZIPPO Reactions | Self-study<br>Card-Sort |

| Activity Steps | |
|---|---|
| | 1) Students are to conduct readings of ATP-31 Chapter 7 (7706 – 7709), ATP-55 Chapter 2 and CFCD 106 MARTI 4000 (50 min)<br>2) Students will be divided into pairs. Each pair will be given a set of 21 cards, containing the 7 Zippos (1-6 and C), the ZIPPO criteria for each ZIPPO, and the ZIPPO reactions for each ZIPPO. Each pair will then be required to arrange their cards into 7 sets, with each set containing the correct ZIPPO criteria and reaction for a given ZIPPO. (10 min)<br>3) The instructor will then ask each pair, one at a time, to provide the their solution for one of the ZIPPOs, correcting them if they are wrong. (10 min)<br>4) The students will then break up their pairs, and will individually be required to write out each set of Zippo criteria and reactions on a piece of lined paper. These sheets will be their study notes for EC 002.02.02. (30 min)<br>Follow-On Activity<br>**Zippo Criteria/Reactions Cards around the room incorrectly** |

| Resources | References |
|---|---|
| 1) Activity Cards | 1) ATP-31 Chapter 7<br>2) ATP-55 Chapter 2<br>3) CFCD 106 MARTI 4000 |

| Notes to Instructor |
|---|
| |



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EO 004.02 – Defend Against Missile Threats

# 004.02.03 – Employ Hardkill

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After 004.02.02 |
| **Time** | **Classification** | **Instructor** |
| 50 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1. What constitutes hardkill, and what hardkill weapons does the RCN employ<br>2. Who chooses MSDs and HPDs, when are they loaded, and when are they not loaded.<br>3. Primary Threat vs. Alpha Threat<br>4. How and when to choose EPMs<br>5. How Auto-Engage Sectors are employed, how they are impacted by HPDs, and who is responsible for them | Best Summary<br>Self-reading |

| Activity Steps | |
|---|---|
| | 1) Instructor to split class into teams of 4 and ask each group to create their best summary of "HARDKILL". They will be provided a flip chart and must create a paragraph covering all the content the group believes to be crucial in determining hardkill. (10 min)<br>2) Teams to present their summary of hardkill to the class. (8 min total)<br>3) Instructor is to then ask the class to go and put a checkmark on their favourite/most complete summary of hardkill to determine the class's fullest definition. (2 min)<br>4) Students are then to read MARTI 4600/4752/4753. (20 min)<br>5) Students are then asked to return to their summary and update the content to be more complete based on their readings. (5 min)<br>6) Students to then present their updated summaries and explain the relevance of what was updated (5 min). |

| Resources | References |
|---|---|
| 1) flip charts & markers<br>2) access to MARTI 's (hard copy or computer based) | 1) MARTI 4600/4752/4753 |

| Notes to Instructor |
|---|
| Recommend hard copy MARTI be provided to students if possible to reduce flash up time of computers for more effective use of class time. |



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EO 004.02 – Defend Against Missile Threats

## 004.02.04 – Employ Softkill

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After 004.02.03 |
| **Time** | **Classification** | **Instructor** |
| 150 min | SECRET | RQ-PO2 NES OP<br>AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) When is Softkill effective and when it isn't<br>2) How ESM systems contributes to Softkill solutions<br>3) How chaff is employed in Softkill and how is it affected by FUFTFI<br>4) How DLF is employed in Softkill<br>5) How electronic jammers are employed in Softkill<br>6) Fly-up, Fly-Through, Fly-Into (What they are, how to verify them, and how FUFTFI is different when fouled by an ADU vs HVU)<br>7) Softkill while manoeuvering<br>8) What are ELNOT and SKTTs | Self-Study<br>Floating Flip-Chart<br>Lecture |

| Activity Steps | |
|---|---|
| | 1) Students are to read MARTI 4785 – Engage – Softkill Coordination for HCM (50 mins)<br>2) Instructor will begin by discussing with the students the importance of softkill against modern ASMs and when it is (and isn't) effective utilizing the appropriate MARTIs as a guide. (15 min)<br>3) Students will divided into 4 equal groups. Four flipcharts will be placed around the classroom with one of one learning objectives 2-5 written along the top of each one. Each group will be given a different colour marker and will write down 1 point that answers the question on each flipchart. (5 min per). After each group is finished, they will rotate to another flip-chart, read what other groups have written down, and then add a point to that flipchart. This will continue until each group has written one point on each flip chart. (20 min)<br>4) The instructor will then go over each of the flip charts, adding any additional information/points they feel are relevant, all while encouraging discussion about the topics amongst the students (25 mins)<br>5) Using the whiteboard and ship models, the instructor will then demonstrate Fly-up, Fly-Through, Fly-Into. (15 min)<br>6) The instructor will conclude the lesson by discussing LOs 7 and 8. (10 min) |

| Resources | References |
|---|---|
| 1) White-board and markers<br>2) Flip charts and markers | 1) CFCD 106, MARTI 4785<br>2) ATP-55 Chapter 2 |



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## Notes to Instructor

Instructors should endeavour to pair up each Lt(N) with a PO2/PO1 if possible.

EO 004.02 – Defend Against Missile Threats

## 004.02.05 – Detect To Engage Sequence

| Conditions | | Sequence |
|---|---|---|
| SEACOT (primary)<br>SEAWOLF (secondary) | | After 004.02.04 |
| **Time** | **Classification** | **Instructor** |
| 400 | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Reactions IAW NCPM 1000<br>2) Reactions IAW MARTI 4751<br>3) Reaction differences <25nm/>25nm (Resolve vs Pop-Up)<br>4) Voice procedures for DTE<br>5) Quality control routine<br>6) Assessing Kills<br>7) Post-Raid Reports | Self-Study<br>EDI |

| | |
|---|---|
| **Scenario** | Mission Scenario |
| **Activity Steps** | 1) Instructor to play the DTE video for the class. (3 min)<br>2) Students to conduct readings : CFCD 106 MARTI 4751, 4785 and NCPM 1000 Chapters 4 & 9 (47 min)<br>3) Using the whiteboard in the trainer, Instructor is to walk the students through the DTE sequence, explaining each step in the process from Resolve to Post-Raid report, answering students' questions as necessary. (30 min)<br>4) Instructor will then demonstrate the DTE sequence at a CMS console. The instructor should demonstrate the possible difference in the DTE sequence by conducting engagements against both a subsonic seas skimmer first alerted to via a threat emitter (pop-up within 25nm) and a supersonic high-diver launched by an aircraft (detectable all the way from launch) (20 mins)<br>5) Students will then practice the DTE sequence by conducting engagements against a variety of missile threats. (300 min) |

| Resources | References |
|---|---|
| 1) DTE video | 1) CFCD 106 MARTI 4751/4785<br>2) NCPM 1000 Chapters 4 and 9 |

| Notes to Instructor |
|---|
| Go Op support manning required IAW NWTP vol 2<br>Each student will rotate through an "instructor station" where they will have devoted support from the instructor and the support manning team equally distributing the allocated time between the students. The remainder of the students will conduct pair guided practical runs. |

# EC 004.02.01 – Defend Against Missile Threats (ZIPPO criteria)

| Scope | Conditions | Sequence |
|---|---|---|
| 004.02 | Classified classroom | After 004.02.05 |
| **Time** | **Classification** | **Instructor** |
| 50 min | SECRET | AWWD Instructor |

| | |
|---|---|
| **Assessment Steps** | Input: Student are provided a 41 mark written test where they must:<br>• Identify all Missile Groups (6)<br>• Define ZIPPO criteria for Groups 1-6 and C IAW CFCD 106 (14)<br>• Describe ZIPPO reactions for ZIPPOs 1-6 and C IAW CFCD 106 and ATP-55 (21)<br>Performance: Students will have 50 min to conduct this written test.<br><br>Output: Trainee must achieve 37/41 to pass this EC. *THIS IS A CRITICAL EC.<br><br>Feedback: Students will receive feedback within 24 hrs of writing the EC, based on corrected EC from instructor. |

| Resources | References |
|---|---|
| 1) EC<br>2) EC DS solution for instructor | 1) CFCD 106<br>2) ATP 31<br>3) ATP 55 |

| Notes to Instructor |
|---|
| No references are to be used during EC 004.02.01. |

| | |
|---|---|
| **Limitations** | This is a theory based assessment to ensure students are able to conduct non-assessable performances from EC 004.02.02. |
| **Impact of Failure** | CRITICAL EC: Further instruction and re-assess. Student may not move on to PC 004 without achieving SAT on EC 004.02.01. |

# EC 004.02.02 – Defend Against Missile Threats

| Scope | Conditions | Sequence |
|---|---|---|
| 004.02 | SEACOT (primary)<br>SEAWOLF (secondary) | After EC 004.02.01 |
| **Time** | **Classification** | **Instructor** |
| 400 min total<br>50 min/student | SECRET | AWWD Instructor |

| | |
|---|---|
| **Assessment Steps** | Input: Student is closed-up with policy assumed and support manning detailed in NWTP 2, where they must execute the duties of the AWWD where they must defend against missile threats.<br><br>Performance: Students will conduct a 40 min run where they must react to a supersonic high diver, supersonic sea skimmer, and a subsonic missile attack. Students must conduct resolve procedures IAW CFCD 106.<br><br>Output: The trainee must achieve 13/17 SATs in each presentation to pass this EC. (x3)<br><br>Feedback: Immediate verbal and written using EC 004.02.02 checklist. 10 min allocated for debrief. |

| Resources | References |
|---|---|
| 1) missile threat scenarios | 1) CFCD 106 |

| Notes to Instructor |
|---|
| Each column represents one Presentation of differing Missile Types. |

| | |
|---|---|
| **Limitations** | |
| **Impact of Failure** | Further instruction and re-assess |



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| Item | EC 004.02 - DEFEND AGAINST MISSILE THREATS | SAT | UNSAT | SAT | UNSAT | SAT | UNSAT | Remarks |
|---|---|---|---|---|---|---|---|---|
| **1. Defend Against Anti-Ship Missile Threat:** | | | | | | | | |
| | | Raid #1 | | Raid #2 | | Raid #3 | | |
| | Zippo report by EWS Acknowledged Correct or Corrected | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | Doctrine loaded by ORO, QA by SWC via Timestamp | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | MTR input by ARRO, QA by SWC | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | Auto-Engage sector activated by ORO, QA by SWC | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | ESSM Tab depressed on Weapons Engagement Panel | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | Aux screen setup showing AWW overview panel and Threat List | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | Confirm HPD HKTT/SKTT correct or SWC corrects, SWC makes ***HARDKILL CORRECT*** call | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | EWS ***SOFTKILL CORRECT*** call made or SWC corrects (includes FUFTFI check and evaluation if reqd) | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | Ensure Pop-Up called | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | selects/hooks a track from Threat List/TACPLOT, makes assessment and EPMs | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | refers to all tracks by FTN or GTN and prioritizes | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | CIWS activated at doctrinal range using voice procedures IAW NCPM 1000 | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | selects Auto-Fire/Approve on each track | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | confirms status of any employed weapon system verbally | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | When no longer ESSM engageable, SWC selects Gun tab on Weapon Engagement Panel | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |
| | makes timely kill assessment, acknowledges Kill Alarm if reqd | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| (kill assessments must be combined with Kill SID or engagement action) | | | | | | | |
| SWC delivers correct post-raid report in the format<br>Raid Complete-Incomplete/Ship's Free-Not Free to Manoeuvre/CIWS Air Ready-AAW Auto-AAW Manual/# SAMs Remaining/Reload MSD-Maintain MSD-Maintain HPD<br>-Load HPD [name] | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | |

**Score:** **Presentation 1: Supersonic high diver/supersonic sea skimmer/subsonic /17**

**Presentation 2: Supersonic high diver/supersonic sea skimmer/subsonic /17**

**Presentation 3: Supersonic high diver/supersonic sea skimmer/subsonic /17**

The trainee must achieve 13/17 SATs in each presentation to pass this EC. (x3)

| Positive Feedback | Improvement Feedback |
|---|---|
| | |

| Overall Assessment – Circle One | | | |
|---|---|---|---|
| **Met Standard** | | **Did Not Meet Standard** | |
| **Assessor**<br>I have discussed the results of this assessment with the candidate. | | **Candidate**<br>I have been debriefed on the result of this assessment | |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

PROTECTED B (when complete)

EO 004.03 – Defend Against Surface Threats

## 004.03.01 – Surface Engagement Plans and FAC/FIAC Defence

| Conditions | | Sequence |
|---|---|---|
| SEACOT (primary)<br>SEAWOLF (secondary) | | After EC 004.02.02 |
| **Time** | **Classification** | **Instructor** |
| 350 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Plan Whippet<br>2) Plan Greyhound<br>3) Plan Grouse<br>4) Plan Snipe<br>5) How to defend against FIAC/FAC | Individual Study<br>EDI |

| | |
|---|---|
| **Scenario** | Mission Scenario |
| **Activity Steps** | 1) Student are to read the following: (100 min)<br>a. MARTI 3000B, 3010, 3012A-I, 3014, 3040 and TACTNOTE 3014<br>b. NCPM 1000 Chapter 5 – ESSM SASS<br>c. CFCD-120 7-14 and 7-24<br>d. APP-01 0950-0951<br>2) Using the instructor console connected to the projector, the instructor will demonstrate various different scenarios that would require the use of each of the plans listed in LOs 1-4 and how the TG may be employed against those threats. (20 mins)<br>3) The instructor will then demonstrate defence against a FAC/FIAC attack, consummating the engagement using SSM, SASS, main armament, and CIWS. As the FAC/FIAC approach the TG, the instructor shall highlight what modes of fire can be employed against small boats (including main armament with secondary procedures), how different weapon systems will be effective against them, and when specific weapon systems become un-useable against a FAC/FIAC target. (30 min)<br>4) Each student will then close-up at a CMS console. Students will practice defending against inbound FAC/FIACs using all weapons systems. (200 min) |

| Resources | References |
|---|---|
| 1) Warfare Commander Requirement | 1) CFCD 106 (CEO)<br>2) CFCD 120<br>3) APP-01 |

## Notes to Instructor

Go Op support manning required IAW NWTP vol 2
Each student will rotate through an “instructor station” where they will have devoted support from the instructor and the support manning team equally distributing the allocated time between the students. The remainder of the students will conduct pair guided practical runs

EO 004.03 – Defend Against Surface Threats

# 004.03.02 – Surface Action Groups (SAG)

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After 004.03.01 |
| **Time** | **Classification** | **Instructor** |
| 200 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) What is a SAG and SAGC<br>2) What are the responsibilities of the SAG Commander (SAGC) and what are SAGC Orders<br>3) How to dispatch a SAG, assume command of a SAG, and issue SAGC orders | Student readings<br>Table Top<br>EDI |

| Scenario | |
|---|---|
| **Activity Steps** | 1) Students are to complete the following readings: (110 min)<br>a. MARTIs 3600 and 3700<br>b. ATP-31 articles 6311, 8500-8512<br>c. ATP-1 (vol 1) article 4157<br>d. ATP-1 (vol2 ) Flag 3 table<br>2) With model ships, the instructor will create a surface picture on the white board, and provide each of the students with worksheets for Plan Whippet and Plan Greyhound engagements. The surface picture should include a friendly TG, an enemy SAG, and a friendly LRPA, and track numbers for each. Using the surface picture, the instructor will conduct a demonstration of planning for a surface engagement using either Plan Whippet or Plan Greyhound, including the dispatching of a SAGC, the development of the plan, and the issuing of SAGC orders. The instructor will explain the tactical thinking behind the plan, demonstrate salvo planning and target assignment, and demonstrate how to fill out the plan worksheets. (35 min)<br>3) Students will then pair-up, and the instructor will create a new surface picture with the same criteria as above. Working together, each student pair will fill out their worksheets for both a Plan Whippet and Plan Greyhound engagement against the new SAG. (20 min)<br>4) Each student will then be asked to present their SAGC orders to the rest of the class. (4 mins per student – 35 mins total) |

| Resources | References |
|---|---|
| 1) Whiteboard, Markers<br>2) Fleet-in-a-Box | 1) CFCD 106<br>2) ATP-31<br>3) ATP-1 (vol 1 and 2) |

| Notes to Instructor |
|---|
| |



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EO 004.03 – Defend Against Surface Threats

## 004.03.03 – Tactical Employment of AHWCS (Theory)

| Conditions | | Sequence |
|---|---|---|
| Classified Classroom | | After 004.03.02 |
| **Time** | **Classification** | **Instructor** |
| 100 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) When to enable/disable HARPOON functions, including DTOT/STOT/Ripple, re-attack, HAFO, flight altitude, terminal Skim vs. pop-up, seeker-head search bias, search-patterns (elliptical, rectangular, bearing-only)<br>2) Tactical consideration of way-points placement vs use of the Auto-Plan<br>3) Selecting engagement plans and salvo sizes | Talk through/Walk through |

| Activity Steps | |
|---|---|
| | 1) The instructor will explain and discuss LOs 1-3 using provided definition sheet and White Board symbols. (100mins) |

| Resources | References |
|---|---|
| 1) Definitions of LOs 1<br>2) Fleet-in-a-box with symbols | 1) HAWSTEM<br>2) MARTI 3012 |

| Notes to Instructor |
|---|
| Support Manning IAW NWTP Vol 2, the surface picture should be robust and complex to force students to consider all aspects of surface engagement planning, including salvo size, dive angle, back-ground shipping, etc. |



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EO 004.03 – Defend Against Surface Threats

## 004.03.04 – Tactical Employment of AHWCS (Practical)

| Conditions | | Sequence |
|---|---|---|
| SEACOT (primary)<br>SEAWOLF (secondary) | | After 004.03.03 |
| **Time** | **Classification** | **Instructor** |
| 250 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Covering, Taking, and ROE<br>2) When to enable/disable HARPOON functions, including DTOT/STOT/Ripple, re-attack, HAFO, flight altitude, terminal Skim vs. pop-up, seeker-head search bias, search-patterns (elliptical, rectangular, bearing-only)<br>3) Tactical consideration of way-points placement vs use of the Auto-Plan (Primary)<br>4) Displaying missile geometry in CMS and verifying flight-corridor and background shipping<br>5) When to enable missile power and select command-launch<br>6) How to report bulldogs away both internally and externally | EDI<br>Fishbowl |

| | |
|---|---|
| **Scenario** | Mission Scenario |
| **Activity Steps** | 1) With the instructor console connected to the projector, the instructor will plan and execute several different engagements against enemy surface units, demonstrating and explaining the aspects of LOs 1-6. (50 mins)<br>2) Each students will then close-up at a CMS console. Using a robust and complex surface picture, the instructor will assign each student with a target surface contact and direct the students to plan an engagement against that target. (10mins)<br>3) When the students are finished planning their engagements, the instructor will lead a review each of the plans by connecting the projector, one at a time, to the each of students' consoles. The instructor will discuss with each student their reasoning behind their various tactical decisions (i.e. which settings they enabled/disabled, where they placed waypoints, etc.), and the remainder of the class will be free to comment and ask questions. Once satisfied with the discussion, the instructor will direct the student to make their BF report, execute the engagement, and report bulldogs away to the class. (90 mins)<br>4) Students will then pair-up. One student will choose a surface contact from the surface picture, and the other student will act as SWC and plan an engagement against the chosen target. When planning is complete, the student acting as SWC will deliver their BF report to the other student, execute their engagement, and report bulldogs away to their partner. Students will then reverse roles and repeat the process while the instructor moves around the room assisting and discussing engagement plans with students. (100 min)<br>5) Students to add to "learning wall" |

| Resources | References |
| --- | --- |
| 1) LEARNING WALL | |

## Notes to Instructor

The surface picture should be robust and complex to force students to consider all aspects of surface engagement planning, including salvo size, dive angle, back-ground shipping, etc. The Learning Wall is a whiteboard or List where students can write down personal lessons learned to benefit the group.

# EC 004.03 – Defend Against Surface Threats

| Scope | Conditions | Sequence |
|---|---|---|
| 004.03 | SEACOT (primary)<br>SEAWOLF (secondary) | After 004.03.04 |
| **Time** | **Classification** | **Instructor** |
| 600 min | SECRET | AWWD Instructor |

| Assessment Steps | |
|---|---|
| | Input: Student is standing-to policy, with support manning detailed IAW NWTP 2, where they must execute the duties of the AWWD/SAGC and defend against surface threats IAW MARTI 4751.<br><br>The instructor will select 10 surface combatants that will surround the TG at ranges between 10-70nm. A TRU will be provided. Support Staff, acting as ASUWC, will provide a Plan Whippet or Plan Greyhound engagement order (with DTOT) for multiple units that the student must participate in. The instructor will then coordinate a FAC/FIAC attack on the ship, and order a take-immediate. Instructor will then designate the student as SAGC. As SAGC, the student will provide a Plan Greyhound/ Plan Whippet engagement order for multiple units (STOT) (i.e. plan the engagement).<br><br>Performance: Students will conduct a 60 min run where they must complete several surface engagements as a member of a SAG, and then act as SAGC. Students will not be required to conduct their planned engagement while acting as SAGC; instead, they will simply be required to prepare and transmit their plan to their SAG.<br><br>Output: Trainee must achieve 25/31 SATs on non-critical factors **AND** all critical factors to pass this EC.<br><br>Feedback: Immediate verbal and written using EC 004.03 checklist. 15 min allocated to reset the scenario and conduct debrief. |

| Resources | References |
|---|---|
| 1) Surface Scenario (10 units)<br>2) Support Manning IAW NWTP Vol 2<br>3) Warfare Commander | 1) MARTI 4751 |

| Notes to Instructor |
|---|
| A successful engagement for FAC/FIAC can be accomplished using Grouse/Snipe/Greyhound. Whichever Plan (Greyhound or Whippet) is used for the first engagement, the other plan must be used for the SAGC order portion of this EC. |

| | |
|---|---|
| **Limitations** | Students are provided a complete surface picture, no air contacts |
| **Impact of Failure** | Further instruction and re-assess |

| Item | EC 004.03 - DEFEND AGAINST SURFACE THREATS | SAT | UNSAT | Remarks |
|---|---|---|---|---|
| **1. Whippet engagement** | | | | |
| SWC refers to all tracks by FTN or GTN and prioritizes | | ☐ | ☐ | |
| Surface tracks have RCL/IDCRIT IAW mission OPTASK SCC/ASuW | | ☐ | ☐ | |
| Salvo assignment correct to contact type | | ☐ | ☐ | |
| Track projection on/off correctly IAW doctrine | | ☐ | ☐ | |
| Pre-search and Terminal flight setting IAW doctrine | | ☐ | ☐ | |
| Autoplan active unless rationale provided | | ☐ | ☐ | |
| SSM launcher assignment accounts for force MLA | | ☐ | ☐ | |
| Waypoints adjusted to avoid hazards | | ☐ | ☐ | |
| DTOT/STOT/Ripple set IAW order and achieved | | ☐ | ☐ | |
| Addresses any Harpoon Alarms/Alerts | | ☐ | ☐ | |
| Sets HAFO IAW tactical situation and target | | ☐ | ☐ | |
| Re-Attack mode IAW doctrine | | ☐ | ☐ | |
| ***Missile geometry displayed in CMS** | | ☐ | ☐ | **Critical** |
| Flight corridor clear or discussed with ORO | | ☐ | ☐ | |
| Harpoon Report IAW: BF/ID/RCL/FTN/SS/PACQ/Engagement Type/TOF/HAFO/Pre-search and Terminal mode settings/Re-Attack setting/Flight corridor/ background shipping | | ☐ | ☐ | |
| Receive “Take” authorization, final check, launch procedure correct OR correct deficiencies | | ☐ | ☐ | |
| Bulldog Report: Bulldog(s) Away/ID/RCL/FTN/SS/TOT | | ☐ | ☐ | |
| Make SSM Safe | | ☐ | ☐ | |
| Setup Re-Attack awaiting outcome of engagement | | ☐ | ☐ | |
| **2. FAC/FIAC (plan Grouse/Snipe/Greyhound)** | | | | |
| Makes initial RCL/IDCRIT designation based on the OPTASK criteria for the threaded scenario | | ☐ | ☐ | |
| Refers to all tracks by FTN/GTN | | ☐ | ☐ | |

| | | | |
|---|---|---|---|
| Ensures all ID conflicts are resolved before approving engagement orders | ☐ | ☐ | |
| Makes threat assessment of all contacts and prioritizes | ☐ | ☐ | |
| QA the Surface picture to ensure all contacts are linked out | ☐ | ☐ | |
| Recommends defensive measures with ROE or in response to self-defence | ☐ | ☐ | |
| Actions "Take" orders IAW Command Direction | ☐ | ☐ | |
| Approves SDEP TEWA plans or uses alternate means to engage contacts | ☐ | ☐ | |
| Recommends alternative engagement if ordered plan not achievable. | ☐ | ☐ | |
| Assess Kill and actions Kill SID or advises ORO when no longer engageable by AWW weapons | ☐ | ☐ | |
| **3. Plan Greyhound engagement as SAGC** | | | |
| Surface plan order format IAW CFCD 106 | ☐ | ☐ | |
| Surface plan order is tactically sound IAW platforms given | ☐ | ☐ | |
| Plan is communicated to participating units | ☐ | ☐ | |

**Score: / 31**

Trainee must achieve 25/31 SATs on non-critical factors **AND** all critical factors to pass this EC.

| Positive Feedback | Improvement Feedback |
|---|---|
| | |

| Overall Assessment – Circle One | | | |
|---|---|---|---|
| **Met Standard** | | **Did Not Meet Standard** | |
| **Assessor**<br>I have discussed the results of this assessment with the candidate. | | **Candidate**<br>I have been debriefed on the result of this assessment | |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

EO 004.04 – Defend Against Air Threats

## 004.04 – Fighter Bomber Attack/Low-Slow Flyer Doctrine

| Conditions | | Sequence |
|---|---|---|
| SEACOT (primary)<br>SEAWOLF (secondary) | | After EC 004.03 |
| **Time** | **Classification** | **Instructor** |
| 400 min | SECRET | AWWD Instructor |

| Learning Objectives | Method |
|---|---|
| 1) Fighter-Bomber attack profiles and weapons employed by Fighter-Bombers<br>2) How to assess jamming from an a/c<br>3) Understand the impact of jamming on ROE<br>4) Probes vs Raids and the important of height finding<br>5) NATO vs RCN FBA doctrine<br>6) How and when to call Air-Threat Warning Red FBA<br>7) FBA Manoeuvring,<br>8) Using the FBA HPD<br>9) How to employ CMS against an FBA<br>10) How to defend against a LSF:<br>a. What platforms could a LSF?<br>b. What are the thresholds for a LSF? (speed, altitude)<br>c. What is the threat posed by a LSF?<br>d. What are the doctrine and tactics for engaging a LSF? | Individual Study<br>EDI<br>Fishbowl |

| Scenario | |
|---|---|
| **Activity Steps** | 1) Students will read (50 min)<br>a. MARTIs 4600<br>b. MARTI 4460<br>c. ATP 31 - article 9215<br>2) With the instructor console connected to the projector screen, the instructor will demonstrate several different FBA attack profiles, including fighter-bombers flying directly toward ownship and conducting rocket/cannon/bomb attacks, as well as a/c conducting attacks without attacking ownship directly (i.e. laser-guided bombs) (15 min)<br>3) The instructor will then demonstrate what jamming looks like in CMS and what it means in terms of ROE and self-defence. The instructor shall also discuss the concept of “probes” and “attack profiles”, and the importance of height finding in assessing hostile intent. (15 mins)<br>4) Instructor will then demonstrate an engagement against an FBA, beginning with the initial resolve, and completing with an SAM and main armament engagement against the threat. The instructor shall describe the FBA HPD and what settings change when it is enabled. (20 mins) |

| | |
|---|---|
| | 5) Students will then pair-up and take turns practicing defending against Fighter-Bombers, with one student acting as SWC, and the other student performing the voice procedures of the EWS, ARRO, ORS and ORO during the resolve and warning process. (250 min)<br>6) During the final portion of the tutorial, the instructor will introduce the concept of a Low-Slow Flyer. Once again using the instructor console connected to the projector, they will have a LSF approach the task group from long range. The instructor will conducted the DTE sequence against the LSF, including issuing of warnings, and during the time, the instructor should lead the students through a discussion of the points of LO 10. (50 mins) |

| Resources | References |
|---|---|
| 1) NCPM 1000<br>2) CFCD 106 MARTI 4600/4460<br>3) ATP-31 9215 | 1) MARTI 4600<br>2) MARTI 4460 (CEO)<br>3) ATP-31 9215 |

| Notes to Instructor |
|---|
| |

EO 004.04 – Defend Against Air Threats

# EC 004.04 – Defend Against Air Threats

| Scope | Conditions | Sequence |
|---|---|---|
| EO 004.04 | SEACOT (primary)<br>SEAWOLF (secondary) | After 004.04 |
| **Time** | **Classification** | **Instructor** |
| 400 min | SECRET | AWWD Instructor |

| | |
|---|---|
| **Assessment Steps** | Input: Student is closed-up with policy air assumed and support manning detailed in NWTP 2, where they must execute the duties of the AWWD to defend against air threats.<br><br>Performance: Students will conduct a 40 min run where they must respond to three air attacks. Each run must include:<br>• 2 engagements against FBA profiles<br>• 1 engagements against LSF<br>• 1 response to warning<br>• 1 probe<br>• Situation where ROE prevent SWC from conducting standard responses<br>• 1 commercial air without IFF<br>Output: Trainee must achieve 22/27 SATs on non-critical factors **AND** all critical factors to pass this EC.<br><br>Feedback: Immediate verbal and written using EC 004.04 checklist. 10 min allocated to reset the scenario and conduct debrief. |

| Resources | References |
|---|---|
| 1) Support Manning | 1) CFCD 106 4000 Series<br>2) NCPM 1000 |

| Notes to Instructor |
|---|
| One FBA profile is single A/C (will come in low and pop-up within 25nm), one FBA profile is multiple (2) A/C (will come in at altitude and dive towards ownship)<br>LSF should respond to warnings and claim to be media A/C when in reality they are threat<br>Response to warning shall be such that it creates discussion WRT ID crit and follow on actions.<br>ROE situation must align with the mission scenario |

| | |
|---|---|
| **Limitations** | Unable to simulate bomb drop |
| **Impact of Failure** | Further instruction and re-assess |

| Item | EC 004.04 – DEFEND AGAINST AIR THREATS | SAT | UNSAT | Remarks |
|---|---|---|---|---|
| **1. Defend Against a Fighter Bomber Attack** | | | | |
| | Ensures resolve procedures conducted on all unknown contacts (and those previously ID with changed profile) | ☐ | ☐ | |
| | Resolve completed with appropriate IDCRIT assigned | ☐ | ☐ | |
| | Warnings initiated IAW delegated authority | ☐ | ☐ | |
| | Calls ATW RED FBA/LSF | ☐ | ☐ | |
| | HPD loaded by ORO, QA by SWC via Timestamp | ☐ | ☐ | |
| | Selects/hooks a track from Threat List/TACPLOT, makes assessment and EPMs | ☐ | ☐ | |
| | Auto-Engage sector activated by ORO, QA by SWC | ☐ | ☐ | |
| | ESSM Tab depressed on Weapons Engagement Panel | ☐ | ☐ | |
| | Aux screen setup showing AWW overview panel and Threat List | ☐ | ☐ | |
| | Confirm HKTT/SKTT correct or SWC corrects | ☐ | ☐ | |
| | Pop-Up called (By any team member, QA by SWC) | ☐ | ☐ | |
| | SWC receives TAKE orders and selects Auto-Fire/Approve on all ordered track(s) | ☐ | ☐ | ****Critical** |
| | SWC confirms status of any employed weapon system verbally | ☐ | ☐ | |
| | CIWS activated at doctrinal range using voice procedures IAW NCPM 1000 | ☐ | ☐ | |
| | When no longer ESSM engageable, SWC depressed Gun tab on Weapon Engagement Panel | ☐ | ☐ | |
| | SWC makes timely kill assessment, acknowledges Kill Alarm if reqd (kill assessments must be combined with Kill SID or engagement action) | ☐ | ☐ | |

| | | | |
|---|---|---|---|
| SWC delivers post-raid report in the following format:<br>Raid Complete-Incomplete/Ship's Free-Not Free to Manoeuvre/CIWS Air Ready-AAW Auto-AAW Manual/# SAMs Remaining/Reload MSD-Maintain MSD-Maintain HPD-Load HPD | ☐ | ☐ | |
| **2. Response to warning** | | | |
| Warning response received and briefed to ORO/AAWC | ☐ | ☐ | |
| Makes justifiable recommendation based on response to warning | ☐ | ☐ | |
| Conducts follow through actions for recommendations (when accepted by ORO) | ☐ | ☐ | |
| **3. Probe** | | | |
| Assesses probe, makes justifiable recommendation based on response to warning/negative response | ☐ | ☐ | |
| Conducts follow through actions for recommendations (when accepted by ORO) | ☐ | ☐ | |
| **4. ROE considerations situation** | | | |
| Brief ORO on ROE considerations regarding employment of weapons and sensors | ☐ | ☐ | |
| Brief ORO/AAWC for response to warning of commercial air without IFF | ☐ | ☐ | |
| **5. Maintain RAP** | | | |
| Ensures resolve procedures conducted on all unknown contacts (and those previously ID with changed profile) | ☐ | ☐ | |
| Resolve completed with appropriate IDCRIT assigned | ☐ | ☐ | |
| Warnings initiated IAW delegated authority | ☐ | ☐ | |
| SWC refers to all tracks by FTN or GTN and prioritizes | ☐ | ☐ | |

**Score: / 27 CRITICALS /1**
The trainee must achieve 22/27 SATs and all CRITICALS to pass this EC.

| | | | |
|---|---|---|---|
| | | | |
| **Positive Feedback** | | **Improvement Feedback** | |
| | | | |
| **Overall Assessment – Circle One** | | | |
| **Met Standard** | | **Did Not Meet Standard** | |
| **Assessor**<br>I have discussed the results of this assessment with the candidate. | | **Candidate**<br>I have been debriefed on the result of this assessment | |
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

PROTECTED B (when complete)

# PO 004 – Consolidation

| Conditions | | Sequence |
|---|---|---|
| SEAWOLF (primary)<br>SEACOT x 2 (secondary) | | After EC 004.04 |
| **Time** | **Classification** | **Instructor** |
| 8000 min | Up to SECRET | RQ MS Instructor |

| Learning Objectives | Method |
|---|---|
| 1. Build and Maintain the RMP<br>2. Manage the AWW Team<br>3. Defend Against Missile Threats<br>4. Defend Against Surface Threats<br>5. Defend Against Air Threats | Practical |

| Activity Steps | |
|---|---|
| | Input: Student is closed-up with policy assumed and support manning detailed in NWTP 2, where they must execute the duties of the AWWD during mission runs.<br><br>Performance: IAW PC 004<br><br>Students should have the opportunity to complete a minimum of 5 developmental runs (with instructor assessing the student with the PC checklist). Any additional time can be used for either group mentorship or additional developmental run.<br><br>During every developmental run, the other students shall be closed-up in the control room at an instructor console, and using SAFSIM, they shall take control of one of the other TG units. They shall be present on external comms, and shall navigate, communicate, and fight their assigned units.<br><br>Output: IAW PC 004<br><br>Feedback: Immediate verbal and written using PC 004 checklist. |

| Resources | References |
|---|---|
| 1) | 1) NCPM 1000 |

## Notes to Instructor

If instructing in SEAWOLF students will need to pair up and take turns as there are limited systems to work on. May need to allocate double the scheduled time due to resource management.

# PC 004 - Execute Duties of an AWW Director

| Scope | Conditions | Sequence |
|---|---|---|
| PO 004 | SEAWOLF (primary)<br>SEACOT x 2 (secondary) | After PO 004 |
| **Time** | **Classification** | **Instructor** |
| 4800 min | SECRET | AWWD Instructor |

| | |
|---|---|
| **Scenario** | IAW selected mission (game) |
| **Assessment Steps** | Input: Student is closed-up with policy assumed and support manning detailed in NWTP 2, where they must execute the duties of the AWWD during mission runs.<br><br>Performance: students will conduct a 3 x 120 min runs, where each run will include:<br>• A Multi Threat Environment<br>• 1 x supersonic missile<br>• 1 x subsonic missile raid<br>• 1 x Fighter Bomber Attack<br>• 1 x Surface to Surface Missile Engagement (OTHT or Snipe)<br>• Use of a Target Reporting Unit<br>• NFS-H or LSF*<br><br>Output: Each section of the checklist must be scored individually. Students must receive a SAT in each section to receive an overall SAT for this PC. Students must achieve SAT on 2/3 assessed runs to achieve this PC.<br><br>Feedback: Immediate verbal and written using PC 004 checklist. 30 min allocated to reset the scenario and conduct debrief. Debriefs are to be conducted with the assistance of the ORO to ensure thoroughness and impartiality.<br><br>*NFS-H and LSF scenarios will not be assessed, but instead used to develop and progress RMP development, warnings, student stress, and complexity in the overall scenario. |

| Resources | References |
|---|---|
| 1) Student Produced TACPAC<br>2) Locally produced CMS setup sheet | 1) CFCD 106<br>2) NCPM 1000<br>3) ATP 1<br>4) ATP 31<br>5) ATP 55<br>6) APP 7 |

| Notes to Instructor |
|---|
| Each run shall include either a strike mission (CTS engagement) or LSF, but not both. Each student should be exposed to a minimum of one of each over the course of their 3 assessed runs. |

| Limitations | |
|---|---|
| **Impact of Failure** | The full PC is 3 x 90 runs. If a student is unsuccessful in attaining SAT on 2/3 runs, any authorized retest requires an additional set of 3 runs, where they must attain SAT on 2/3.<br>IAW NPTORD 4500-54 |

| Item | PC 004 - Execute Duties of an AWW Director | SAT | UNSAT | REMARKS |
|---|---|---|---|---|
| **1. Direct the AWW Team** | | | | |
| CMS configured IAW locally produced set-up sheet or battle orders | | ☐ | ☐ | |
| Confirm, TDAs and Special Points are input IAW OPORD/Scenario/DIMS/Intel | | ☐ | ☐ | |
| MSD selected by the ORO IAW doctrine and verified by the SWC | | ☐ | ☐ | |
| Communication nets set-up IAW COMPLAN | | ☐ | ☐ | |
| Briefed AWW team on Alpha threat, mission priority, primary threat, and team expectations | | ☐ | ☐ | |
| Briefed all delegations, ensuring duties and responsibilities are understood | | ☐ | ☐ | |
| Provided coherent direction to the team and set clear priorities | | ☐ | ☐ | |
| Execute Warfare Commander direction IAW Threat | | ☐ | ☐ | |
| Used voice procedures IAW doctrine | | ☐ | ☐ | |
| Used SAR format when briefing ORO | | ☐ | ☐ | |
| Briefed ORO on threats before they become imminent threats | | ☐ | ☐ | |
| Employed threat data IAW promulgated OOB | | ☐ | ☐ | |
| **2. Build and Maintain the Recognized Maritime Picture** | | | | |
| Ensured all contacts are updated with all ID and multi-link fields | | ☐ | ☐ | |
| Conducted assessment of all contacts using RCL/IDCRIT IAW OPTASK messages | | ☐ | ☐ | |
| Provided direction and guidance to AAW team with appropriate Brevity Words | | ☐ | ☐ | |
| Resolved and assigned RCL/IDCRIT for all air contacts IAW OPTASK Messages | | ☐ | ☐ | |
| All Suspect/PROB and higher warships and all hostile a/c are reported to Warfare Commander IAW OPTASK | | ☐ | ☐ | |
| Ensured all ESM rackets are associated IAW Doctrine | | ☐ | ☐ | |

| | | | |
|---|---|---|---|
| Sanitized all AOPs | ☐ | ☐ | |
| RCL/IDCRIT are applied to all air contacts with IFF in air lanes | ☐ | ☐ | |
| **3. Tactically Employ Assigned Air Assets** | | | |
| Informed SAC of priorities for employment of ASUW air assets | ☐ | ☐ | |
| Air Asset advised of any known SAGs that may present a threat and confirm COI/CCOI has been relayed/intel held by AC | ☐ | ☐ | |
| Briefed Command on status of air asset at join, assumption of mission, alteration of mission | ☐ | ☐ | |
| Gave positive direction to SAC on SAM Standoff (STOF) IAW doctrine | ☐ | ☐ | |
| **4. Employ Warning Organization** | | | |
| Warnings initiated IAW delegated authority | ☐ | ☐ | |
| Warning response received and briefed to ORO and Warfare Commanders | ☐ | ☐ | |
| Makes justifiable recommendation based on response to warning | ☐ | ☐ | |
| Conducts follow through actions for recommendations (when accepted by ORO) | ☐ | ☐ | |
| **5. Defend Against SUPER-sonic Anti-Ship Missile Threat** | | | |
| Ensure Pop-Up called and external | ☐ | ☐ | |
| Zippo or Threat assessment/confirmation for perceived threat | ☐ | ☐ | |
| MTR input by ARRO, QA by SWC | ☐ | ☐ | |
| ESSM Tab depressed on Weapons Engagement Recommendation Panel | ☐ | ☐ | |
| Aux screen setup showing AWW overview/ Entity Recognition and Threat List | ☐ | ☐ | |
| ***HARDKILL CORRECT.*** Applied EPM with highest match; verified weapon arcs and availability; confirm plans generating. | ☐ | ☐ | |
| ***SOFTKILL CORRECT.*** Jockey course appropriate with available DLF. | ☐ | ☐ | |
| ASMD Course is viable (arcs open) | ☐ | ☐ | |
| Uses Threat Lanes to cycle between threats. Uses TACPLOT to verify “own package” if multiple raids/units targeted | ☐ | ☐ | |

| | | | |
|---|---|---|---|
| Refers to all tracks by FTN or GTN and prioritizes | ☐ | ☐ | |
| PHALANX activated at doctrinal range | ☐ | ☐ | |
| Selects Auto-Fire/Approve on each track if reqd | ☐ | ☐ | |
| Confirms weapons engagement/ if fault occurred reacts to alternate procedures | ☐ | ☐ | |
| When no longer ESSM engageable, SWC selects Gun tab on Weapon Engagement Panel | ☐ | ☐ | |
| Makes immediate engagement assessment, acknowledges Kill Alarm if reqd (MSDF kill assessments must be combined with Kill SID or engagement action) | ☐ | ☐ | |
| SWC delivers post-raid report IAW current MARTI format | ☐ | ☐ | |
| **6. Defend Against SUB-sonic Anti-Ship Missile Threat** | | | |
| Ensures Pop-Up called and external | ☐ | ☐ | |
| Zippo or Threat assessment/confirmation for perceived threat | ☐ | ☐ | |
| MTR input by ARRO, QA by SWC | ☐ | ☐ | |
| ESSM Tab depressed on Weapons Engagement Panel | ☐ | ☐ | |
| Aux screen setup showing AWW overview panel and Threat List | ☐ | ☐ | |
| ***HARDKILL CORRECT.*** Applied EPM with highest match; verified weapon arcs and availability; confirm plans generating. | ☐ | ☐ | |
| ***SOFTKILL CORRECT.*** Jockey course appropriate with available DLF. | ☐ | ☐ | |
| ASMD course is viable (arcs open) | ☐ | ☐ | |
| Uses Threat Lanes to cycle between threats. Uses TACPLOT to verify “own package” if multiple raids/units targeted | ☐ | ☐ | |
| Refers to all tracks by FTN or GTN and prioritizes | ☐ | ☐ | |
| PHALANX activated at doctrinal range | ☐ | ☐ | |
| Selects Auto-Fire/Approve on each track if reqd | ☐ | ☐ | |

| | | | |
|---|---|---|---|
| Confirms weapons engagement/ if fault occurred reacts to alternate procedures | ☐ | ☐ | |
| When no longer ESSM engageable, SWC selects Gun tab on Weapon Engagement Panel | ☐ | ☐ | |
| Makes immediate engagement assessment, acknowledges Kill Alarm if reqd (MSDF kill assessments must be combined with Kill SID or engagement action) | ☐ | ☐ | |
| SWC delivers post-raid report IAW current MARTI format | ☐ | ☐ | |
| **7. Conduct Surface-to-Surface Missile Engagement** | | | |
| Surface tracks have RCL/IDCRIT IAW mission OPTASK SCC/ASuW | ☐ | ☐ | |
| Salvo size correct | ☐ | ☐ | |
| Track projection on/off correctly IAW doctrine | ☐ | ☐ | |
| Pre-search and Terminal flight setting IAW doctrine | ☐ | ☐ | |
| Autoplan active unless rationale provided | ☐ | ☐ | |
| SSM launcher assignment accounts for force MLA and/or ship's current heading | ☐ | ☐ | |
| Waypoints adjusted to avoid hazards | ☐ | ☐ | |
| DTOT/STOT/Ripple set IAW order and achieved | ☐ | ☐ | |
| Addresses any Harpoon Alarms/Alerts | ☐ | ☐ | |
| Sets HAFO IAW tactical situation and target | ☐ | ☐ | |
| Re-Attack mode IAW doctrine | ☐ | ☐ | |
| Missile geometry displayed in CMS | ☐ | ☐ | |
| Flight corridor clear or discussed with ORO | ☐ | ☐ | |
| Harpoon Report IAW: BF/ID/RCL/FTN/SS/PACQ/Engagement Type/TOF/HAFO/Pre-search and Terminal mode settings/Re-Attack setting/Flight corridor/ background shipping (to ORO and External) | ☐ | ☐ | |
| Receive "Take" authorization, final check, launch procedure correct OR correct deficiencies | ☐ | ☐ | |
| Bulldog Report IAW: Bulldog(s) Away/ID/RCL/FTN/SS/TOT | ☐ | ☐ | |
| Make Kill Assessment after TOT expires and DAMASS received | ☐ | ☐ | |

| 8. Defend Against Fighter Bomber Attack | | | |
|---|---|---|---|
| Pop-Up called by ARRO (if appropriate) or by SWC if first sighted | ☐ | ☐ | |
| Assesses FBA attack and calls “Scorpion” or “Viper” as applicable for the assessed threat | ☐ | ☐ | |
| Selects/hooks a track from Threat List/TACPLOT, makes assessment and EPMs | ☐ | ☐ | |
| ESSM Tab depressed on Weapons Engagement Recommendation Panel | ☐ | ☐ | |
| Aux screen setup showing AWW overview panel and Threat List | ☐ | ☐ | |
| Ensures weapon arcs remain open and if required Gun Unsafe | ☐ | ☐ | |
| SWC refers to all tracks by FTN or GTN and prioritizes | ☐ | ☐ | |
| SWC selects Auto-Fire/Approve on each track after receiving TAKE order if reqd | ☐ | ☐ | |
| PHALANX activated at doctrinal range | ☐ | ☐ | |
| When no longer ESSM engageable, SWC selects Gun tab on Weapon Engagement Panel | ☐ | ☐ | |
| Confirms weapons engagement/ if fault occurred reacts to alternate procedures | ☐ | ☐ | |
| SWC makes immediate engagement assessment, acknowledges Kill Alarm if reqd (MSDF kill assessments must be combined with Kill SID or engagement action) | ☐ | ☐ | |
| SWC delivers post-raid report IAW current MARTI format | ☐ | ☐ | |
| CRITICALS (must not conduct any of the following) | | | |
| Conducts a HARPOON engagement and Blind impacts on any unintended contact | ☐ | ☐ | |
| Conducts a situation to PIP over a contact (Merchant, Civilian Airliner, Consort, Air Asset) during an ESSM engagement | ☐ | ☐ | |
| Did not Adhere to the CO/ORO direction during a Weapon Engagement | ☐ | ☐ | |
| | | | |

**Scoring:**
Each section of the checklist must be scored individually. Students must receive a SAT in each section to receive an overall SAT for this PC.

| Section | Minimum Score | Actual Score | |
|---|---|---|---|
| 1 | 10/12 | **/12** | **Sat ☐ / Unsat ☐** |
| 2 | 7/8 | **/8** | **Sat ☐ / Unsat ☐** |
| 3 | 3/4 | **/4** | **Sat ☐ / Unsat ☐** |
| 4 | 3/4 | **/4** | **Sat ☐ / Unsat ☐** |
| 5 | 13/16 | **/16** | **Sat ☐ / Unsat ☐** |
| 6 | 13/16 | **/16** | **Sat ☐ / Unsat ☐** |
| 7 | 14/17 | **/17** | **Sat ☐ / Unsat ☐** |
| 8 | 11/13 | **/13** | **Sat ☐ / Unsat ☐** |
| CRITICALS | 3/3 | **/3** | **Sat ☐ / Unsat ☐** |

Overall: **Sat ☐ / Unsat ☐**

| Positive Feedback | Improvement Feedback |
|---|---|
| | |

**Overall Assessment – Circle One**

| Met Standard | Did Not Meet Standard |
|---|---|
| **Assessor**<br>I have discussed the results of this assessment with the candidate. | **Candidate**<br>I have been debriefed on the result of this assessment |

| | | | |
|---|---|---|---|
| **Assessor** | Name: | Date: | Signature: |
| **Candidate** | Name: | Date: | Signature: |

# CHAPTER 3: ASSESSMENT OF TRAINEE

## GENERAL

1. All Performance Checks (PCs) are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of PRACTICAL or WRITTEN assessment and the trainee must pass all POs to successfully complete this course.

## ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. Each EO must have at least one EC. EC results do not factor into a trainee's successful completion of the PO.

3. Performance Checks (PCs). PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

## RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be completed IAW the standards delineated in this QSP and IAW CFITES Volume 7 (Evaluation).

## PROGRESS MONITORING

5. Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Volume 6 Conduct of Instruction Programmes.

6. Continuous monitoring of trainee progress is required to provide the following:

   a. Early warning of trainee difficulties;
   b. Feedback on the effectiveness of training;
   c. Information for the Training Review Board (TRB) (if required); and
   d. Information to assist in the completion of the CF 377 Course Report.

7. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment (TE) SOs or SOPs.

8. PC failures and supplemental PC failures are to be handled IAW TE SOs and SOPs. Supplemental ECs and PCs shall not be given no less than 24 hours of the trainee being informed of their EC or PC failure.

## TRAINING REVIEW BOARD

9. The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54 Training Review Board Process.

## COURSE REPORTS

10. Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51.

11. Trainees will be graded pass or fail. NPTORD 4500-51 Course Reporting and Certificates, provides direction and guidelines for Letter Grading and Ranking.

12. Narratives will indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager. Training limitations and/or incomplete training must also be included in the course report. A record of trainee counselling is to be maintained by the TE.

# Task List

| NWO Master Task List, Annex A: A-PD-055-002/PP-001 | | TRAIN/NO TRAIN |
|---|---|---|
| **DUTY AREA A - NAVIGATION AND SEAMANSHIP** | | |
| AT0090 | Maintain ship's logs and records | NT |
| **DUTY AREA B - NAVAL OPERATIONS** | | |
| BT0030 | Plan an anti-surface operation | T |
| BT0040 | Plan anti-air warfare operation | T |
| BT0060 | Coordinate anti-air warfare operation | NT |
| BT0070 | Coordinate anti-surface warfare operation | NT |
| BT0120 | Draft firing orders | T |
| BT0130 | Prepare operational messages | T |
| BT0140 | Prepare evaluation operational reports | NT |
| BT0150 | Supervise anti-air weapons firing analysis | NT |
| BT0160 | Supervise anti-surface weapons firing analysis | NT |
| BT0180 | Coordinate the control of weapons | T |
| BT0190 | Coordinate the employment of weapons | T |
| BT0200 | Coordinate the control of sensors | T |
| BT0210 | Coordinate the employment of sensors | T |
| BT0220 | Coordinate the control of aircraft | T |
| BT0230 | Coordinate the employment of aircraft | T |
| BT0250 | Contribute to the Recognized Maritime Picture (RMP) | T |
| BT0260 | Supervise the compilation of the anti-air warfare picture | T |
| BT0270 | Supervise the compilation of the anti-surface warfare picture | T |
| BT0290 | Implement anti-air warfare tactics | T |
| BT0300 | Implement anti-surface warfare tactics | T |
| BT0320 | Advise command on anti-air warfare safety | T |
| BT0330 | Advise command on anti-surface warfare safety | T |
| BT0380 | Coordinate the maintenance of records for operations analysis | T |
| BT0390 | Determine the effectiveness of anti-air weapons and sensors | T |
| BT0400 | Determine the effectiveness of anti-surface weapons and sensors | T |
| BT0490 | Organize intelligence | T |
| BT0520 | Direct electronic warfare | T |
| BT0540 | Identify systems performance deficiencies, malfunctions or failures | T |

| | | |
|---|---|---|
| **NESOP Master Task List, Annex A: A-PD-055-002/PP-002** | | **TRAIN/NO TRAIN** |
| **Duty Area A - Communications** | | |
| A00010 | Authenticate communication transmissions | T |
| A00020 | Perform communication checks | T |
| A00030 | Configure communication terminal for operation | T |
| A00060 | Communicate using tactical voice procedures | T |
| **Duty Area B - Operations** | | |
| B00010 | Classify radar contacts | T |
| B00060 | Conduct Intelligence Gathering | NT |
| B00080 | Operate combat systems (CS) equipment | T |
| B00090 | Evaluate intelligence information | T |
| B00100 | Detect radar contacts | NT |
| B00130 | Adhere to Emission Control (EMCON) restrictions | T |
| B00170 | Perform pre-sail checks on equipment | NT |
| B00190 | Prepare EMCON Table | NT |
| B00200 | Prepare operational training exercises | T |
| **Duty Area C - Electronic Warfare (EW)** | | |
| C00040 | Apply promulgated tactical info as it applies to internal and external organizations | T |
| C00050 | Assess jamming | T |
| C00110 | Coordinate ship (ES), (EA) and (EP) operations | T |
| C00450 | Perform Area of Interest (AOI) monitoring | T |
| C00460 | Recommend tactical employment of sensors | T |
| C00520 | Direct ship ES, EA and EP operations | T |
| C00530 | Interpret Electronic Warfare orders | T |
| C00540 | Perform threat analysis | T |
| C00580 | React to changes in the EW environment | T |
| C00650 | Interpret EMCON policies | T |
| C00670 | Track High Interest Vessels | T |
| **Duty Area D - Fire Control (FC)** | | **T** |
| D00010 | Operate Anti Surface Warfare (ASuW) weapons and sensors | |
| D00060 | Operate Anti Air Warfare (AAW) weapons and sensors | T |
| D00090 | Supervise FC equipment operator functional checks | NT |
| D00110 | Supervise FC system tracking and assessments | NT |
| D00120 | Coordinate radar anti-jamming procedures | T |
| D00130 | Supervise FC systems operation | T |
| **Duty Area E - Above Water Warfare (AWW)** | | |
| E00030 | Assess AWW threat information | T |
| E00040 | Coordinate ASuW | T |
| E00050 | Coordinate compilation of RMP | T |
| E00060 | Coordinate Local AAW (LAAW) | T |
| E00070 | Tactically employ aircraft assigned to force | T |
| **Duty Area F - Operational & Analysis Centers** | | |

| | | |
|---|---|---|
| F00010 | Amplify track information | T |
| F00080 | Conduct/Participate in technical evaluations, tests and trials | NT |
| F00120 | Conduct/Participate threat analysis including the collection, analysis and dissemination of electronic data | T |
| F00180 | Identify Maritime, Air, and Land based platforms | T |
| F00190 | Maintain Recognized Maritime Picture (RMP) | T |
| F00210 | Operate Command, Control, Communications, Computers and Intelligence (C4I) systems | T |
| F00230 | Provide information requested by Request For Information (RFI) | NT |
| **Duty Area G - Training and Education** | | |
| G00010 | Assist in individual training events | T |
| G00020 | Conduct individual training events | T |
| G00040 | Operate simulator/trainer systems | NT |
| G00050 | Plan / design training exercises | T |
| G00070 | Assess individual training performance | T |
| G00090 | Conduct team training events | T |
| G00110 | Update training scenarios | T |
| G00140 | Assess team training performance | T |
| G00180 | Assist in team training events | T |
| G00230 | Input training scenarios | NT |

# Record of Proceedings

Dates: 13 -24 May 2019

Location: CFB Stadacona, S15

Working Group Members:

| Surname | Initials | Rank | MOSID | Unit |
|---|---|---|---|---|
| Liebersbach | A.R. | Lt(N) | 00207 | NTDC (A) |
| Cushing | R.A. | CPO2 | 00115 | NTDC (A) |
| Murray | M.P. | PO1 | 00115 | NTDC (A) |
| Sparkes | J. | PO1 | 00115 | NFS(A) |
| Hillier | C.M. | PO1 | 00115 | NFS(A) |
| Baker | J. | PO1 | 00115 | ST(A) |
| Jawornicki | B. | SLt | 00211 | NTDC (A) |

Working Group Objective:

The objective of this working group is to amend the training currently being instructed. A QSP was previously designed and developed that didn't make it past pilot stage and NTDC (A) approval. This new QSP was written with a structure that emulates the performance of the job with a focus on job specific tasks.

Specification Task – No Train Decisions:

| Task # | Task Statement | Substantiation |
|---|---|---|
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |

Discussions / Decisions:

1. Inclusion of naval fire support
2. Assessment of naval gunfire support not/is conducted in PC 004. TDO recommended to assess Naval Fire Support gunnery or strike missions in PO 004 as it is already assessed in PO 001.
3. Written assessment for zippo
4. LSF and FBA assessment in PC 004
5. Training individual and team training related tasks

Guidance to Development and/or Instructional Staff:

6. All courseware identified throughout in resources must be developed and delivered to the TE prior to course roll out.

Course History:

7. Based on the QSP that was initially piloted:
8. NTDC (A) WG (Spring 2017) changed the following:

a. The MARS AWWD/NESOP RQ-PO1 course prior to this review was partially performance based, but measured using an unsustainable metric. Current trainers meet required standard for spiral 1 requirements. Although there is ongoing development of CMS and trainer software and functionality, it will only enhance the designed training package:

| | **Min/coast** | **Optimal/coast** | **Max/coast** |
|---|---|---|---|
| **Students** | 4 | 6 | 8 |
| **Periods** | 480 | 480 | 480 |
| **Special Periods** | 80 | 80 | 80 |
| **Total Periods** | 560 | 560 | 560 |
| **Training Days** | 80 | 80 | 80 |
| **Reduction** | 30% | 30% | 30% |

b. The training design limits the ability to reduce the course duration based on student load; however, the overall EC/PC designs dramatically shrink the required amount of evaluation time and therefore limits the impact evaluations have on core training.

c. The future intent of combined PO 012 PC evaluation for both coasts simultaneously will provide both time savings and staff time savings. This methodology will also allow for increased output if required, increasing 50% from 24/yr. to up to 32/yr. if required.

d. Principle design elements have removed the subjective legacy course evaluation system and concept organization and replaced it with a task/function based organization with legible and observable metric evaluations.

e. Flexibility has been integrated into the evaluation structure to provide optimal course loading in the event of a student failure. The number and scope of POs/PCs ensures that a student failure in one module allows for the RTU and re-loading for that same module on the following serial. This also allows for the possibility of a hybrid course where 4 students all requiring return to the course at the same module could be scheduled for a course beginning from that module and proceeding without the additional staff and school infrastructure needs of a full course.

f. This review has achieved all the milestones set out by CO NTDC (A) and CNPTG. The AWW section has attained a 30% reduction, modularized for future platforms, made practical and task based, made TEL compatible, and increased the knowledge base and threshold for the course.

9. NTDC (A) CAAR (Spring 2018) change the following;

a. Following the completion of the QSP pilot course, it was established that more Mission Phase runs for each trainee would be beneficial. Course was extended by 5 days.

b. Following the removal of the IWSQ examination and Weapon Certification phase on this course, it was strongly recommended by both the trainees and the NFS (A) instructors as well as other external sources that it be reinstated. NTDC(A) agreed with that recommendation;

    i. PC 003 has been amended and is now divided in two sections, section II being the IWSQ examination; and

    ii. PO 004 has been amended to be a practical AWW Sensors and Weapon Procedures including a practical PC.

c. Both the trainees and NFS (A) instructors indicated that PO 004 Advise Command on Above Water Warfare Manoeuvering benefits were minimal. NTDC (A) replaced Advice Command on Above Water Warfare Manoeuvering by Conduct Above Water Warfare Sensors and Weapon Procedures.

# Assessment Plan

| EC/PC | Scope Covered | Purpose | How | Pass/Fail Criteria |
|---|---|---|---|---|
| EC 001.01 | EO 001.01 | To assess student's competence in Directing the Use of AWW Weapons, Sensors and Associated Equipment | Practical | Student must obtain 32/39 SATs and all Criticals IAW EC 001.01 Checklist |
| EC 001.02 | EO 001.02 | To assess student's competence in Conducting Strike Missions | Practical | Student must obtain 4/5 SATs and all Criticals IAW EC 001.02 Checklist |
| PC 001.01 | PO 001 | To assess student's competence in Operating AWW Weapons and Sensors | Written | Student must obtain 80% IAW PC 001 list |
| PC 001.02 | PO 001 | To assess student's competence in Operating AWW Weapons and Sensors | Practical | Student must obtain 36/44 SATs and all Criticals IAW PC 001 Checklist |
| EC 002.01 | EO 002.01 | To assess student's competence in Planning and Designing Training Exercises | Practical | Student must obtain 10/13 SATs IAW EC 002.01 Checklist |
| EC 002.02 | EO 002.02 | To assess student's competence in Preparing Operational Training Exercises | Written / Practical | Student must obtain 8/10 SATs IAW EC 002.02 Checklist |
| EC 002.03 | EO 002.03 | To assess student's competence in Advising Command on AAW/ASuW Safety | Written / Practical | Student must obtain 8/10 SATs IAW EC 002.03 Checklist |
| PC 002 | PO 002 | To assess student's competence in Ensuring the Operational Readiness of AWW Teams | Written / Practical | Student must obtain 9/12 SATs IAW PC 002 Checklist |
| EC 003.01 | EC 003.01 | To assess student's competence in Performing Threat Analysis | Written / Practical | Student must obtain 8/10 SATs IAW EC 003.01 Checklist |
| EC 003.02 | EC 003.02 | To assess student's competence in Preparing Operational Messages | Written / Practical | Student must obtain 3/3 SATs IAW EC 003.02 Checklist |
| PC 003 | PO 003 | To assess student's competence in Planning AWW Operations | Written / Practical | Student must obtain 7/8 SATs IAW PC 003 Checklist |
| EC 004.01 | EO 004.01 | To assess student's competence in Maintaining the RMP | Written / Practical | Student must obtain 18/23 SATs IAW EC 004.01 Checklist |
| EC 004.02.01 | EO 004.02 | To assess student's competence in Defending Against Missile Threats | Written / Practical | Student must obtain 37/41 SATs IAW EC 004.02.01 Checklist |
| EC 004.02.02 | EO 004.02 | To assess student's competence in Defending Against Missile Threats | Written / Practical | Student must obtain 13/17 SATs IAW EC 004.02.02 Checklist |
| EC 004.03 | EO 004.03 | To assess student's competence in Defending Against Surface Threats | Written / Practical | Student must obtain 25/31 SATs and all Criticals IAW EC 004.03 Checklist |

| | | | | |
|---|---|---|---|---|
| EC 004.04 | EO 004.04 | To assess student's competence in Defending Against Air Threats | Written / Practical | Student must obtain 22/27 SATs and all Criticals IAW EC 004.04 Checklist |
| PC 004 | PO 004 | To assess student's competence in Executing the Duties of an AWW Director | Written / Practical | Students must receive a SAT in each section IAW EC 004.04 Checklist and pass 2/3 runs. |

# TEL Development Opportunities

| Title | Type | Description of Requirement | Lesson | SOR # |
|---|---|---|---|---|
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# Scheduling Information

| Scheduling Information | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Lesson | | Location | IT Assets | | | | | Footprint | | | Instructor | | |
| | | | | | | | | | | | | | |
| # | Time | Location | DWAN | CSNI | SSA | Proj | TEL | Class | WB | Desk | Type | Qualification | Sec Cl |
| 001.01.01 | 150 | Classroom | | | 8 | | - | SEC | 1 | 8 | MIL | AWWD | SEC |
| 001.01.02 | 250 | Classroom | | | | 1 | | SEC | 1 | 8 | MIL | RQ-PO2<br>NCIOP+<br>AWWD | SEC |
| 001.01.03 | 200 | Classroom | | | | 1 | | UNC | 1 | 8 | MIL | TDL 300 | SEC |
| 001.01.04 | 150 | Classroom | | | | 1 | | SEC | 1 | 8 | MIL | RQ-PO2<br>NESOP | SEC |
| 001.01.05 | 150 | Classroom | | | | 1 | | SEC | 1 | 8 | MIL | RQ-PO2<br>NESOP | SEC |
| 001.01.06 | 200 | Classroom | | | | 1 | | SEC | 1 | 8 | MIL | RQ-PO2<br>NESOP | SEC |
| 001.01.07 | 50 | SEACOT | | | | 1 | | SEC | 1 | 8 | MIL | RQ-PO2<br>NESOP | SEC |
| 001.01.08 | 200 | Classroom | | | | | | SEC | 1 | 8 | MIL | AWWD | SEC |
| 001.01.09 | 200 | Classroom | | | 8 | | | SEC | 1 | 8 | MIL | AWWD | SEC |
| 001.01.10 | 150 | Classroom | | | | 1 | | SEC | 1 | 8 | MIL | As Req | SEC |
| 001.01.11 | 150 | Classroom | | | | | | SEC | 1 | 8 | MIL | AWWD | SEC |
| 001.01.12 | 200 | Tour | | | | | | UNC | | | MIL | AWWD | SEC |
| 001.01.13 | 50 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| 001.01.14 | 200 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| 001.01.15 | 100 | SEACOT | | | | | | SEC | | | MIL | AWWD<br>RQ-MS<br>NESOP | SEC |
| 001.01.16 | 450 | SEACOT | | | | | | SEC | | | MIL | AWWD<br>RQ-MS<br>NESOP | SEC |
| 001.01.17 | 200 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| 001.01.18 | 150 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| EC 001.01 | 800 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |

| Scheduling Information | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Lesson | | Location | IT Assets | | | | | Footprint | | | Instructor | | |
| | | | | | | | | | | | | | |
| # | Time | Location | DWAN | CSNI | SSA | Proj | TEL | Class | WB | Desk | Type | Qualification | Sec Cl |
| 001.02.01 | 300 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| 001.02.02 | 200 | Class/Tour | | | 8 | | | SEC | | 8 | MIL | MCC SO Tgt | SEC |
| 001.02.03 | 150 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| EC 001.02 | 350 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| PC 001A | 50 | Classroom | | | | | | SEC | | 8 | MIL | AWWD | SEC |
| PO 001.01 | 3200 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| PC 001.02 | 1600 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| **PO 002** | | | | | | | | | | | | | |
| 002.01.01 | 150 | Classroom | | | | 1 | | UNC | 1 | 8 | MIL | ORO | SEC |
| 002.01.02 | 50 | Classroom | 8 | | | 1 | | UNC | 1 | 8 | MIL | AWWD | SEC |
| 002.01.03 | 50 | Classroom | | | | 1 | | UNC | 1 | 8 | MIL | AWWD + Target Cell | SEC |
| 002.01.04 | 50 | Classroom | 8 | | | 1 | | UNC | 1 | 8 | MIL | STA/ NFR/ Fleet | SEC |
| 002.01.05 | 100 | SEACOT | | | | | | SEC | | | MIL | AWWD/ RQ-PO1 NCIOP | SEC |
| 002.01.06 | 100 | Classroom | 8 | | | 1 | | UNC | | 8 | MIL | AWWD | SEC |
| EC 002.01 | 400 | Classroom | | | 8 | 1 | | SEC | | 8 | MIL | AWWD | SEC |
| 002.02.01 | 300 | Classroom | | 8 | | | | SEC | | 8 | MIL | AWWD | SEC |
| 002.02.02 | 250 | Classroom | | 8 | | 1 | | SEC | | 8 | MIL | AWWD | SEC |
| 002.02.03 | 100 | Classroom | 8 | | | | | UNC | | | MIL | AWWD | SEC |
| 002.02.04 | 50 | Classroom | | | | | | UNC | | 8 | MIL | AWWD | SEC |
| EC 002.02 | 200 | Classroom | | | 8 | | | SEC | | 8 | MIL | AWWD | SEC |
| 002.03.01 | 100 | Classroom | | | 8 | | | UNC | | 8 | MIL | AWWD | SEC |
| 002.03.02 | 150 | Classroom | | | 8 | 1 | | UNC | | | MIL | AWWD | SEC |
| EC 002.03 | 550 | Classroom | | | 8 | 1 | | SEC | | | MIL | AWWD | SEC |
| PC 002 | 600 | Classroom | | | 8 | 1 | | SEC | | 8 | MIL | AWWD | SEC |
| **PO 003** | | | | | | | | | | | | | |
| PO 003 | 150 | Classroom | | 8 | | | | SEC | | 8 | MIL | AWWD | SEC |
| 003.01.01 | 200 | DMOC | | 8 | | | | SEC | | 8 | MIL | AWWD | SEC |

| Scheduling Information | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Lesson | | Location | IT Assets | | | | | Footprint | | | Instructor | | |
| | | | | | | | | | | | | | |
| # | Time | Location | DWAN | CSNI | SSA | Proj | TEL | Class | WB | Desk | Type | Qualification | Sec Cl |
| 003.01.02 | 200 | Classroom | | | | 1 | SEC VTC (west) | SEC | 1 | 8 | MIL | TRINITY | SEC |
| 003.01.03 | 100 | Classroom | | 8 | | | | SEC | | 8 | MIL | AWWD | SEC |
| 003.01.04 | 300 | Classroom | | 8 | | | | SEC | | 8 | MIL | AWWD | SEC |
| 003.01.05 | 100 | DMOC | | 8 | | | | SEC | | 8 | MIL | AWWD | SEC |
| 003.01.06 | 50 | Classroom | | | 1 | 1 | | SEC | | 8 | MIL | AWWD | SEC |
| EC 003.01 | 400 | Classroom | | | 8 | 1 | | SEC | | 8 | MIL | AWWD | SEC |
| 003.02.01 | 150 | Classroom | | | 8 | | | SEC | | 8 | MIL | AWWD | SEC |
| 003.02.02 | 100 | Classroom | 8 | | | 1 | | SEC | | 8 | MIL | JAG | SEC |
| 003.02.03 | 150 | Classroom | | | 8 | | | SEC | | 8 | MIL | AWWD | SEC |
| 003.02.04 | 100 | Classroom | | 8 | | | | SEC | | 8 | MIL | AWWD | SEC |
| 003.02.05 | 100 | Classroom | | 8 | | | | SEC | | 8 | MIL | AWWD | SEC |
| 003.02.06 | 150 | Classroom | | | 8 | | | SEC | | 8 | MIL | AWWD | SEC |
| 003.02.07 | 100 | Classroom | | | 8 | | | SEC | | 8 | MIL | AWWD | SEC |
| EC 003.02 | 200 | Classroom | | | 8 | | | SEC | | 8 | MIL | AWWD | SEC |
| PC 003 | 600 | Classroom | | | 8 | 1 | | SEC | | 8 | MIL | AWWD /ORO | SEC |
| PO 004 | | | | | | | | | | | | | |
| 004.01.01 | 200 | Classroom | | | 8 | 1 | | SEC | 1 | 8 | MIL | ACSO/MFC | SEC |
| 004.01.02 | 50 | Classroom | | | | 1 | | SEC | 1 | 8 | MIL | ACSO | SEC |
| 004.01.03 | 150 | Classroom | | | | 1 | | SEC | 1 | 8 | MIL | AWWD | SEC |
| 004.01.04 | 150 | Classroom | | | 8 | 1 | | SEC | 1 | 8 | MIL | AWWD | SEC |
| 004.01.05 | 200 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| 004.01.06 | 200 | SEACOT | | | | | | SEC | | | MIL | RQ-PO2 NCIOP | SEC |
| 004.01.07 | 1000 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| EC 004.01 | 400 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| 004.02.01 | 200 | Classroom | | 8 | | 1 | | SEC | | 8 | MIL | AWWD | SEC |
| 004.02.02 | 100 | Classroom | | | 8 | | | SEC | | 8 | MIL | AWWD | SEC |
| 004.02.03 | 50 | Classroom | | | 8 | 1 | | SEC | 1 | 8 | MIL | AWWD | SEC |

| Scheduling Information | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Lesson | | Location | IT Assets | | | | | Footprint | | | Instructor | | |
| | | | | | | | | | | | | | |
| # | Time | Location | DWAN | CSNI | SSA | Proj | TEL | Class | WB | Desk | Type | Qualification | Sec Cl |
| 004.02.04 | 150 | Classroom | | | 8 | | | SEC | 1 | 8 | MIL | AWWD/ RQ-PO2 NESOP | SEC |
| 004.02.05 | 400 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| EC004.02.01 | 50 | Classroom | | | | | | SEC | | 8 | MIL | AWWD | SEC |
| EC004.02.02 | 400 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| 004.03.01 | 350 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| 004.03.02 | 200 | Classroom | | | 8 | | | SEC | 1 | 8 | MIL | AWWD | SEC |
| 004.03.03 | 100 | Classroom | | | 8 | | | SEC | 1 | 8 | MIL | AWWD | SEC |
| 004.03.04 | 250 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| EC004.03.05 | 600 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| 004.04 | 400 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| EC004.04 | 400 | SEACOT | | | | | | SEC | | | MIL | AWWD | SEC |
| PO 004 | 8000 | SEAWOLF | | | | | | SEC | | | MIL | AWWD | SEC |
| PC 004 | 4800 | SEAWOLF | | | | | | SEC | | | MIL | AWWD | SEC |

# Plan, Train, Do

| PO | Plan | Train | Execute |
|---|---|---|---|
| **001** | | Operate AWW Weapons and Sensors;<br>Written Weapon System Cert Exam | Weapon System Cert Practical |
| **002** | Ensure the Operational Readiness of AWW Teams;<br>Plan an AWW Exercise | | |
| **003** | Plan AWW Operations | | |
| **004** | | Execute Duties of an AWW Director | Execute the Duties of an AWW Director |

# Recommended Schedule

| Time | Day 1 | Day 2 | Day3 | Day 4 | Day 5 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Expectations /Admin ISSO ,IT, Command Intro | Direct the AWW Team(Theory)<br>001.01.01 | Employ Tactical Data Links<br>001.01.03<br>Guest (RQ-PO 2 NCIOP ) | Employ Decoys<br>001.01.05<br>(EW Instructor) | Operate Guns (Theory)<br>001.01.08 |
| Period 2<br>0850-0940 | | Employ RADAR theory and IFF<br>001.01.02<br>Guest(RQ-PO2 NCIOP) | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | **BAT** | Employ RADAR theory and IFF<br>001.01.02<br>Guest(RQ-PO2 NCIOP) | Employ Tactical Data Links<br>001.01.03<br>Guest (RQ-PO 2 NCIOP) | Employ EA Systems<br>001.01.06<br>(EW Instructor) | Operate Guns (Theory)<br>001.01.08 |
| Period 4<br>1040-1130 | | **PT** | **PT** | **PT** | |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | **TAT** | Employ RADAR theory and IFF<br>001.01.02<br>Guest(RQ-PO2 NCIOP) | Employ Electronic Support systems<br>001.01.04<br>(EW Instructor) | Employ EA Systems<br>001.01.06<br>(EW Instructor) | Operate Surface to Surface Missiles(Theory) |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Direct the AWW Team(Theory)<br>001.01.01 | Employ RADAR theory and IFF<br>001.01.02<br>Guest(RQ-PO2 NCIOP) | Employ Electronic Support systems<br>001.01.04<br>(EW Instructor) | Employ EA Systems<br>001.01.06<br>(EW Instructor) | Operate Surface to Surface Missiles(Theory)<br>001.01.09 |
| Period 8<br>1510-1600 | Direct the AWW Team(Theory)<br>001.01.01 | Employ Tactical Data Links<br>001.01.03 Guest (RQ-PO2 NCIOP) | Employ Decoys<br>001.01.05<br>(EW Instructor) | Employ Electro-Optic/Infra-red Sensors<br>001.01.07<br>(EW Instructor) | |

## Recommended Schedule

| Time | Day 6 | Day 7 | Day 8 | Day 9 | Day 10 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Ballistic Theory<br>001.01.10<br>(Guest From CFMWC/Artillery)or AWWD Instructor | Equipment Familiarization Tour<br>001.01.12<br>DDA/NAD/west coast Gun Bay | Operate CMS<br>001.01.14 | Operate Guns<br>001.01.16 | Operate Surface to Air Missiles<br>001.01.17 |
| Period 2<br>0850-0940 | | | Operate FC RADARS<br>001.01O<br>AWWD /RQ-MS Instructor | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Ballistic Theory<br>001.01.10<br>(Guest From CFMWC/Artillery)or AWWD Instructor | Equipment Familiarization Tour<br>001.01.12<br>DDA/NAD/west coast Gun Bay | Operate FC RADARS<br>001.01O<br>AWWD /RQ-MS Instructor | Operate Guns<br>001.01.16 | Operate Surface to Surface Missiles<br>001.01.18 |
| Period 4<br>1040-1130 | **PT** | | **PT** | **BAT** | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Operate Surface to Air Missiles<br>001.01.11 | Direct the AWW team<br>001.01.13 | Operate Guns<br>001.01.16 | Operate Guns<br>001.01.16 | Operate Surface to Surface Missiles<br>001.01.18 |
| Period 6<br>1320-1410 | | Operate CMS<br>001.01.14 | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Operate Surface to Air Missiles<br>001.01.11 | Operate CMS<br>001.01.14 | Operate Guns<br>001.01.16 | Operate Surface to Air Missiles<br>001.01.17 | Direct the AWW team weapons ,Sensors and Associated equipment<br>EC 001.01 |
| Period 8<br>1510-1600 | **TAT** | | | | |

## Recommended Schedule

| Time | Day 11 | Day 12 | Day 13 | Day 14 | Day 15 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Direct the AWW team weapons ,Sensors and Associated equipment EC 001.01 | Direct the AWW team weapons ,Sensors and Associated equipment EC 001.01 | Naval Gun Fire Support 001.02.01 | Targeting 001.02.02 Guest MCC SO Targeting) | Conduct Strike Mission EC 001.02 |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Direct the AWW team weapons ,Sensors and Associated equipment EC 001.01 | Direct the AWW team weapons ,Sensors and Associated equipment EC 001.01 | Naval Gun Fire Support 001.02.01 | Targeting 001.02.02 Guest MCC SO Targeting) | Conduct Strike Mission EC 001.02 |
| Period 4<br>1040-1130 | **PT** | **TAT** | **PT** | **BAT** | |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Direct the AWW team weapons ,Sensors and Associated equipment EC 001.01 | Direct the AWW team weapons ,Sensors and Associated equipment EC 001.01 | Naval Gun Fire Support 001.02.01 | Strike Missions 001.02.03 | Conduct Strike Mission EC 001.02 |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Direct the AWW team weapons ,Sensors and Associated equipment EC 001.01 | Direct the AWW team weapons ,Sensors and Associated equipment EC 001.01 | Naval Gun Fire Support 001.02.01 | Strike Missions 001.02.03 | Conduct Strike Mission EC 001.02 |
| Period 8<br>1510-1600 | | | Targeting 001.02.02 Guest MCC SO Targeting) | **PT** | Operate AWW weapons and Sensors PC 001A (written) |

## Recommended Schedule

| Time | Day 16 | Day 17 | Day 18 | Day 19 | Day 20 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Intro to MTP<br>002.01.01<br>Guest(OROCO/ORO CCD Manager) | SETT/SORTI/SAF SIM<br>002.01.05<br>(Guest RQ-PO01 6B) | Plan/Design Training Exercises<br>EC 002.01 | Drafting Request for Service Support Messages<br>002.02.01 | Drafting TABORD's<br>002.02.02 |
| Period 2<br>0850-0940 | | Serial CONOPS Brief<br>002.01.06 | | | Contribute to FLEX Development<br>002.02.03 |
| 0940-0950 | | | | | |
| Period 3<br>0950-1040 | Intro to MTP<br>002.01.01<br>Guest(OROCO/ORO CCD Manager) | Serial CONOPS Brief<br>002.01.06 | Plan/Design Training Exercises<br>EC 002.01 | Drafting Request for Service Support Messages<br>002.02.01 | Contribute to FLEX Development<br>002.02.03 |
| Period 4<br>1040-1130 | **PT** | **TAT** | **PT** | **BAT** | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Combat Readiness Requirements<br>002.01.02 | Plan/Design Training Exercises<br>EC 002.01 | Plan/Design Training Exercises<br>EC 002.01 | Drafting TABORD's<br>002.02.02 | Draft Firing Orders<br>002.02.04 |
| Period 6<br>1320-1410 | References for Planning Training Exercises<br>002.01.03<br>(Guest Target Cell Instructor) | | Drafting Request for Service Support Messages<br>002.02.01 | | Prepare operational Training Exercises<br>EC 002.02 |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Ship/Fleet Training<br>002.01.04<br>(Guest CCFL/P or Sea Training/NFR) | Plan/Design Training Exercises<br>EC 002.01 | Drafting Request for Service Support Messages<br>002.02.01 | Drafting TABORD's<br>002.02.02 | Prepare operational Training Exercises<br>EC 002.02 |
| Period 8<br>1510-1600 | SETT/SORTI/SAF SIM<br>002.01.05<br>(Guest RQ-PO01 6B) | | | | |

## Recommended Schedule

| Time | Day 21 | Day 22 | Day 23 | Day 24 | Day 25 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Prepare operational Training Exercises<br>EC 002.02 | Advise Command on Anti-Air/Anti-Surface Warfare<br>EC 002.03 | Advise Command on Anti-Air/Anti-Surface Warfare<br>EC 002.03 | Ensure Operational Readiness of AWW Teams<br>PC 002 | Ensure Operational Readiness of AWW Teams<br>PC 002 |
| Period 2<br>0850-0940 | Weapons Firing Safety<br>002.03.01 | | | | Consolidation<br>PO 001 |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Weapons Firing Safety<br>002.03.01 | Advise Command on Anti-Air/Anti-Surface Warfare<br>EC 002.03 | Advise Command on Anti-Air/Anti-Surface Warfare<br>EC 002.03 | Ensure Operational Readiness of AWW Teams<br>PC 002 | Consolidation<br>PO 001 |
| Period 4<br>1040-1130 | **PT** | | **PT** | **BAT** | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Creating a Pre-Firing Brief<br>002.03.02 | Advise Command on Anti-Air/Anti-Surface Warfare<br>EC 002.03 | Ensure Operational Readiness of AWW Teams<br>PC 002 | Ensure Operational Readiness of AWW Teams<br>PC 002 | Consolidation<br>PO 001 |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Creating a Pre-Firing Brief<br>002.03.02 | Advise Command on Anti-Air/Anti-Surface Warfare<br>EC 002.03 | Ensure Operational Readiness of AWW Teams<br>PC 002 | Ensure Operational Readiness of AWW Teams<br>PC 002 | Consolidation<br>PO 001 |
| Period 8<br>1510-1600 | **TAT** | | | | |

## Recommended Schedule

| Time | Day 26 | Day 27 | Day 28 | Day 29 | Day 30 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 |
| Period 4<br>1040-1130 | **PT** | **TAT** | **PT** | **BAT** | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 |
| Period 8<br>1510-1600 | | | | | |

## Recommended Schedule

| Time | Day 31 | Day 32 | Day 33 | Day 34 | Day 35 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Operate AWW weapons and Sensors<br>PC 001B |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Operate AWW weapons and Sensors<br>PC 001B | Operate AWW weapons and Sensors<br>PC 001B |
| Period 4<br>1040-1130 | **PT** | **TAT** | **PT** | **BAT** | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Operate AWW weapons and Sensors<br>PC 001B | Operate AWW weapons and Sensors<br>PC 001B |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Consolidation<br>PO 001 | Operate AWW weapons and Sensors<br>PC 001B | Operate AWW weapons and Sensors<br>PC 001B |
| Period 8<br>1510-1600 | | | | | |

## Recommended Schedule

| Time | Day 36 | Day 37 | Day 38 | Day 39 | Day 40 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Operate AWW weapons and Sensors<br>PC 001B | Operate AWW weapons and Sensors<br>PC 001B | Operate AWW weapons and Sensors<br>PC 001B | Introduction to the Mission<br>PO 003<br>INTRO | TRINITY Support<br>003.01.02<br>(Guest from TRINITY) |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Operate AWW weapons and Sensors<br>PC 001B | Operate AWW weapons and Sensors<br>PC 001B | Operate AWW weapons and Sensors<br>PC 001B | Introduction to the Mission<br>PO 003<br>INTRO | TRINITY Support<br>003.01.02<br>(Guest from TRINITY) |
| Period 4<br>1040-1130 | **PT** | **TAT** | **PT** | **PT** | |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Operate AWW weapons and Sensors<br>PC 001B | Operate AWW weapons and Sensors<br>PC 001B | Operate AWW weapons and Sensors<br>PC 001B | Sources of Intelligence Data<br>003.01.01 | Drafting RFI's<br>003.01.03 |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Operate AWW weapons and Sensors<br>PC 001B | Operate AWW weapons and Sensors<br>PC 001B | Operate AWW weapons and Sensors<br>PC 001B | Sources of Intelligence Data<br>003.01.01 | Determine Effectiveness of Weapons and Sensors<br>003.01.04 |
| Period 8<br>1510-1600 | | | **BAT** | Sources of Intelligence Data<br>003.01.01 | |

## Recommended Schedule

| Time | Day 41 | Day 42 | Day 43 | Day 44 | Day 45 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Determine Effectiveness of Weapons and Sensors<br>003.01.04 | Perform Threat Analysis<br>EC 003.01 | Interpret Command Structure and Strategic Guidance<br>003.02.01 | Interpret Operational Messages<br>003.02.03 | Identify and Classify Contacts<br>003.02.06 |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Determine Effectiveness of Weapons and Sensors<br>003.01.04 | Perform Threat Analysis<br>EC 003.01 | Interpret Command Structure and Strategic Guidance<br>003.02.01 | Interpreting OPTASK AWW/AD<br>003.02.04 | Drafting OPTASK Supp Messages<br>003.02.07 |
| Period 4<br>1040-1130 | **PT** | | **PT** | **BAT** | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Determine Effectiveness of Weapons and Sensors<br>003.01.04 | Perform Threat Analysis<br>EC 003.01 | **TAT** | Interpreting OPTASK AWW/AD<br>003.02.04 | Drafting OPTASK Supp Messages<br>003.02.07 |
| Period 6<br>1320-1410 | Compile OOB/ORBAT<br>003.01.05 | | Rules of Engagement<br>003.02.02 | Interpreting OPTASK ASuW/SCC<br>003.02.05 | Prepare Operational Messages<br>EC 003.02 |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Compile OOB/ORBAT<br>003.01.05 | Perform Threat Analysis<br>EC 003.01 | Rules of Engagement<br>003.02.02 | Interpreting OPTASK ASuW/SCC<br>003.02.05 | Prepare Operational Messages<br>EC 003.02 |
| Period 8<br>1510-1600 | Briefing Threat<br>0003.01.06 | | Interpret Operational Messages<br>003.02.03 | Identify and Classify Contacts<br>003.02.06 | |

## Recommended Schedule

| Time | Day 46 | Day 47 | Day 48 | Day 49 | Day 50 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Prepare Operational Messages<br>EC 003.02 | Plan AWW Operations<br>PC 003 | Employ Air assets and CAP<br>004.01.01 (Guest LRPA ACSO or RCAF MFC) | Resolve Procedures and Warnings<br>004.01.04 | ID and Data Fusion<br>004.01.06<br>(Guest RQ-PO2 NCIOP) |
| Period 2<br>0850-0940 | Plan AWW Operations<br>PC 003 | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Plan AWW Operations<br>PC 003 | Plan AWW Operations<br>PC 003 | Employ Air assets and CAP<br>004.01.01 (Guest LRPA ACSO or RCAF MFC) | Resolve Procedures and Warnings<br>004.01.04 | ID and Data Fusion<br>004.01.06<br>(Guest RQ-PO2 NCIOP) |
| Period 4<br>1040-1130 | **PT** | | | **BAT** | |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Plan AWW Operations<br>PC 003 | Plan AWW Operations<br>PC 003 | Over the Horizon Targeting<br>004.01.02<br>(Guest LRPA ACSO) | Communications<br>004.01.05 | **PT** |
| Period 6<br>1320-1410 | | | Utilize ESM Information<br>004.01.03<br>EW Instructor | | RMP Management<br>004.01.07 |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Plan AWW Operations<br>PC 003 | **TAT** | Utilize ESM Information<br>004.01.03<br>EW Instructor | Communications<br>004.01.05 | RMP Management<br>004.01.07 |
| Period 8<br>1510-1600 | | **PT** | | | |

## Recommended Schedule

| Time | Day 51 | Day 52 | Day 53 | Day 54 | Day 55 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | RMP Management<br>004.01.07 | RMP Management<br>004.01.07 | RMP Management<br>004.01.07 | Maintain Recognized Picture<br>EC 004.01 | Zippos<br>004.02.02 |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | RMP Management<br>004.01.07 | RMP Management<br>004.01.07 | RMP Management<br>004.01.07 | Maintain Recognized Picture<br>EC 004.01 | Employ Hardkill<br>004.02.03 |
| Period 4<br>1040-1130 | **PT** | **TAT** | **PT** | | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | RMP Management<br>004.01.07 | RMP Management<br>004.01.07 | Maintain Recognized Picture<br>EC 004.01 | Anti-Ship Missiles<br>004.02.01 | Employ Softkill<br>004.02.04<br>(RQ-PO2 NESOP or AWWD) |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | RMP Management<br>004.01.07 | RMP Management<br>004.01.07 | Maintain Recognized Picture<br>EC 004.01 | Anti-Ship Missiles<br>004.02.01 | Employ Softkill<br>004.02.04<br>(RQ-PO2 NESOP or AWWD) |
| Period 8<br>1510-1600 | | | | | **BAT** |

## Recommended Schedule

| Time | Day 56 | Day 57 | Day 58 | Day 59 | Day 60 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Detect to Engage Sequence<br>004.02.05 | Detect to Engage Sequence<br>004.02.05 | Defend Against Missile Threat EC 004.02.02 | Surface Engagement Plans and FAC/FIAC Defence<br>004.03.01 | Tactical Employment of AHWACS Theory<br>004.03.03 |
| Period 2<br>0850-0940 | | Defend Against Missile Threats<br>EC 004.02.01 | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Detect to Engage Sequence<br>004.02.05 | Defend Against Missile Threat EC 004.02.02 | Defend Against Missile Threat EC 004.02.02 | Surface Engagement Plans and FAC/FIAC Defence<br>004.03.01 | Tactical Employment of AHWACS Practical<br>004.03.04 |
| Period 4<br>1040-1130 | **PT** | **TAT** | **PT** | **BAT** | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Detect to Engage Sequence<br>004.02.05 | Defend Against Missile Threat EC 004.02.02 | Surface Engagement Plans and FAC/FIAC Defence<br>004.03.01 | Surface Action Groups<br>004.03.02 | Tactical Employment of AHWACS Practical<br>004.03.04 |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Detect to Engage Sequence<br>004.02.05 | Defend Against Missile Threat EC 004.02.02 | Surface Engagement Plans and FAC/FIAC Defence<br>004.03.01 | Surface Action Groups<br>004.03.02 | Tactical Employment of AHWACS Practical<br>004.03.04 |
| Period 8<br>1510-1600 | | | | | |

## Recommended Schedule

| Time | Day 61 | Day 62 | Day 63 | Day 64 | Day 65 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Defend Against Surface Targets<br>EC 004.03 | Defend Against Surface Targets<br>EC 004.03 | FBA/LSF Doctrine<br>004.04 | Defend Against Air Threat<br>EC 004.04 | Consolidation<br>PO 004 |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Defend Against Surface Targets<br>EC 004.03 | Defend Against Surface Targets<br>EC 004.03 | FBA/LSF Doctrine<br>004.04 | Defend Against Air Threat<br>EC 004.04 | Consolidation<br>PO 004 |
| Period 4<br>1040-1130 | **PT** | **TAT** | **PT** | | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Defend Against Surface Targets<br>EC 004.03 | Defend Against Surface Targets<br>EC 004.03 | FBA/LSF Doctrine<br>004.04 | Defend Against Air Threat<br>EC 004.04 | Consolidation<br>PO 004 |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Defend Against Surface Targets<br>EC 004.03 | FBA/LSF Doctrine<br>004.04 | FBA/LSF Doctrine<br>004.04 | Defend Against Air Threat<br>EC 004.04 | Consolidation<br>PO 004 |
| Period 8<br>1510-1600 | | | **BAT** | | |

## Recommended Schedule

| Time | Day 66 | Day 67 | Day 68 | Day 69 | Day70 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 4<br>1040-1130 | **PT** | **TAT** | **PT** | **BAT** | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 8<br>1510-1600 | | | | | |

## Recommended Schedule

| Time | Day 71 | Day 72 | Day 73 | Day 74 | Day75 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 4<br>1040-1130 | **PT** | **TAT** | **PT** | **BAT** | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 8<br>1510-1600 | | | | | |

## Recommended Schedule

| Time | Day 76 | Day 77 | Day 78 | Day 79 | Day 80 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 4<br>1040-1130 | **PT** | **TAT** | **PT** | **BAT** | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 8<br>1510-1600 | | | | | |

## Recommended Schedule

| Time | Day 81 | Day 82 | Day 83 | Day 84 | Day 85 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 4<br>1040-1130 | **PT** | **TAT** | **PT** | **BAT** | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Consolidation<br>PO 004 |
| Period 8<br>1510-1600 | | | | | |

## Recommended Schedule

| Time | Day 86 | Day 87 | Day 88 | Day 89 | Day 90 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Execute Duties of an<br>AWW Director<br>PC 004 | Execute Duties of an<br>AWW Director<br>PC 004 | Execute Duties of an<br>AWW Director<br>PC 004 |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Execute Duties of an<br>AWW Director<br>PC 004 | Execute Duties of an<br>AWW Director<br>PC 004 | Execute Duties of an<br>AWW Director<br>PC 004 |
| Period 4<br>1040-1130 | **PT** | | **PT** | **BAT** | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Consolidation<br>PO 004 | Consolidation<br>PO 004 | Execute Duties of an<br>AWW Director<br>PC 004 | Execute Duties of an<br>AWW Director<br>PC 004 | Execute Duties of an<br>AWW Director<br>PC 004 |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Consolidation<br>PO 004 | **TAT** | Execute Duties of an<br>AWW Director<br>PC 004 | Execute Duties of an<br>AWW Director<br>PC 004 | Execute Duties of an<br>AWW Director<br>PC 004 |
| Period 8<br>1510-1600 | | **PT** | | | |

## Recommended Schedule

| Time | Day 91 | Day 92 | Day 93 | Day 94 | Day 95 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 |
| Period 4<br>1040-1130 | **PT** | **TAT** | **PT** | **BAT** | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 | Execute Duties of an AWW Director PC 004 |
| Period 8<br>1510-1600 | | | | | |

## Recommended Schedule

| Time | Day 96 | Day 97 | Day 98 | Day 99 | Day 100 |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 |
| Period 4<br>1040-1130 | **PT** | **TAT** | **PT** | **BAT** | **PT** |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 |
| Period 6<br>1320-1410 | | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 | Execute Duties of an AWW Director<br>PC 004 |
| Period 8<br>1510-1600 | | | | | |

## Recommended Schedule

| Time | Day 101 | Day 102 | | | |
|---|---|---|---|---|---|
| Period 1<br>0800-0850 | Execute Duties of an AWW Director<br>PC 004 | **BAT Grad** | | | |
| Period 2<br>0850-0940 | | | | | |
| 0940-0950 | **Break** | | | | |
| Period 3<br>0950-1040 | Execute Duties of an AWW Director<br>PC 004 | PT | | | |
| Period 4<br>1040-1130 | | | | | |
| 1130-1230 | **Lunch** | | | | |
| Period 5<br>1230-1320 | Execute Duties of an AWW Director<br>PC 004 | | | | |
| Period 6<br>1320-1410 | **PT** | | | | |
| 1410-1420 | **Break** | | | | |
| Period 7<br>1420-1510 | **TAT PCD** | | | | |
| Period 8<br>1510-1600 | | | | | |

## ANNEX H – COURSE HISTORY

This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

**Note: This annex shall NOT be deleted**. This annex must be annotated with every update / modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Detail as to the nature and reasons for amendments are to be provided here.

18-Nov 2020 v.2.01 - Chapter 3 added Assessment of Trainee.

18-Nov 2020 v 2.01 - Changes made to PC 004.refinement of marking criteria for PO 004 HPD removed from checklist also explicit direction on the completion of a re-test should the candidate fail two out of three runs but sufficient time remains for a re-test.

15-Jul-2021 v 2.01 - Course updated with mite code and approval dates. No changes to content. Version uploaded to NPTG Sharepoint.

18-Nov-2021 v 2.02- Updated MARTI references IAW CFCD 106 Change 20 doctrine overhaul

30-May-2022 v 2.03- Changes in doctrine necessitated a change to PC 004 Check List to facilitate assessors to capture new methods and brevity enforced through CFCD 106 Change 20. NFS staff from both schools were solicited for input to improve the format and incorporate required changes. Changes limited to word usage as applies to brevity, removal of actions no longer executed, rewording of Hardkill/Softkill directives, post raid expectations, and a re-alignment of scores to reflect those changes.